

Mus 101 Introduction to Music Theory

Numbered Transcript

MUS 101 • Samuel Andreyev

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Lecture 1: The Language of Music

At a glance — Andreyev on the language of music.

- 1.1 Languages and cultures change and disappear all the time. We all know this. Sometimes they disappear because a culture is being oppressed. Sometimes it's an act of will. Sometimes it's a matter of historical contingency. Sometimes things simply change. Sometimes we just forget.
- 1.2 The materials and tools and techniques, the body of shared knowledge of the Western musical tradition, is not literally a language, but it's a codified body of practices that at this point is at least 2,000 years into its evolution. And it could disappear.
- 1.3 It might not look likely from our present vantage point, but if you think about how long it took for us to get here—to develop all the instruments, the scales, the keys, the rhythms, the whole body of materials upon which this art form depends— it was by no means an obvious development. It took place over an extremely lengthy span of time, and it required the input of countless musicians and performers. This is our great inheritance.
- 1.4 But it is possible that, of course, it won't last forever and that it could disappear not through an act of vandalism, not through being oppressed by anybody, but simply because we've forgotten. We haven't retained it. We haven't continued to transmit these skills.
- 1.5 The object of this course is to give you a grand tour through these tools, these traditions, these techniques, so that you'll have a broad understanding of what's involved in writing music.
- 1.6 Now, I gave a course earlier here at Peterson Academy on an Introduction to Musical Composition. One of the things I want to do from the outset is to differentiate between the two courses. So musical composition is a more applied course, you might say. It tells you in practical terms how you might go about making a piece, what the steps are, what some of the challenges are.
- 1.7 This course is a little bit different. In order to explain the particular nature of this course, I have to step back a little bit, and we have to talk about the term "music theory," to start with. The theory of music is kind of an unusual appellation, in a sense, because what we're going to talk about has got nothing in common with a scientific theorem. This is not exactly like that. Again, this is a body of past practices, which, of course, persist into the present.
- 1.8 But it's important to look at it that way in the sense that this is not prescriptive knowledge. It doesn't tell you what to do. It tells you what has been done. It's descriptive. It's not necessarily going to help you constitute a creative practice, but it's

a matter of learning the grammar, the syntax, the building blocks of language in a way that allows musicians, performers, and audiences to communicate with each other.

- 1.9 Each one of us is both the inheritor and the guardian of this trove of knowledge, of this tradition, which depends on all of us for its continuation. If tomorrow we were to stop teaching the basics of harmony, if we were to stop teaching counterpoint, it's quite conceivable that this body of knowledge could come to an end. That's the spirit in which I approach this. This is a deeply valuable inheritance. It's important to look at this not as a collection of dry facts, but again, as an attempt to describe the phenomenon of music such that it can be transmitted.
- 1.10 We mentioned that languages, of course, change all the time, and that this course doesn't tell you what to do. But it's very useful for having an idea of what might lie at the center of this tradition. And a big part of that, a big part of what distinguishes the Western musical tradition, is two things.
- 1.11 The first is that a great deal of it has to do with notation. It has to do with writing down music, such that it can be reproduced in accordance with a written text in multiple different situations. There's a certain permanence of the musical idea this way. So that's one thing. It's largely a written tradition.
- 1.12 And the other thing is that Western music has tended to emphasize polyphony over other dimensions. That's something that I get into in more detail in my course on the History of Western Music. But I want to remind you of those two things because it's important in coming to terms with the particular nature of music theory as it's practiced in the West.
- 1.13 A lot of this is predicated on the practice of writing down music. Of course, we're all accustomed to the way that sheet music looks, the way that notation looks. We have some degree of familiarity. Even people who have never sat down at an instrument can recognize staff notation, a clef, and a note. We might think that this is completely obvious. How else could you do it?
- 1.14 But in fact, these systems, these symbols, these notations, took thousands of years to develop, and there's nothing obvious about it at all. If you think about what music notation attempts to do, it's attempting to take things that are by nature ephemeral—it's literally vibrations in the air—and attempts to give them a fixed form. That's actually very difficult to do. It took an immensely long time to sort out the various problems.
- 1.15 What we have today is by no means perfect. There are all kinds of things that are very difficult to convey through any written expression, any written form in music. But it's a broadly applicable and broadly useful language of symbols.
- 1.16 We're going to look at rhythm, how to notate it, how to understand rhythm. We're going to look at pitch, which involves intervals, both melodic and harmonic. We're going to look at the five-line staff, how to read it, what this whole language of symbols

means. We're going to look at matters of harmony, and also just practical ear training exercises, so that each of us can hopefully become a better musician.

- 1.17 Remember that this gives you a very, very wide berth. You can do anything with this. There's absolutely no stylistic a priori in any of this material. The proof of that is that as you look at the music of the 20th century, it's extraordinarily broad. It encapsulates all kinds of different practices and styles, each of which has at least something to do with this body of knowledge.
- 1.18 To take a couple of examples, one from literature, one from music. The novelist, James Joyce, was commenting on the furore that was kicked up by the publication of his book, *Finnegans Wake*, which is a famously difficult book to read involving multiple neologisms and a very elastic use of language. It's very, very densely written. He would take words from different languages and combine them to make a new word and so on. He responded to this by saying, "I don't know what they're so upset about. They can have their English language afterwards, when I'm done with it."
- 1.19 Likewise, the German composer, Stockhausen, Karlheinz Stockhausen, was asked by a BBC journalist why he felt it was necessary to break things, why it was necessary to break the rules of music. Stockhausen said, "I haven't broken anything. I'm just adding to it. I've added a whole new room to the mansion," so to speak. So think of this as being a central body of knowledge that allows you, if you wanted to, to expand outwards or to remain right in the center. But this is a very broad enterprise.
- 1.20 So we're going to start at the very beginning. We're going to start with the basics of music notation, and we're going to understand how that works. Remember that there are lots of people who are successful in music who don't learn these things.
- 1.21 I gave the example in the Composition course of Paul McCartney, who can neither read nor write music and hasn't stopped him from being one of, if not the most, successful musicians of the 20th century. But that's partly because he sits on top of this 2000-year tradition that he's absorbed through his fingertips unconsciously through osmosis, if you like.
- 1.22 Knowing this material can help to speed things up for most people, I would say. I can't think of any circumstance in which it could possibly be a detriment to learn this. If your approach is more intuitive, if you're not accustomed to thinking music in this way, it's not going to cause you any harm. The additional knowledge can only be helpful. That would be the spirit with which I would approach it.
- 1.23 Let's start at the very beginning. We're going to look at the five-line staff. We're going to start with the notion of the staff. The staff in conventional music notation has five lines. Now, that being said, historically, if you go back, you'll see staves with two lines, three lines, four lines. You have to go back quite a way; it's farther back than a thousand years to see this.

- 1.24** But this was a technical problem that had to be solved, which is: How can you take something as nebulous as pitch? In other words, what is the frequency of a tone? Or in common parlance, how high or how low is it? How do you write that down? How do you give it a fixed form? That turned out to be a very, very complicated problem.
- 1.25** The solution, which started to stabilize approximately a thousand years ago, was to have this ruled staff, where you can have a spatial metaphor for the disposition of sounds. The farther you are towards the bottom of the staff, the lower the sound, and the higher you go, the higher the sound. That's a spatial representation of sound, if you like.
- 1.26** Now, that being said, you might also have staves, and this is increasingly common, that are intended not for notating pitch, but simply for notating rhythms, in which case you don't need five lines. One line is enough if you're just writing down a rhythm. If you look at percussion notation, as an example, you'll often just see a single line like this with a little clef at the beginning, and then you'll see rhythmic values on it. Because the pitch doesn't need to change, we don't need anything more than one line.
- 1.27** Let's look at the basics of staff notation. The first thing you need to know is that to the left of a staff is a symbol that looks like this, and this is called a clef. The clef tells you what register you're in. Is it high? Is it low? Is it medium? The origin of these different clefs, of which there are many, was to specify a vocal range. There are four basic human voice types: soprano, alto, tenor, and bass. Depending on which type of voice you had, you might want to situate a different clef on the staff in order to reflect that different range.
- 1.28** This is because the total range of pitches that it's possible to reproduce with the human voice is pretty broad. A baritone can usually sing down to an F [plays low F]. A bass might go a little bit lower than that [plays low E-D]. A basso profondo, which is a very rare voice type, might go down to [plays low C] or even lower. Then a soprano can sing up to about [plays high C] there, maybe a little higher [plays high D]. If you're a coloratura soprano, a virtuoso, then you might go [plays high F] up to the high F, which, to hear that sung by a human voice, is really something.
- 1.29** That's a pretty big range. But remember that our staff here has only five lines on it. How are you going to fit all those notes onto one staff? Well, the answer is you don't. You use different clefs depending on the range that you're working with. If that's the case for instruments, it also holds true for voices.
- 1.30** The clef that you're probably most likely to encounter is the treble clef. We're going to use the tone middle C as our reference point to get oriented in all these different clefs. The reason this is called middle C is that it lies directly in the middle of the piano keyboard [plays middle C]. That's middle C.
- 1.31** If you've ever taken a piano lesson, that's one of the first things you learn because you look at a piano keyboard and it's like this sea of keys. You don't know which octave

you're playing in. It's very difficult at first to get your bearings at a keyboard because there's so many notes. So that's your orientation point, is middle C.

- 1.32** Once you know where that is, you can find everything else much more easily. It's the same with staff notation. With each different clef, middle C is positioned differently on the staff. In the case of the treble clef, middle C is here. You'll notice that it doesn't actually fit inside the clef. But I have to draw an additional line below the staff in order to put my middle C down.
- 1.33** This additional line, which looks like this—it's just a little fragment of an additional staff line—is called a ledger line. Ledger lines are used sparingly in order to notate pitches or tones that lie below or above what the staff can hold—there'll be more on that in a minute. So this tone here, one ledger line below the lowest line of the treble staff, is middle C. Now, that's for the treble clef.
- 1.34** Now, if I write in the bass clef, the bass clef looks like this. Now, with both of these staves, there's a little visual clue, again, to help you get oriented so that you know where you're at. We talked about middle C and how its position changes depending on the staff. But there's also one other thing.
- 1.35** You notice that the bass staff has this little nub here. If I zoom in a little bit, you can see it there, which is situated on the second line, reading from the top. Then on top of that, there are these two dots above and below it. The purpose of these symbols, and the reason the clef looks like this, is to draw your attention to this second line. If you place a note on that second line like that, then you get an F.
- 1.36** The old name for this clef, and actually the current name in French and other languages, is the F clef or the "clé de fa." More on that later. Likewise, if I return to the treble clef, the French term for this is the "clé de sol," or the G clef. These terms are not broadly used in English anymore, but it's useful to understand their historical origins. The G clef or the treble clef wraps around the second line from the bottom. If you put a note there, you get a G, hence the name G clef.
- 1.37** Those are the two most widely used clefs. The bass clef, which, as its name indicates, allows you to write notes from middle C downwards, and middle C on the bass staff looks like this. Now, you'll notice something. I'll write that down. Middle C on the bass clef also requires a ledger line or a supplementary line above the staff in order to be written down. These two tones are one and the same. This and this. But they're notated in different ways.
- 1.38** On the treble staff, the middle C is right at the bottom. You could think of it as being where that staff begins its range. And on the bass clef, it's at the top, and they're the same. Those are the treble and bass clefs, and there are two others that we need to know.

- 1.39** Now, this is kind of a funny-looking clef. Once again, you'll see that there's a particular design to this clef, which is intended to help us get our bearings. In the case of this clef, which is called commonly the alto clef, or in French, the "clé d'ut trois," the key in which C is situated on the third line. In the alto clef, middle C is found right in the middle of the staff on the third line.
- 1.40** Finally, there's a very similar-looking clef called the tenor clef, or in French, the "clé d'ut quatre," which looks like this. Almost the same, but it's a little bit higher on the staff, and middle C is located here. So these four different staves—sorry, these four different clefs, allow us to situate a range of a voice or an instrument in a way that it fits conveniently on the staff, whether it's a very low range, very high range, or something in the middle. The alto and tenor clefs have a more medium range, in between soprano and bass.
- 1.41** Different instruments use different clefs for their musical notation. If you're playing a cello, or a double bass, or a tuba, or a trombone, or some other low instrument, in all likelihood, your part is going to be notated using the bass clef. If you're playing a high instrument like a flute, a clarinet, or a violin, most likely it will be notated on the treble clef. If you play the viola, or sometimes the trombone, your part will be notated on the alto clef or possibly the tenor clef. With the idea, again, of making the notation easier to read so that the part fits comfortably on the staff.
- 1.42** Okay, so let's return to this five-line treble staff and how we can get oriented on it. Remember that middle C, again, requires a ledger line, and that this symbol here curls around the G, which is situated here [plays C–G–C]. There's middle C [plays C], and there's the G above it [plays G].
- 1.43** Now, fortunately, we have mnemonic aids to help us remember where the notes lie on the staff. Some of these are very silly, and some of them are less so. But the one I learned when I was a kid was F-A-C-E, "Face". The staff has both spaces and lines, which is to say that a note can be situated between two lines on a space like this, or it can be situated on a line with the line going through the note head.
- 1.44** If we're reading the spaces, then reading from the bottom to the top, we get F-A-C-E, "Face". If we're reading the lines, we get E-G-B-D and F. Now, the mnemonic for that, there's quite a few of them, that I learned, was "Every Good Boy Deserves"—okay, you can decide what the last one is. How about Favors?? Whatever helps you to remember it. And so by remembering these two mnemonics, you can find a note head on a staff and remember what it is, when you're getting started. Eventually, this becomes second nature, and you don't have to think about it anymore.
- 1.45** And likewise with the bass staff, starting from the bottom space, we get A-C-E-G. And starting from the bottom line, we get G-B-D-F and A. For some reason, I don't have a hilarious mnemonic for these. I don't know why. They're less common. Maybe I didn't

learn them for whatever reason. You can invent your own, in any case. But for now, we'll just stick with the treble clef just for the sake of simplicity. So F-A-C-E-E-G-B-D-F.

- 1.46** Once again, that gives you a fairly limited range. If I'm reading notation in the treble staff and I start with the bottom line, which gives me E above middle C [plays E], then I can play E-G-B-D-F [plays E-G-B-D-F]. And if I read the spaces starting on F, I get [plays F-A-C-E] F-A-C-E. That gives you just over an octave [plays E-F-G-B-C-D-E-F]. Pretty limited range. Unless you're playing an instrument that only has six or seven notes on it, you're going to need more than that. That's where, again, we come back to the ledger lines.
- 1.47** So ledger lines, the idea is to use them as little as possible because they make the notation a little bit harder to read. Of course, professional musicians are accustomed to them, and it's not a problem, but it can make the learning process a little bit slower. So we try to avoid writing lots and lots of ledger lines unless it's necessary and there's no way to avoid it.
- 1.48** So just as the middle C in the treble clef requires one ledger line below the staff, you can also have ledger lines that appear above the staff. So remember that we have F-A-C-E. If we keep going, then we get G, that's above the staff there. If we want to go above that, we need a ledger line. Just like on the staff, the note can go above the line like this, or the line can go through it. And of course, we can have as many ledger lines as we want, in principle, although it gets very hard to read.
- 1.49** Now we should talk a little bit about the nomenclature of tones. Now, you have to bear in mind that music theory is a conception of sharing this knowledge that is located in a specific culture. In other words, it's not done universally in every country. If you were to talk about music theory in a French conservatory, that term doesn't have a wide currency the way it does in the English-speaking world.
- 1.50** The Germans have a term "Musiktheorie" that corresponds to music theory, but it's not taught as a separate subject as such in France. Instead, the body of knowledge that's grouped within music theory is divided into different subjects that have a more practical and less theoretical orientation, such as the notion of "écriture," which is music writing, and that immediately situates it as more of a practical than an abstract skill.
- 1.51** In North America, when you say, "music theory," it's understood that you mean, generally speaking, the rudiments of music, understanding staff notation, understanding how chords are put together, understanding the terminology, effectively. It's a body of knowledge that covers everything that you can do without ever touching an instrument, basically.
- 1.52** But again, different cultures will have a more theoretical leaning or a more practical leaning, depending on how their musical pedagogy has evolved. In Germany, you can

do a doctoral degree in music theory. That doesn't exist in France. So some of the way that these disciplines are taught very much depends on the culture.

- 1.53** When I talk about the nomenclature of the tones, that is also something that varies from country to country. If you're in Italy, for example, or France, or Spain, most likely when you begin your musical education, you're going to be subjected to a form of torture called "solfège" or "solffeggio." What that is, is a way of memorizing intervals and tones through the use of note names. These note names are very, very old. They're at least a thousand years old.
- 1.54** We're going all the way back to the early medieval era in which, as a mnemonic aid, theorists devised a set of six syllables called the medieval hexachord, which was used to help singers remember the notes that they would have to sing. And those six syllables are ut, re, mi, fa, sol, la. Ut, re, mi, fa, sol, la. Now, those syllables were chosen not by accident, but actually, they form an acrostic in Latin if you take the first letter of the six. On top of that, each one has a distinctly different sound.
- 1.55** Now, later on, centuries later, "ut" was replaced with the syllable "do," simply because there was a theorist who preferred the sound of that. And later on, a seventh syllable was added, which is "si" which is often called "ti" in North American nomenclature. So that's very common in the Latin countries: do, re, mi, fa, so, la, si, or ut, re, mi, fa, so, la, si, or ti.
- 1.56** In the German-speaking world, however, there's a completely different tradition where the notes are simply assigned the names of letters. We have C, D, E, F, G, A, and confusingly, H, which breaks the series. We don't have to get into all the historical and theoretical reasons for why this is the case and why it evolved that way. But "ha" or "H" is the German equivalent of our "B" or B-natural. So all these different nomenclatures, but all with the same intention, which is to help us remember the tones.
- 1.57** And so if you're learning solfège, you're learning to associate notation with the sound that you can produce with your voice reliably and consistently. Rather than do, re, mi, fa, sol, we go C, D, E, F, G, A, B. These are the basic tones of a seven-note or diatonic scale, which forms the basis of most of our musical tradition. Once you get to B, this series just repeats itself—C, D, E, F, G, A, B, C, D, E, F, G, A, B, et cetera.
- 1.58** Okay, so now that we've understood that on a staff, we have both spaces and lines, we can start writing out a simple scale. I'll show you what that looks like. If I start on middle C in the treble clef, I'm going to have to go up to the next space or line in order to produce the next tone in the series. That's our middle C. Then we go up to the space above C, which sits on a line. We're alternating lines and spaces, and then I just keep going. And that gives me C, D, E, F, G, A, B, C. So that's a complete scale with the bottom note repeated at the top [plays C scale]. C, D, E, F, G, A, B, C.
- 1.59** Now, the nice thing about the Italian and French systems is that when you're learning these materials, you're required to sing them. You don't simply understand the

notation in an abstract sense, but you immediately have to externalize it. You associate the tone with a syllable, [plays C scale] do re mi fa sol la si do, and so on. In some ways, it's a difficult discipline because you can go extremely far with this, where you can learn to very rapidly sightread a piece of sheet music with all of the syllables in place.

- 1.60 To some degree, that's a matter of culture and taste and different historical traditions. A lot of North Americans are not accustomed to that system and find it quite challenging when they first encounter it. But again, for others, especially if you begin early in life, it can become second nature.
- 1.61 That's how you situate notes on a staff with spaces and with lines. Now, we've looked at the different clefs. We've looked at the different ranges of the clefs, and we've looked at some mnemonic aids that help you to get situated. We've also looked at ledger lines. Now we're going to continue a little bit.
- 1.62 We're going to talk about the grand staff, which looks like this. Now, music written for keyboard instruments or instruments that have a particularly large range, like a piano—a piano covers more than seven octaves. You can't fit all of that on a single staff, and it's not practical to change clefs all the time, especially since you might need to notate a very low sound and a very high sound simultaneously, in which case you need another system. If you look at piano notation, you're going to see something called a grand staff.
- 1.63 A grand staff consists of both a treble clef on top and a bass clef on the bottom. You need two staves, most of the time, to notate piano music. Remember, this and this are the same. They're both middle C. That's very convenient because it means the middle of the piano is situated exactly in between the two staves. Then, of course, you can expand upwards from there or downwards from there.
- 1.64 We've looked at percussion clefs. Remember that in percussion writing where you don't need to differentiate between pitches, you don't need five lines, you can just have one line. We've looked at the grand staff. There's also something called transposing clefs, which you probably won't need right away, but we'll just take a second to look at what those are.
- 1.65 A transposing clef looks like a regular treble or bass clef, except that there's going to be a little 8 above it or below it, or a 15 above it or below it. The 15 means quindicesima alta, which means two octaves higher than written. If I put a middle C down on a staff like this with the 15 added to the [top], then the sound is not going to be middle C, but it's going to be the C two octaves higher [plays high C].
- 1.66 That's very convenient if you're writing for instruments that are extremely high or extremely low, where you don't want to use millions of ledger lines all the time. You still want it to fit on the staff. But you also want to keep it easy to read by using a familiar clef, like the treble clef or the bass clef. If 15 uses the Italian nomenclature,

quindicesima, the 8 means, octava alta, in other words, one octave higher. And the 15 is two octaves higher or lower.

- 1.67** Okay, so now we're going to have a look at notes and rhythms. We've looked at the basics of writing down pitches associated with either syllables or with note names. That allowed early theorists and composers to solve a very difficult problem, which is, of course, how do you take something as difficult to conceptualize as a tone or a pitch and write it down in some kind of a fixed form?
- 1.68** The next problem that had to be solved was: How do you notate rhythm? In some respects, that might be an even harder problem. The solution that eventually was come up with was completely transformative. But we have to back up a little bit here in order to understand the implications of it.
- 1.69** If you look at very, very early music—if you look at early medieval music— you won't see anything resembling the rhythmic notation that we use today. Instead, they used things called rhythmic modes, which would be specified in the score, which would tell you that the rhythms are meant to be played with this or that type of rhythm— either an equal rhythm, or an unequal rhythm, or some little rhythmic figure that would persist throughout the entire piece. That was fine if your piece was just going [sings a rhythmic figure] and it never changed, but it also had some limitations.
- 1.70** In very early musical notation, it's very ambiguous in some cases, exactly how the rhythms were meant to be performed. A lot of this would have been not written down because it was obvious to everybody at the time how it fit into their performance practice. Typically, you don't write things down that everybody knows. It's not necessary. But of course, if that tradition has been broken, or if we no longer understand what it was meant to sound like, then it's very difficult to reconstitute it after the fact.
- 1.71** Eventually, hundreds of years after the development of the five-line staff, a system called mensural notation took hold. The basic concept behind mensural rhythmic notation is this: you start with a large beat unit, and you divide it by halves. This beat unit is called the breve. Now, here again, there are different nomenclatures that are used in different countries. In North America, this is referred to as a double whole note. In the United Kingdom, you would say breve.
- 1.72** This is the longest note value that you can produce conventionally using this notation. Then you divide it by half. If you divide a breve into half of its duration, you obtain a semibreve, or a whole note. So one breve is equivalent to two semibreves. Then dividing further, we obtain a minim, or a half note in North American nomenclature. Just as you would expect, you can fit two half notes in the time of one whole note.
- 1.73** Continuing, you notice that the appearance of the notes is starting to change. The very long note values are an oval. The minim, or the half note, is an oval with a stem attached to it. And as we keep going, the note head becomes not open, but rather

closed. We have a black note head starting with the crotchet, or the quarter note. Continuing the series, we come to the quaver, which is the same as the whole note, except that it has this little flag on it. This is also referred to in North America as an eighth note.

- 1.74** In some ways, the North American system is a little easier to remember because it's simply multiples. Double whole note, whole note, half note, quarter note, eighth note, 16th note, 32nd note, 64th note, 128th note, 256th note, and so on. You can keep dividing them into smaller and smaller and smaller rhythmic values, more or less indefinitely, although you do run into practical limitations with that.
- 1.75** If we keep going, we obtain the semiquaver, or 16th note. As you can see, we've added a second flag to the note. Of course, as you would expect, next we come to the demisemiquaver. These note names do get increasingly ridiculous as we get smaller and smaller. The next after that is the hemidemisemiquaver. This is also referred to as a 32nd note. And so that gives you an idea. So again, each of these rhythmic values halves each time. Now, that's for the notation of a single note.
- 1.76** If you want to group them together, as soon as you get to rhythmic values smaller than the quarter note or the crotchet, then when they are grouped together, you have something called a beam. The beam is a ligature. It's a visual symbol that allows you to see the grouping of the beats a little bit more easily. It makes them easier to read. In the case of two eighth notes, rather than having two flags, you would typically join them together with a beam, and likewise with 16th notes. If we go down to 32nd notes, it starts to look like that. Those are for notes, which tell us that a sound is meant to be played.
- 1.77** But in music, you're not necessarily playing all the time. In order to notate silences, we have a whole other repertoire of symbols, and these symbols are called rests. And rests obey the same basic logic as notes, note values, which is that you start with a large value. This is the breve rest or double whole note rest. Then we have the whole note rest. There's a bit of a subtlety here. This is something that is very easy to mix up. The whole note rest hangs downwards from the second line, and the half note rest sits on the third line. They're almost identical symbols, but one of them hangs down and the other sits on the floor, as it were. This is a half note rest.
- 1.78** The breve rest is a notch. It's a thin, narrow notch that lies between the second and third lines. Now, as we keep going down, we obtain the quarter note rest, which is a kind of a squiggle, and the eighth note rest. I'm going to stick to North American parlance for now. The 16th note rest. And so on and so forth.
- 1.79** That's quite a lot of ground to cover already if you're not familiar with music notation. It does take some time to become used to all of this. If you were taking a course in beginning music theory or solfège, or some other related discipline, then you would

typically drill all of this, and the teacher would give you simple little notations with assignments either to sing or to play what you see.

- 1.80** I'm going to suggest we stop there for this lecture. In the next one, we're going to put all this into practice. We're going to have little examples of actual notation. We'll show you what they look like, and we're going to have to look at other concepts, such as the notation, not just of abstract pitches and rhythms, but of time signatures. We're going to take a break for now and resume with the notation of meter and rhythm. Are there any questions so far?
- 1.81** So, earlier, you said that the music notation from the Middle Ages was a bit limited compared to now. Do you think the notation we have now is limited compared to what we will have in the future?
- 1.82** Well, it already is limited because now, of course, we have digital audio workstations, we have electronic music. For at least 80 years now, composers have been working with electronic instruments— instruments that can produce not just the discrete tones that you see on a piano keyboard, but a whole continuum of pitches.
- 1.83** Once you get into electronic sounds, then this system of notation is absolutely inadequate. If you're getting into extremely precise fractional durations, people involved in media scoring, for example, will often think in terms of milliseconds, as opposed to quarter notes and 16th notes and eighth notes.
- 1.84** You could argue that MIDI, which is a language that allows musical instruments and computers to talk to each other, to have a common language, is in a sense a new form of notation. If you are involved in making soundtracks, for example, or many different kinds of electronic music, or commercial music, then you have to come to terms with what MIDI looks like, with how that's represented visually, and that's a whole other language. You could argue, again, that MIDI allows you to encode quite a lot more information than this system does, which is nearly always relative in the way that it conveys information.
- 1.85** For example, if I write down a quarter note on a piece of staff paper, that doesn't tell me anything about duration. The duration only comes into it once the notation is embedded in a whole expressive context and a time signature, a style, an era, a performance practice. Just a quarter note or a whole note on a page by itself tells me nothing.
- 1.86** But in MIDI, and no doubt MIDI will eventually be replaced by something else, but in MIDI, you can write down a sound, give it a numerical value from 1 to 127, which tells you the intensity of the note or the velocity of the attack. You can also specify the duration to a precision of milliseconds. That's a form of writing, in a sense, because you're fixing information in a way that can be reproduced. It's not necessary for everything. For concert music, you can't give a violinist a screenshot of a ProTools file.

It's not going to help them. But if you're working in the electronic domain, then that might become a native tongue to you.

- 1.87** Is there a link between the different way countries learn how to write music and the fact that, for instance, the Germanic countries have emphasis on the orchestral, whereas the French more on the soloist?
- 1.88** I'm tempted to say no. The reason is that multiple different countries use the solfège system, and their musical cultures look very different from each other. If you compare the musical life of Spain and Italy, it's very different. Likewise, the German system and the Anglo-Saxon system are not identical, but they're much closer together, and the way music is practiced in Germany as opposed to the way it's practiced, say, in the USA, is not the same.
- 1.89** I suspect it has more to do with historical, cultural, possibly linguistic developments, and a whole complicated series of contingencies about the way musical life is organized in different countries. But you're absolutely right to assert that the Germans have tended to prioritize collective music making, in the form of orchestras, and a very robust amateur practice of ordinary people learning instruments to a high level.
- 1.90** Whereas the French system is more steeply hierarchical in its nature. It's more oriented towards a professional education with the goal of producing high-level musicians. Also, it's much more oriented towards the production of soloists. These are broad generalizations. You'll find examples of both countries doing the opposite of that. But as a general tendency, I think that that broadly holds true.
- 1.91** I guess one of the assumptions underlying the way that the system of notation works is separating the frequencies into semitones. Obviously, the piano that we have as a visual reference is also broken up into semitones. I guess I'm just wondering if the system that we use is based on semitones as a consequence of an approach to the physical aspects of playing—if it came from something like a keyboard instrument or afretted system or something like that—or if the approach to break it into semitones came first as a way to just organize frequencies.
- 1.92** Yeah, the two things are clearly interrelated. The musical culture, the instruments that are played and developed within that culture, and the need for a particular type of notation are all intrinsically linked. If you look at other cultures and the way that they've developed, you can compare them and see that they've often evolved along very different lines, depending on the way music fits into their society, the instruments they have, what their performance tradition looks like, and so on.
- 1.93** For example, if you look at the music of Bali or Java, they don't have a notated tradition for their music. The parts that are played in a gamelan orchestra are learned note by note by the beginner from a teacher directly, and they're memorized. The patterns, the rhythmic patterns, and the pitch patterns in those pieces are dizzyingly complicated, but they don't use notation.

- 1.94** Likewise, tuning systems change from one region of the world to another. Through a very lengthy historical process, we've settled upon something called equal temperament. We probably don't have time to get into a detailed examination of tuning systems and their implications. But basically what equal temperament means is that on this piano, the interval, the difference in pitch between each successive tone is identical.
- 1.95** If you take any two random notes on the keyboard and play them one after the other, the interval, the gap between those two notes will be the same. But that's not the case in every musical tradition. You can test this out very easily by listening to examples of folk music from Eastern Europe. Listen to traditional Balkan music—music of Bulgaria, of Macedonia. What you'll find is that it sounds very different to our Western ears because they don't use the same scales. Their scales may use intervals that we've never learned how to sing. They can do them perfectly in tune, but we don't have that in our ear.
- 1.96** Just as it's very, very difficult for a non-native speaker to pick up Mandarin and reproduce all the tones accurately—whereas for a native speaker it's effortless— it's the same with different tuning systems as well. The notation, or the lack thereof, reflects these different systems and traditions.
- 1.97** Finally, there are also musical cultures that rely much more heavily on percussion and rhythm than on pitch. Different cultures emphasize and prioritize different musical parameters, to the exclusion of others. You might have a culture, as is the case in large parts of sub-Saharan Africa, where the rhythmic repertoire is extraordinarily developed, but the music is either unpitched—so it's primarily preoccupied with rhythm—or if it does use pitch, it might use a very limited number of tones.
- 1.98** Likewise, Indian music—this is a very broad generalization because it's a huge country, obviously, and divided between a northern musical tradition and a southern one, which are very distinct from each other— but broadly speaking, there's little to no harmony in their music. There's a virtuosity of pitch and hundreds of different scales that they use, and different modes which have different tuning systems—far beyond anything we have. Their rhythms are complicated, but there's no harmony—or there might just be a drone.
- 1.99** Music has a very broad range of uses and different cultural traditions. Depending on how those traditions have evolved, then you're going to see very different notations or no notation at all. As it happens in the West, the development of our musical tradition has relied very heavily on the capacity to write things down, which allows you to have polyphony. It allows you to write down multiple different independent parts and to have each of them evolve differently.
- 1.100** It's very hard to do if you had to memorize every line. You could only really do that if you had tightly-knit communities that spent a substantial portion of their lives

memorizing these things, and then transmitting them to the next generation—with, of course, the possibility that at any time that link could be broken if the knowledge were not passed down. But this way, a composer in 1450 can write down his idea, and 550 years later, we can read it.

Lecture 2: The Time of Music

At a glance — The time of music — rhythm and notation.

- 2.1 Now we're going to get deeper into music notation, specifically rhythm and how rhythm is notated. Remember earlier, we were looking at the system of mensural notation, which again was not an obvious idea; it had to be invented. So how do you capture something as nebulous as a duration in a fixed form? That's what it attempted to do.
- 2.2 Now, you might have thought to yourself that that's a good system because it's logical, and you can divide everything by half, and you get progressively smaller values. But it's also very limited in some ways because there are lots of rhythms that aren't articulated in terms of exact multiples of durations. In fact, generally speaking, rhythm doesn't work that way.
- 2.3 So it's a very abstract system that allows you to represent a series of proportional durations. It doesn't tell you exactly what the duration is, but it tells you the proportions between one value and another. But it still doesn't communicate anything about rhythmic feel, or a meter, or the relationship between the body and the gesture and the rhythm that we're hearing. This is where we have to get into time signatures, meter, and duration, which are three distinct but interrelated concepts.
- 2.4 Let me start by making a distinction. You can have rhythms that are subtractive or divisive, where you start with a big unit such as a breve, and keep dividing it in half—cutting it in half—until you get finer and finer bits of time, if you like.
- 2.5 You could also do things in the opposite direction, which is how a lot of Eastern European music works. So rather than subtracting time from a larger unit, you might start the other way around by adding little beats together to make larger beat units. If you listen again to the music of Macedonia as a good example, Turkey is another example; what you'll find is they don't subdivide a big unit, but rather they have beats of unequal duration, and those beats are not exact multiples of each other.
- 2.6 When we learn subtractive rhythms, we're thinking in terms of exact pulses [demonstrates beats at regular intervals: ta-ka, ta-ka, ta-ka, ta-ka], where the pulses are all equally spaced and they are exact multiples of each other. But what if you didn't have equal beats, but you had short beats and long beats, and instead of being exactly proportional to each other, they were more to do with physical gesture, and they were more predicated on the time it takes to perform an action of some kind?
- 2.7 We have an example of that actually in our Western tradition, which is the Viennese waltz. I've probably already mentioned this in the History of Western Music course, but I'm going to briefly recapitulate this because it's a very important point in coming to terms with rhythm.

- 2.8** If you listen to a Viennese waltz played by the Vienna Philharmonic Orchestra—take any random waltz by Strauss or any composer in that tradition, and listen to it carefully—you'll see that the beats, the three beats in a waltz, are not equal, it's not [demonstrates beats at regular intervals: pom-pom-pom, pom-pom, pom]. If you were to play a waltz like that, they would look at you as though you were from another planet because that's not how we do it at all. In the Viennese waltz, the beats are unequally spaced. The second beat is a bit early, and the third beat is a bit late [demonstrates beats at irregular intervals: pom-pom, pom, pom-pom, pom]; it's completely irregular.
- 2.9** Now, if you look at a notated waltz, it's going to be written in a time signature, which is $3/4$, and on the page, it's going to look regular, but they don't play it that way. The reason they don't play it that way is because the waltz is, of course, a dance, and it has a step that involves turning, and it takes longer to do that. So the beats are unequally spaced in order to reflect the time it takes to perform these different physical motions.
- 2.10** Now, all of these things are difficult or impossible to take into account in a notation. These are things that are culturally encoded, let's say, that belong to our collective cultural memory. The notation cannot possibly take all of those things into account. It's of necessity incomplete and abstracted to a degree. This is the case in pretty much every culture.
- 2.11** Again, in the music of the Balkans or in parts of Turkey, what you'll find are beats that are like this [demonstrates irregular beats], which can then be subdivided. Now, that's hard to notate. It's hard to notate because you have a short beat and a long beat, and you make phrases by adding them together. Then you can subdivide them in different ways. It's a fundamentally different approach. Technically, if you were to try to notate it, which ethnomusicologists have tried to do using various different systems, it gets complicated very, very quickly.
- 2.12** This is the case also with all kinds of popular traditions. How would you notate a blues song? How would you notate swing rhythms? How would you notate ragtime? It's not straightforward. There are matters of feel, and that's simply a matter of physical embodiment—things that are not abstract, they're not intellectual, they are embodied. Notation is not very good at encoding that.
- 2.13** However, we still have to solve the problem of trying to write these things down so that a piece of music can be fixed in some form and transmitted through time and across different geographical locations. Before that was the case, if you wanted to have a composition, a piece of music to be played somewhere else, then you had to physically go there and teach the singers or whoever how to do it. Obviously, notation makes that a lot easier.
- 2.14** So what do you do about the problem of rhythm? The fact that there's something very intangible about this. It has to do with feel, it has to do with gesture, it has to do with

the body. Well, the solution that eventually arose was to use something called a time signature.

- 2.15** The time signature is something that you'll see on the staff right at the beginning. Remember, we said that the clef goes at the left of the staff in order to tell us how to get situated. Where does middle C fall on that staff?— maybe it's here, maybe it's here, maybe it's somewhere in the middle. That allows us to orient ourselves with respect to our instrument and the range of the instrument on the staff.
- 2.16** But then we also need something to orient us rhythmically. Now, if I were to just sit and write a sequence of rhythmic values like this [writes musical notes on the staff], I could go on forever just writing values like that—two quarter notes, two eighth notes, two 16th notes, an eighth note, four more eighth notes, etc—then that allows me to write some rhythms, but it doesn't tell me anything about how these rhythms are meant to be interpreted, how they relate to a gesture, how they relate to a phrase.
- 2.17** So I can just write endless successions of rhythmic durations, proportional rhythmic durations like that, but it doesn't really help me to read it necessarily because I won't be able to imagine this succession of values in relationship to a gesture or to a phrase.
- 2.18** In order to do that, I need to have a system that allows me to set up a pattern, a pattern of weak and strong beats, or a pattern of accented and unaccented beats. This is fundamental to any notion of rhythm that is connected to the body. Dance has to do with periodic rhythms. Periodic rhythms set up a pattern and repeat that pattern. You can think of this as a predictable grid that continues indefinitely for as long as the piece is going. What you're actually playing or singing can line up with that grid, or it can be out of phase with it, and it can push against it, but the grid allows us to understand what the underlying pattern is.
- 2.19** You can think of it again as a set of weak and strong beats. For example, I might write a time signature that looks like this [writes time signature on the staff], and you'll see that right after the clef. If you look at the screen there, you'll see there's a treble clef, and immediately to the right of it, the numbers 4 and 4. That's a 4/4 time signature.
- 2.20** Time signatures have a numerator, which is the upper number, and a denominator, which is the lower one. The upper number tells you how many beats there are in the bar. The lower number tells you what is the basic beat unit to which you will be referring in the piece. In other words, what are we counting? Are we counting quarter notes as the basic beat unit, or eighth notes, or 16th notes, or half notes, or whole notes?
- 2.21** This 4/4 time signature tells me that the basic beat unit is the quarter note, and there will be four quarter notes per bar of music. And so once we get to four, we've completed a measure—or a bar of music, depending on whether you're using British or American English; you might say a measure or a bar, but they're interchangeable terms.

Now, once we've got four quarter notes, we can't fit any more quarter notes into a 4/4 meter.

- 2.22** Now, I mentioned that there is an accent pattern in these different meters that tells the player what to emphasize and what not to emphasize and what shape to give to a phrase. It looks something like this—imagine that we are plotting a little graph in terms of intensity. The first beat is called the downbeat, and that relates to a physical pattern. If you're conducting a piece of music, you'll notice the conductor is doing patterns with their arms. Sometimes the patterns are quite wild and hard to follow. But the pattern is you go down on the first beat, over for the second beat, over to the other side for the third beat, and then up again for the fourth beat. And then down again.
- 2.23** The downbeat is the first beat of the bar, and it's the strongest beat. If we were to plot a graph, it might look something like this—the first beat is the strongest, the second beat is a weak beat, the third beat is stronger than the second beat, but not as strong as the first, and the fourth beat is the weakest beat. You can plot that in terms of a simple graph that just shows you.
- 2.24** Now, of course, nobody actually plays that way. You don't think, "Oh, yes, okay, it has to be exactly here on the graph." Nobody does that. It's simply an abstract representation to illustrate the point that although on paper we have four equal beats, in reality, they have different degrees of stress. Now, depending on the nature of the rhythmic figure that you're playing, of course, you might have something very different from that.
- 2.25** For example, in a lot of Baroque and Classical music, if you have a short beat immediately followed by a long held beat, it's common for the long held beat to be accented more than the first, even if the first beat is a downbeat. So, for example, in Schubert, you can have lines like this [plays the piano]. Now, that's in 4/4, but it starts like this [plays the piano]. Now, the first note I'm playing is on the downbeat. It's the first note of the bar, but it's immediately followed by a long note. It's customary and instinctive for performers to emphasize the long note, even though that breaks with the pattern.
- 2.26** So the notes are not emphasized equally in rhythm, but according to a meter. It's the same thing in classical poetry. Poetry also uses metric—what are called metric feet—which are a way of telling the reader effectively, or rather, allowing the poet to set up patterns of stresses, stressed and unstressed syllables. Of course, there are all kinds of different poetic feet, such as dactyls and spondees and so on.
- 2.27** There's a whole repertoire in poetic writing of figures that parallel but are not the same as the rhythms that we have in music. Remember that rhythm in music is, of course, an extension of rhythm in speech. These are closely interrelated ideas. You'll see parallels very clearly between poetry and rhythmic meter.

- 2.28** Now we have to talk about the different types of time signatures that exist because there are many of them besides just 4/4. Remember that the numerator tells us how many beats, and the denominator tells us what is the beat unit. Of course, you can modify that. As an example, rather than 4/4, I might change the top number to 3. So now I have a 3/4 time signature. In 3/4, rather than 4 beats per bar, I'm going to have 3, and that would look like this. Then, once I get to 3, I need a bar line, which is a vertical line that crosses the staff from top to bottom and tells me that another bar or another measure has to begin.
- 2.29** Now, the bar line is not merely a convenience to make it easier to read. The bar line also tells you at a glance what the phrase is like, what the character of the rhythm is going to be, where the downbeat is, and what beat is emphasized. If we didn't have that, then there would be no difference at all between 3/4 and 4/4. You could just keep writing beats forever, and you wouldn't be able to distinguish between one rhythm and another. The thing that allows you to distinguish between 3/4 and 4/4 is the accent pattern. So 3/4 gives us a basic waltz pattern.
- 2.30** So this example of 4/4 versus 3/4 allows me to make a preliminary distinction in meters between binary and ternary rhythms. And as the name indicates, a binary rhythm divides into groups of two beats: 2, 4, 8, 16, and so on. And ternary rhythms divide into groups of three beats. So 3/4, for example, or 6/4, or 3/8, or 6/8—and we'll get there in a minute. So we have binary rhythms and we have ternary rhythms.
- 2.31** A third category—which we probably don't have to worry too much about for now because it's much less common—but a third category combines both of those into a single meter. These are called asymmetric meters. They include rhythms such as 5/4, 7/4, and so on. Now, these are actually quite common in folk traditions. They're much less common in Classical and Baroque music, or unheard of, in fact, until you get a little bit farther into the 19th century.
- 2.32** We call them asymmetrical or sometimes compound meters, because even though it looks like we have five equal beats in a measure of 5/4, in fact, we have to subdivide into patterns of 2 + 3 or 3 + 2. It's called asymmetrical because those beat units are of different duration: 2 + 3 or 3 + 2, as opposed to 2 + 2 or 3 + 3 + 3 + 3; we have unequal subdivisions. The binary or ternary division is an important one to understand because you might have two measures that, in theory, can contain the same number of pulses, but in which the pattern is divided up very differently.
- 2.33** Let me give you an example. If I write a measure of 3/4—remember, again, that the downbeat or the first beat in the bar, which is this one, is going to be emphasized [counts 3/4 time signature].
- 2.34** Now, if we go backwards again to what we were talking about earlier today, where we had equal subdivisions of beats. Remember that one quarter note or one crotchet is equal to two eighth notes or two quavers. So if I wanted to, I could fill this bar with six

eighth notes instead of three quarter notes, and that would look like this. So instead of [counts 3/4 time signature], I've got [counts different 3/4 time signature]. Now I've moved from three quarter notes to six eighth notes.

- 2.35** Now, let's have some fun. We're going to change the denominator from 4 to 8. We're going to change the numerator from 3 to 6. Now, we have a totally different time signature. This time signature tells me that we have six beats per bar, and the beat unit is the eighth note as opposed to the quarter. So then you might think, "Okay, well, that's going to be the same. It's going to look like this."
- 2.36** Remember, this is 3/4, and you can fit six eighth notes into a bar of 3/4, or you can also fit six eighth notes into a bar of 6/8. So what's the difference? The difference is that 6/8 is not a ternary rhythm, but rather a compound binary rhythm. What that means is it divides into two beats, each of which is subdivided into three. It's written like this. So yes, there are six eighth notes in a measure of 6/8, but they're grouped together differently with a different pattern of stresses. So that would look like this.
- 2.37** Now, notice that the beaming in my measure of 6/8—and I'm going to put this in red so that we can see it—the pulse is in two groups of three. The beaming in my 3/4 bar was in three groups of two. So that's the difference between [counts 3/4 time signature] and the 6/8 example [counts 6/8 time signature].
- 2.38** So in reality, our 6/8 bar has two main pulses, each of which is divided into three pulses. That's a binary rhythm that is a compound meter. It's taking two subdivided beats and putting them together. That's called a compound binary because we have one main beat, two main beats, and then it repeats. So a multiple of two.
- 2.39** As you would imagine, there's also compound ternary meters. Compound ternary meters will have a pattern of three beats subdivided into three in this case. That could look like this: 9/8. In 9/8—you can probably follow the logic at this point—three beats, each of which is divided into three [counts 9/8 time signature]. So there we've got three beats.
- 2.40** Now, of course, you can have even more complicated meters if you want, and many musical traditions and many composers do use them. You can put any arbitrary number of beats in a bar. You could, for example, have a meter that looks like this: 15/16. Now, we're getting on the esoteric side of things at this point. We don't have to spend a great deal of time there, but the point is that in a 15/16 measure, the composer has to tell us through his notation how those beats are meant to be divided up, because 15 is a large number. It's not obvious exactly how that's meant to be divided.
- 2.41** It could look like this—in this bar, we have five equal beats, each of which is subdivided into three [demonstrates 15/16 time signature]. Or—this is also possible—a bar of 15/16, divided like this, and you can see here that it's the same number of 16th notes. If you look at the screen there, there's fifteen 16th notes in both cases. In the top example, they're grouped into five pulse groups. In the lower example, they're grouped

into three groups of pulses [demonstrates 15/16 time signature]. The question is, at a glance, the performer has to understand what the pulse groups are, how many of them are there per bar, and then how are the pulse groups divided up.

2.42 So that's a little tour of time signatures. To recapitulate what we've looked at. You have binary meters: 2/4, 4/4, 2/8, 2/16—anything where the top number is a multiple of two. Then you have ternary meters, where the top number is a multiple of 3—3, 6, 9, 12, and so on. On top of that, you can have compound binary or compound ternary meters in which the basic beat is further subdivided into smaller pulses.

2.43 We also looked at how it's possible to have asymmetrical meters in which the pulse groups connect together in groups of two or three. Now, in that case, again, it's necessary to make it visually clear to the performer whether we're meant to group it in 2 + 3 or 3 + 2 or 2 + 3 + 2 or 3 + 2 + 3, or whatever it might be.

2.44 So that would give you, for example, here's a measure of 5/8. Now, it could be like this, or it could look like this. In the first example, we have [counts 5/8 time signature in a 2 + 3 pattern]. Now, there's two beats per bar subdivided in the first beat in two and subdivided in three equal beats in the second part of the bar [counts the pattern again]. In the second example, we do the opposite pattern: 3 + 2 [counts 5/8 time signature in a 3 + 2 pattern]. So those are the asymmetrical meters, and in many of these folkloristic traditions, that pattern will change many times over the course of the beat, sometimes in dizzyingly complicated arrays of pulses.

2.45 The reason we put the clef and the time signature on the left-hand side of the staff is so that you can pick up the score and immediately understand both where the range of the part lies and also you can immediately understand what is the basic metrical pattern. At a glance, you can then much more rapidly decode the music. Those are two very important symbols to come to grips with.

2.46 Now, we also need to look at how these different rhythmic figures are meant to be written out. The reason for that is there's a bit of a subtlety here, which is for some reason not often taught when beginning music students are coming to grips with notation, and that is the stems and the beams.

2.47 Now, remember that as the written notations get smaller, you add first stems or a long vertical line to the bottom of the note head or the top of it, depending on where it sits on the staff. Then, as they get even smaller, you might add flags, and then the flags can be grouped together to form beams or ligatures. Let's just take a minute to show how those are meant to be drawn.

2.48 Let's say I'm in the treble clef—now, remember, the acronym is F-A-C-E. Let's say that we've got an E on the staff, and we want it to be a quarter note. The rule is that the stem of a quarter note should be one octave long. That means the bottom of the stem is going to come to a rest where the lower E would be if it were one octave lower than

the top note. So the stem should be one octave long. Likewise, if I have a G here, then the stem is going to go all the way up to where a G would appear one octave higher.

- 2.49** That rule only changes when you get up to the upper ledger lines. If I'm writing a C here, two ledger lines above the treble staff, at that point, the stem is still one octave long. But as I add additional ledger lines, the rule is the stem always touches the middle line, no matter how high the note goes. So if you're dealing with very high or very low notes where multiple ledger lines are necessary, then the stem is always going to touch the middle line of the staff. If your note is located within the staff, then the stem is one octave long—it touches the span of one octave.
- 2.50** Now, as for beams and ligatures, let's imagine we have a rhythmic figure that looks like this. Again, it's going to be one octave long for the stem. Then we connect the notes together with a horizontal line called a beam. Beams are typically thicker than stems in notated music. You're going to need two beams in order to make it visually clear that these are 16th notes.
- 2.51** Now, in the case of more complicated rhythms, you might break this beam and make what's called a secondary beam. That would look like this. And notice that the 16th note beam does not go all the way through, but rather it's broken, so we have three subgroupings. You might or might not do that, depending on what you want to convey to the performer. If there's an ambiguity as to how the beats are meant to be subdivided, then you might want to have secondary beams.
- 2.52** In other instances, it's necessary to have secondary beams in order to make the notation easier to read. For example, if you have a stream of 30-second notes, which require three beams, then a group of eight of them starts to get really difficult to read. If you were to just draw the beam straight through like that, it starts to become visually confusing. You can't really see at a glance where you are within that figure, especially if the notes are changing.
- 2.53** In the examples I've written so far, it's just been a bunch of repeated notes. But if I were to write something like this, then in order to make it easier visually to parse this figure, I would break it up and have a secondary beam like this. Now I can see much more easily how to play this figure; otherwise, it would just be a disorienting mass of notes with no particularly easy way to read them.
- 2.54** Hopefully, it's clear by this point that all of these notations are a compromise between the need for precision and flexibility and the need to present a language of symbols that is clear because a lot of music in the professional world is sight-read, which is to say that the performers will show up to a rehearsal, and they may not have had a chance to look at the music ahead of time.
- 2.55** If you're playing on a Hollywood soundstage, for example, or if you're a gig musician in Los Angeles and you show up for a concert, you might be given the music a few minutes beforehand or the day before, and you might not have had time to read

through it. Those musicians develop a skill of rapidly scanning and sight-reading. In order for them to be able to do that, the notation has to be clear; it has to be very precise the way it's written. And of course, they have to be highly trained in rapidly scanning a piece of music and instantly turning the symbols that they see on paper into a physical gesture.

- 2.56** So this system of notations we have is actually quite powerful because it allows us to do a lot and to write it down with precision and to communicate very effectively between the intentions of a composer and the performer who must realize the sounds that are implied or encoded within this notation. It can't do everything, though.
- 2.57** Now we come to a peculiarity of our notation system, which is, again, unlike many other musical traditions, the pulse groups are typically exact multiples of each other. The 16th note is exactly half as long as an eighth note, and the eighth note is exactly half as long as a quarter note, and so on. Well, what if you want to have note values that don't just divide by half? What if you want some other division? How do you write that?
- 2.58** In poetry, for example, you have triplets, and that's three equal pulses in the time of one beat. You need a notation for that because it might not be convenient or appropriate to change the time signature every time you want to change between beat subdivisions. Here we come to the concept, which is hilariously named, of irrational rhythms or irrational note values.
- 2.59** The rational note values, again, are the mensural system where it's one to two, two to four, and so on. Very, very simple. But we needed a way to account for other types of rhythms as well. One of the most common is the triplet. Let's look at what that looks like.
- 2.60** Here I'm drawing a 4/4 time signature, and I'm going to draw three equal beats to start with: three quarter notes. Now, let's imagine that in the fourth beat, instead of having either a quarter note or two eighth notes, I want to have three beats; I want that beat to be subdivided into three. Now, if I just write this, we have a mathematical problem, which is I now have a bar with 11 eighth notes in it—sorry, I now have a bar with nine eighth notes in it. How many eighth notes can you fit into a 4/4 measure? Eight. So that doesn't square; it doesn't quite work. What I want is not [counts uneven subdivision] because that gives me an irregular beat.
- 2.61** What I want is [counts even subdivision]. I want the last beat to be the same duration as all others, but to be divided into three instead of two. Now, if I didn't do that, once again, I'd have [counts uneven subdivision], and that's not what I want. I want [counts even subdivision], so the pulses are the same. In order to notate that, I need a triplet. The triplet is notated with a little numeral three underneath the last beat. That tells me that I meant to play three pulses in the time of one quarter note.

- 2.62** The phenomenon of irregular or irrational beat subdivisions that have to fit into a broader metric pattern, but only temporarily, is known as a tuplet. The tuplet allows me to write values that don't fit squarely into my time signature, and they're incredibly useful. Now, that allows me, among other things, to alternate between groups of two and groups of three if I want to do that. Then you can get into rhythms that are a little bit more interesting.
- 2.63** For example, here's a bar of 4/4. The first beat is a quarter note, the second beat is two eighths notes, the third beat is three eighths notes, expressed as a triplet, and the fourth beat is two eighths notes. That gives me [demonstrates rhythmic pattern]. Now I'm alternating groups of one beat, two beats, and three beats. Then you can start to have a much broader range of possible rhythmic expression. We're no longer confined to exact multiples, but you can have so-called irrational values that break with the underlying beat subdivision.
- 2.64** And of course, you can go quite far with that, as some composers have done. Large aspects of the romantic piano style, for example, the style of Liszt, of Chopin, of Charles-Valentin Alkan, A-L-K-A-N, who's a 19th-century French composer, of extraordinarily virtuoso piano music, and many other figures besides. A lot of that music is predicated on these very complex and very subtly nuanced rhythms.
- 2.65** In Nocturnes by Chopin, if you open the scores up, you'll see what looked like phenomenally complicated rhythmic subdivisions or polyrhythms, where on the upper staff, the pianist will have to play 17 or 23 or 25 16th notes in the time of 12 or 16. Complicated ratios like that. You look at the page and get completely, like, "How do I do that? How do you possibly make a subdivision like that?" Well, the answer is it gets turned into a gesture. The pianist isn't thinking in terms of, "I have to subdivide 17 into the time of 12." You work on it, of course. You work with the two hands separately, then you put them together, and then you turn it into a gesture. It encourages the performer to create something very fluid, something very subtle.
- 2.66** So tuplets allow us then to have a very broad range of extraordinarily subtle rhythmic divisions. In a bar of 3/4, I could write a figure that looks like this, and now I've got a quintuplet: I've got five equal beats in the time of four 16th notes. Why would you ever need to do that?
- 2.67** Well, in speech, we do it all the time without ever thinking about it. We do triplets, we do quintuplets. Take a word like university—five syllables, university. It allows you to have these pulse groups that don't divide neatly up into two or four and still fit them into the phrase. A lot of what this is attempting to do is to take things that we instinctively do in speech and allow us to notate them in a musical context.
- 2.68** We still haven't gotten to the bottom of what's possible in rhythmic notation because let's say that you're in 4/4, you're not going to just write four equal beats per bar in every single bar because, well, that would be extraordinarily monotonous very quickly.

And real music almost never does that unless you're aiming for a special effect of some kind.

- 2.69** That was the case in the Bach prelude that we looked at in the composition course, where he sets up a little rhythmic figure and just repeats it over and over and over again throughout the piece. But most of the time, what you're likely to do is have phrases that have a certain flow to them. You might not have every pulse be articulated, and you might not have an absolutely regular array of pulses within the bar.
- 2.70** For example, I've just written out two bars of music, and you'll notice that this contains not just pulses, but also rests. The notation adds up to 4 beats per bar, but I'm using a mix of different rhythmic values and rests, and that gives me, ta, I'm not articulating all the pulses. I have some rests in there, and I have rhythmic values of different duration.
- 2.71** Now, sometimes you may need to extend the rhythmic value beyond the bar line. You're holding a note, and you want it to continue for longer than what's possible within a bar. Then you need something called a tie. I have a first bar that ends with a half note, and the half note has a tie that connects it to the first beat of the following bar. What that little symbol tells me is that when I arrive at the second bar, I'm not meant to rearticulate that note or play that note again. Instead, I'm holding it over from the previous bar. That's going to sound like this [demonstrates sustained note carried across the bar line]. I'm not actually articulating a note on the downbeat of bar two, but I'm holding it over from the end of the previous bar.
- 2.72** Now we're starting to approach a pretty complete idea of some of the things that you can do in rhythmic notation. This is by no means an exhaustive list—it would take multiple eight-hour courses to even begin to scratch the surface of all the subtleties of rhythm and all the things you can do within it, and all the different styles and dances and patterns and so on. But this is an initial foray into the possibilities of rhythm.
- 2.73** Remember, we have note values that tell us not an absolute duration, but a relative duration. The absolute duration is determined, among other things, by the tempo, by the context, by the style, and so on. Then we have time signatures. Time signatures allow us to break up the flow of beats into a regular pattern that we can parse much more easily, a pattern of stresses. We had binary and ternary time signatures, and compound binary and compound ternary time signatures.
- 2.74** Then we looked at the ways that you can combine different rhythmic values within a single bar and how you can allow for irrational values that are not exact multiples of the beats in the bar. Finally, we looked at ties, which allow you to prolong a rhythmic value beyond a bar line or beyond a written duration.
- 2.75** We're going to end with one more thing, which is dotted rhythms. We don't need to spend a great deal of time on this, but it's important to understand the concept. If I have any rhythmic value—it can be a quarter note, half note, even a whole note—I can

extend the value of that duration by half by adding a dot to the right of the note, and that looks like this. And these are called dotted rhythms.

- 2.76** So a dotted quarter note is equal to three eighth notes. Why three? Because a "normal" quarter note is equivalent to two eighth notes. If I extend it again by half, then that gives me three eighths notes. Likewise, a dotted half note is equivalent to three, not two, but three quarter notes, and a dotted whole note is equivalent to three half notes.
- 2.77** Now, where that starts to get useful is in places like this—remember, we looked at meters like 6/8 or 9/8? Let's return to that for a second. If I have a bar of 6/8, and I don't necessarily want all six pulses to be articulated, maybe I just want two pulses to be articulated in a bar of 6/8. The dot tells me that I've got two beats, each of which is three eighths notes long [counts two beats, each subdivided into three]. So the dotted quarter, again, is equal to three eighths.
- 2.78** And finally, the dot allows you to have unequal rhythms. In French, there's a term which is *valeurs égales* or *valeurs inégales*. So a beat can be divided into equal values. In the language of popular music, that would be straight eighths: each beat is divided into two equal beats that are equidistant to each other [demonstrates equal rhythm].
- 2.79** In unequal values, one of them is longer than the other. You have a long beat, followed by a short pulse [demonstrates unequal rhythm]. So that is an unequal pattern, and it's notated with a dot. Now, it all adds up mathematically because the dotted eighth note in the example I've just written is equivalent to three 16ths. Then, at the end of the beat, we have a fourth 16th. So each of those beats is made up of four 16th notes. The first one is three times longer than the last one.
- 2.80** Now, obviously, I have to emphasize this once again—it would be a mistake to think that performers are constantly going, "One, two, three, four, one, two, three, four—" No. They don't count every single pulse within the 16th notes. It's a matter of feel, and it becomes instinctive after a while. But that rhythm there would be played [demonstrates long-short "skipping" rhythm]. You can think of that as a skipping rhythm. If you're skipping, you're on one leg for longer than the other.
- 2.81** All of these rhythms—and there's an immense repertoire of them—but they're directly derived from physical things: the momentum of walking, the momentum of running, skipping, jumping, riding a horse, turning around in circles. These basic patterns of physical movement that are fundamental to how we interact with the world are manifested in these different time signatures and rhythmic patterns that underlie this musical tradition. We'll stop there for now for rhythm. I've only begun to scratch the surface. We're going to regroup in a few minutes. Thank you very much.
- 2.82** You expressed that music notations are an attempt at crystallizing the ephemeral sounds into permanent writings. Is there any link between music notations and sign language, for instance?

2.83 Well, in the case of sign language, or Braille, or written text or music, what you have is a language of symbols, or in the case of sign language, of gestures. So a symbol can take many different forms—it can be written down, it can be something tactile, or it can be a gesture, provided that the gesture or the symbol or whatever it is, has an agreed-upon meaning that is shared between the person relaying the message and the person receiving the message. That's all. Whether you do that through smoke signals or through whole notes and half notes or through subtle bodily gestures, in a sense, is not the main question. There just has to be a commonality of reference between both parties. That's all.

Lecture 3: Pitch and Intervals

At a glance — Pitch and intervals.

- 3.1 We've been taking a tour through the basic rudiments of music, through the basis of the language that allows performers and composers and indeed, audience members to communicate with each other and to understand each other and to have a common written basis. We've looked at the essentials of the staff notation, where notes are located on the staff and clefs. We've looked at rhythm, we've looked at time signatures.
- 3.2 What we're going to do now is get a little bit deeper into pitch and the possibilities of notating pitch and intervals. So far, what we've done is we've taken a five-line staff and we've learned where the notes go, we've learned the names of the notes, and we've learned some mnemonics, some little acronyms to help remember where those notes are located.
- 3.3 We have to go a little bit farther now. As anyone who's ever glanced at a piano keyboard knows, we have both black keys and white keys. When I play a simple C major scale on the keyboard, I'm not playing every single tone on the keyboard. If I were, it would sound like this. What I just played there was a chromatic scale. The first scale that I played was called a diatonic scale.
- 3.4 Diatonic, in this context, refers to a seven-note scale. It doesn't contain every semitone—every black key and every white key. It contains a pattern of whole tones and semitones in a fixed order. The chromatic scale, however, consists of an equal sequence of semitones. Every single tone is present. Every tempered tone, that is to say, in other words, the tones that it is possible to play on a piano keyboard.
- 3.5 Of course, if you look at other musical traditions and other instruments, it may not be laid out as a sequence of discrete steps, but more as a continuum. You can do this very easily with your voice. Obviously, your voice doesn't have a keyboard. You can produce any frequency, any pitch within the range that you're able to sing. You can sing very, very small fractional intervals that are smaller than a semitone, which is the smallest interval you can play on a piano. But for now, we're just going to concern ourselves with whole tones, or whole steps, and semitones.
- 3.6 If I want to play anything other than the white keys on a keyboard, then I need to have some visual language of symbols that allows me to explain what I'm doing—that takes the form of accidentals. What accidentals are is they take one of the seven diatonic tones of a major or a minor scale—and we'll explain more about scales later on—but for now, all you need to know is if we take the seven basic tones that we described earlier: do, re, mi, fa, sol, la, si, accidentals allow us to alter that pitch either raising it or lowering it or canceling out a previous accidental. I'm going to show you what that looks like.

- 3.7** We need a whole new set of symbols for this. Remember that if I want to write middle C on a treble clef here, a treble staff, then I need a ledger line, and there's my middle C. If I want to play D, then I just go to the next space or line up, and so goes the sequence. However, I can also place a sign before the note that tells me if it's a natural, a sharp, or a flat. And these symbols look like this—natural [$\natural$], sharp [$\sharp$], and flat [$\flat$].
- 3.8** The natural sign and the flat sign are both derived from the letter B. In fact, if you know the French terminology—I know we keep coming back to this—the French term for flat is *bémol* or a "soft B." The French term for natural is *bécarre*, which means a "square B" or a "hard B." That also refers to—we don't want to get too deep into the history of this, but the seventh tone in the sequence do re mi fa sol, la, si, or B, has a somewhat ambiguous status.
- 3.9** The quality of that interval could be different depending on the scale you're using, depending on the era in which you are writing, your geographical region, and so on. Sometimes it sounded like this—this is a very crude demonstration of the idea, but just to illustrate it [plays a scale with hard B], and sometimes you would hear this [plays a scale with soft B].
- 3.10** In order to differentiate between the different ways that you could have this seventh note, this B, we developed these symbols over time, the soft B and the hard B, if you like. Then those symbols became abstracted into what we have today. The flat symbol looks a little bit like a B, and the natural symbol looks a little bit like a square B, like this.
- 3.11** Now, the sharp symbol indicates that whatever note it lies before is raised by a semitone. The flat means that whatever note it appears before is played one semitone lower. The natural sign means you play it as a white note. You cancel out any preceding alteration. The reason you would need to cancel is that the general rule broadly applied is that accidentals hold for an entire bar.
- 3.12** So if I write a D in my piece—imagine we're in 4/4 here; I've got C, D, E, D $\sharp$, D $\sharp$, E, D $\natural$, C—I have to write the natural symbol there in order to cancel out the sharp that we see on the first beat of the second bar. If I didn't have that natural sign, then the performer would assume that the D on the third beat would still be a D $\sharp$. The natural sign allows you to clear up ambiguities, and it allows you also to cancel an accidental when necessary.
- 3.13** To recapitulate: if I want to write a G, let's say on the treble staff, then if it looks like this, you assume in the absence of any accidentals that it's a G $\natural$. If I add a sharp before it, that gives me a G $\sharp$. If I put a flat, now I've got a G $\flat$.
- 3.14** Here's my G $\natural$. If I lower it by a semitone, I obtain a G $\flat$. If I raise it by a semitone, I get G $\sharp$. All of these notes are called G, but the accidentals allow me to alter it—to raise it or to lower it. Then that leads to an inevitable question, which is: If you can write a G $\flat$ and the note below G is F, then you can also write an F $\sharp$, and they're the same.

- 3.15** If I have an F here, I can also write an F $\flat$ and an F $\sharp$. Now, in terms of just pure frequency, all other things being equal, without any context around the notes, this F $\sharp$ sounds identical to this G $\flat$. In both cases—start on an F, add a sharp, it becomes F $\sharp$. Now, if I take the G and lower it, I get that note. It's the same in either case. Why would you spell it F $\sharp$ or G $\flat$?
- 3.16** Well, we'll have to get into that a little bit later because we're getting into questions of harmonic functionality, but as a brief explanation and a simplified one, it partly depends on your direction of travel. If you're going down, then you use flats. If you're going up, then you want to use sharps. We'll use that as a heuristic for now.
- 3.17** For example, if I wanted to write an ascending line like this, this gives me a little chromatic sequence. Remember, we talked about the difference between chromatic and diatonic scales. Chromatic scales include all the semitones—all the black keys, all the white keys on the keyboard. If I start on an E and go up chromatically, then I can write that as E, F, F $\sharp$, G.
- 3.18** If I do this in the opposite direction, I start on the G, then I'm likelier to spell it G, G $\flat$, F, E. It's the same thing played backwards, but because the direction of travel is different, I'm going to use a flat to lower the G as opposed to an F $\sharp$ to raise the F. That's a concise introduction to intervals—sorry, to accidentals. But that doesn't bring us to intervals just yet, which is the next thing we need to discuss.
- 3.19** I mentioned that these scales—the major and minor scales—are the two most widely used ones in the Western tradition, and that has occurred over a very lengthy process of historical refinement. These scales are based on a pattern of seven notes, as I mentioned, in a fixed order. So if we take a C major scale, we have the following sequence [plays C major scale].
- 3.20** Now, again, if you look at the pattern of black and white notes on the keys, you'll see that in order to go from C to D, I have to skip over the C $\sharp$. I'm just going to play C to D and then D to E. But I'm not going to play [plays semitone sequence C, C $\sharp$, D, D $\sharp$, E]; I'm going to skip those.
- 3.21** When you have two notes played one after the other, if it's a sequence like that, then you have what's called a melodic interval, and that is the distance, or the difference, between the first note you played and the second one. If I take two tones and play them simultaneously, you obtain a harmonic interval. So you have to distinguish between melodic intervals, one after the other, and harmonic intervals, which are two tones played simultaneously.
- 3.22** Now, in the order of tones I play, the interval is determined by the distance, if you like, between the first note and the second. In the case of C to D, that is called a whole step because we're going from one natural note to the adjacent one and skipping the semitone in between. If I went from C to C $\sharp$ or C to D $\flat$, that interval would be called a

semitone. If I look at the resulting pattern, that gives me tone–tone–semitone–tone–tone–tone–semitone.

- 3.23** There's a pattern of intervals that I can remember. Let's write out that scale. The first interval from C to D is a whole tone—we're going to abbreviate that as T. Then, from D to E is another whole tone. Now, from E to F, there's no black key in between, so that's the smallest interval you can play—that gives us a semitone. E to F gives us a semitone—we'll abbreviate that as S. Then F to G is a whole tone. G to A is a whole tone. A to B is a whole tone. Then, if we continue to C, then we get another semitone. But the easiest way to remember this sequence is tone–tone–semitone–tone–tone–tone–semitone.
- 3.24** Now, if I want to start on any other tone on the keyboard and play a scale, all I have to do is remember that sequence, and I can reproduce exactly the same scale. That's called a transposition. So it's the same sequence of intervals, but starting on a different note. Here's my C major scale, and here's a D major scale. Exactly the same sequence of intervals, but starting on a different note. That gives us a little introduction to the concept of transposition.
- 3.25** But of course, we haven't made tens of thousands, hundreds of thousands of pieces using only a single scale. One of the things you learn very early on when you're learning an instrument and learning about music theory is that we use both the major scale and something called the minor scale.
- 3.26** The minor scale, as its name implies, has a different sequence of tones and semitones than the major scale. The easiest way to form a minor scale is to begin on what's called the relative minor of the major scale that you've just played. Remember, if we're in C major [plays C major scale], I can form the relative minor by starting one-third lower than the C that I began with. So, I'm going to start on A rather than starting on C, and that gives me this [plays A minor scale].
- 3.27** Like the C major scale, the A minor scale only uses the white keys on the keyboard. But the sequence of tones and semitones is different. We call this the relative minor because C major and A minor share the same key signature—that's a concept that I'll be explaining a little bit later. If there's anything you're not following up to now, don't worry. We're going to cover a lot of this stuff again.
- 3.28** C major has no sharps or flats. We're only playing white keys on the keyboard. Therefore, the key signature is zero sharps and zero flats—pretty straightforward. If I move down to A minor, A minor also has no sharps and no flats, but starting on a different tone and with a different sequence. So, A minor looks like this. Here, the sequence is tone–semitone–tone–tone–semitone–tone–tone.
- 3.29** Why do we call them major and minor scales? Why is this C major [plays C major scale], and why is this A minor [plays A minor scale]? The major and minor refer to the size or the quality of the third interval in this scale. Before I explain different qualities of

intervals, we have to look at what the intervals are. Let's look at that for a minute. Let's start with C. Just a plain ordinary C. We're not going to worry about sharps or flats for now.

- 3.30** Now, if I move to the next tone up directly—C to D—remember, that gives you a whole tone. This is also called an interval of a major second. If I start on the C and move up to an E directly without any notes in between, then that gives me the interval of a major third. If I start on C and go up to F, that gives me an interval called a perfect fourth—or P4, the abbreviation. Continuing the series from C to G gives me a perfect fifth. From C to A gives me a major sixth. From C to B gives me a major seventh.
- 3.31** Now, so far, you'll have noticed that some of the intervals are called major, some of them are called minor, and some of them are called perfect. This connects back to what I was speaking about in my composition program, or forward to it if you're seeing them in the other order, which is that for historical and acoustic reasons, some intervals are referred to as perfect—the unison, the fourth, the fifth, and the octave are considered to be perfect consonances. They are simple consonances. They are very simple ratios of each other. Another way to think about this is that they're simple, or they're considered perfect because they're very easy to relate to the lower note of the interval. So think of these as simple or perfect consonances.
- 3.32** Then we have major intervals, minor intervals. Major second, major third, major sixth, major seventh. Now, you can also play minor seconds. For example, C to D $\flat$. If I take my C to D interval there and add a flat sign on the D, this gives me a minor second. Likewise, if I take it C to E and put a flat symbol there, that gives me a minor third. Remember, this is the major third [plays C–E], and this is the minor third [plays C–E $\flat$].
- 3.33** The A minor scale—if I take the first and the third note in that scale and play them together, I obtain a minor third. Major and minor refer to the quality of interval between the first note of the scale and the third, among other things, but that's the most immediate distinction. So, so far now, we've looked at how intervals can be major, minor, or perfect. There are other possibilities, too, but we'll get to those a little bit later.
- 3.34** The other thing we need to know is that in any scale—any diatonic scale, that is—the notes are assigned names, of course, and if you create intervals between one note and any other note in the scale, then you can describe it in terms of a perfect fourth, a perfect fifth, and so on. But you can also assign the notes of a scale Roman numerals, or degrees: I, II, III, IV, V, VI, VII, and so on, such that, if we're in C major, and I play an A, I can refer to the A as being the sixth degree of the C major scale. I can refer to the E as being the third degree. You can think of this as almost like steps on a ladder. You can imagine it as being an almost spatial movement between one step and another.
- 3.35** If I'm playing my scale and I play only adjacent tones in the scale, then that is called stepwise motion. I'm moving by step, which is to say I'm only moving between adjacent

tones. Now, the instant I have an interval that's larger than two adjacent notes, I'm moving by leap because I'm skipping over some of the notes in between. And generally speaking, melodies, in order to be interesting, will contain a mixture of both steps, stepwise motion, and leaps.

3.36 Music is obviously a very complex phenomenon, and harmony in particular. Part of the reason that harmony can be difficult to get a handle on, intervals can be difficult to get a handle on, is that they change depending on the context. Remember how I said that you can spell that tone there as F $\sharp$ or as G $\flat$, depending on the direction of travel. C, D, E, F, F $\sharp$, or a C, B, A, G, G $\flat$. Both are possible. That's called an enharmonic spelling. Enharmonic means that the sounding pitch is the same, but it's spelled in two different ways: F $\sharp$ or G $\flat$.

3.37 Now, what today counts as an enharmonic spelling would have actually sounded quite different during the Baroque. The reason is that this equal temperament that I described earlier, where all the semitones on a piano are tuned to the same ratio; they're all the same interval. But on older instruments, before equal temperament became the norm, various different systems existed simultaneously, which allowed for either perfectly tuned thirds or perfectly tuned fifths, which necessitates some amount of compromise because you can't have both in the same scale for acoustic reasons.

3.38 What we've opted for is a totally artificial system, which is a compromise, where if we base our tuning system on acoustics, on the actual physical properties of vibrating strings or air columns in a pipe, then you need a very specific ratio to produce a fifth that's exactly perfect, that's exactly in tune. The octave has to be exactly double the frequency of the lower note, for example, and so on and so forth. The third, in order to be precisely in tune, needs to be in a specific frequency ratio between the bottom note and the upper note.

3.39 However, if you try to make a scale out of only acoustically perfect intervals, what will happen is your octave won't be in tune with itself. Your scale will be completely lopsided, and you won't be able to reproduce it in higher octaves. If I start here, I can just keep going the entire length of the keyboard, and it all works. But if you try to do that with exactly perfect intervals, the notes would be subtly rising and getting out of tune with itself. It wouldn't work.

3.40 In older music, they tried to get around this problem by making some intervals a little smaller than they needed to be in order to try to fit them within perfectly in tune octaves. The compromise that we've landed upon is making all the intervals very slightly out of tune. That's what we have here. The fifths on this keyboard are not exactly acoustically perfect, nor are the thirds, nor are the fourths. They're slightly flat, but the octaves are precisely in tune.

3.41 What that means is on a keyboard like this, you can play in any key. I can play in C major, I can play in D major, I can play in A flat, I can play in G $\flat$ major, I can play that

scale, that C major scale or the A minor scale, starting on any key on the instrument and working my way up and down, and all of it will be in principle in tune.

3.42 If you try to do that on an early instrument, like a 17th-century harpsichord, the way it would have been tuned at the time, you would find that as you venture from keys like C major or A major into keys with lots of sharps or flats, like A $\flat$ or D $\flat$, that the farther you get from C major, the more out of tune it gets. You can't actually play in every key on those instruments. You can try it, but it's going to be extraordinarily out of tune once you get too far from C major. It's a compromise.

3.43 On those keyboards, if you're playing in a key that has a relatively small number of sharps or flats, then the intervals—the octaves and the fifths that are meant to be perfect—will be exactly in tune, but there are going to be some funky minor seconds and minor thirds that are going to sound really strange to your ears. So here we have something, again, which is homogenous. It's not perfect. It's very slightly out of tune, but that gives us the flexibility of playing in a broad range of different keys.

3.44 Let's continue with intervals. Now, intervals for beginning musicians can be tricky to get a handle on because at a certain point in your training, you have to be able to not only remember what each interval sounds like, but you have to be able to reproduce it with your voice. If you wake me up at 3:00 in the morning, I have to be able to sing an augmented fourth. I have to be able to do that. I have to remember the interval, I have to be able to externalize it, and I have to, of course, also be able to write it down. That takes a certain amount of your training to be able to do that effortlessly.

3.45 We've looked at major intervals and minor. We've looked at perfect fourths, fifths, octaves, and unisons. Now, there are a couple of other concepts we need to know. Let's make this simple. I'm going to start with the C above middle C on my treble staff here, and I'm going to put a D next to it. If I keep it like this, remember, I obtain a major second. Now, if I add a flat, remember, this gives me a minor second. This is the major second [plays C–D]. This is the minor second [plays C–D $\flat$].

3.46 Now, if I put a sharp next to the C, I also obtain a minor second. However, this is where it gets a little bit weird. If I start with C and go to a D $\sharp$, I get this [plays C–D $\sharp$]. Then you might say, "Well, that sounds like a minor third. Isn't that the same thing?" It's not the same thing. The reason it's not the same—I mean, it's the same acoustically, but it's not the same thing functionally because I've spelled these notes C and D. The fact that the D is sharp gives me an interval larger than a second, but spelled C to D, and C to D is always a second. Even if it's D $\sharp$, D $\flat$, C $\sharp$, C $\flat$, it's always a second because the size of the interval, if you like, is determined by the distance between two adjacent tones in my scale—C and D in this case.

3.47 Now, if I alter them, I alter the quality of the interval, but I don't alter the interval, if that makes sense. I'll try to explain. If I move from C to D $\sharp$, I've now got an interval that's larger than a major second. The next step up from a major second is an

augmented second. Augmented, meaning even larger than a major second; it's the next step up.

- 3.48** Now we're getting into some seriously weird theoretical territory because it's possible to go even farther. This symbol that I've just placed next to the D in my next example is a double sharp [D^{##}]. And a double sharp, as you may have guessed, is one step higher than sharp. And that gives me this [plays C - D^{##}]. So now you're getting seriously irritated. You're like, "Well, that's obviously not a second. That's a third. That is now a major third." Well, yes, except I've spelled it C to D^{##}, and C to D^{##} is a double augmented second.
- 3.49** This is where we get to one of the more confusing or counterintuitive aspects of music theory, which is that there's a gap between the acoustic reality of what you're hearing and the actual function of the interval, depending on how it's written and depending on the harmonic context. If I spell that C to D^{##}— which you might think nobody would ever do that, but it actually does occur—then I get a double augmented second.
- 3.50** If I go the other direction, C to D^b, then we have a minor second. If I go C to D^{##}, which looks like this [D^{##}], then I obtain a diminished second. The diminished second sounds like this [plays C-D^{##}]¹—yeah, it sounds like I'm playing the same note twice, and acoustically, that is what I'm doing. But again, I've spelled it as C to D^{##}, and that gives me this—still a second because of the way I've spelled it.
- 3.51** You get these strange occurrences—this is another one: B[#] to C, where the two notes sound identical, but I spelled the first one as B[#], so I'm raising the B by a semitone, and that gives us the same tone as C. The point is to understand that there's a distinction, again, between the function and the acoustic reality of what you're hearing. The reason this is important is that, depending on the key you're in, the same frequencies will be spelled, will be written, in different ways. Depending on the nature of the harmonic motion, you might spell an interval in many different ways.
- 3.52** I want to return now to the A minor scale that we were looking at previously. Remember that the quality of the third, which in this case is minor, in an A minor scale, goes some ways to determining whether it's a major or a minor scale. It's not the only difference, but that's the most immediately noticeable one. Now, in reality, if you're writing a piece in A minor, you're not only going to use the white keys all the time. Part of the reason for that is that there is a very special phenomenon in tonal music—in other words, music that's written in a key in the Western system, which is called the leading tone. This is a very, very important concept. Here we're getting into questions of harmonic and melodic functionality.
- 3.53** Let me write out that A minor scale again. The bottom note here—the first note of the scale, the first scale degree, if you like—is A. Then, when I get back to the top again at the end, I get back to another A. Now, the interval between the tone immediately below it and the first note of the scale, in this case, is a major second. That ends up

being a bit of a problem. The reason is this phenomenon called the leading tone. The leading tone is the note that lies directly below the first note of the scale.

- 3.54** Now, let's back up and go back to C major for a second. There's the leading tone. The leading tone has this peculiar property: it really wants to go up. It wants to lead you back to the beginning of the scale. That's why it's called the leading tone. And in major scales, the leading tone is a semitone below the first note or the first scale degree. That first note has a name, which is the tonic—the tonic being the strongest or the first note in the scale. The leading tone is directly below the tonic and wants to lead back up to it.
- 3.55** The fact that you have this leading tone and this tonic gives you this strong sense of orientation and attraction and stability when you do and back up on the tonic again. The minor scale gives you a leading tone that is a whole step lower than the tonic—G to A. That gives you a much lesser degree of upward push, let's say, this desire to resolve, this desire to lead us back to the tonic.
- 3.56** This gives rise to something called the A melodic minor [plays A melodic minor scale ascending], which raises the sixth to F# and the seventh to G# in order to preserve that sense [plays A-F#-G#-A] of something sharply leading you back to the tonic again. When you go back down in the opposite direction, you're going away from the tonic, but you don't have that motion towards the tonic again, so you lose the F# and the G# [plays A melodic minor scale descending]. The A minor scale has the peculiar property that you don't necessarily have the same notes going down as going up.
- 3.57** One of the things you want to do if you're coming to terms with music and you're starting to learn an instrument, you're starting to get familiar with all of these concepts, is you learn to, first of all, identify the intervals by ear so that if I wake you up at 3:30 in the morning and go like this [plays C-G], you're like, "Oh, perfect fifth," or [plays G#-C#] "perfect fourth," or [plays C-C#] "that's a minor second," or [plays Bb-D] "major third." You can instantly recognize both the size of the interval and its quality, and you can reproduce it with your voice. That is fundamental to musicianship—to have that capacity. That's something that you do by training your ear. At the end of this course, tomorrow, we're going to look at some exercises to work on that.
- 3.58** The other thing you learn is you start thinking in terms of sequences of intervals, which makes it easier to transpose. You start learning about moving around between different keys, between C major and D major and F [major]. You can play the same scales and the same chord patterns, but starting on a different note.
- 3.59** All of this, of course, takes a great deal of practice. What you want to do is start with very, very simple little exercises, just playing scales in different keys, memorizing the patterns. Eventually, what happens is they become gestural; you're not thinking of them intellectually as you're playing them. You're not thinking, "Okay, C goes to D, D goes to—" No, you remember it as a sequence, a physical sequence. It becomes encoded in your muscle memory and you don't think about it.

- 3.60** But it takes a while to develop that fluency, just as it takes a while to learn a foreign language so that you're not constantly thinking about grammatical constructions and syntax and vocabulary. But like anything, when you're acquiring a new skill, it's usually best to not torture yourself by trying to go too fast or trying to cram all of this information in your head in a short period of time.
- 3.61** All the more so, depending on your aptitude and your interest in music, you might find this extraordinarily tedious to learn. You might find it effortless and natural, but in any case, you want to make sure that when learning skills like this, that you go at a pace that allows you to progress, but without it being torture. If it's torture, you won't keep doing it. You're going to give up and do something else because why torture yourself? But if you can make it where it's stimulating and enjoyable and it's stretching you a little bit, but not to the absolute limit of your capability all the time, then you're much likely to retain it, and eventually it becomes second nature.
- 3.62** So, in this lecture, we've looked at intervals, we've looked at qualities of intervals, and we've looked at some basics of how you can put together major and minor scales. As with all of the things that we're looking at, of course, we could go into much more detail, much more depth, but I think that's a reasonable introduction to the topic.
- 3.63** We've looked at how in minor, you can have the A natural minor scale, which is simply formed from the same tones as the C major scale. You can alter it to obtain a harmonic minor scale or a melodic minor scale, depending on the context and depending on, of course, what type of piece you're writing.

Lecture 4: Keys and Harmony

At a glance — Keys and harmony.

- 4.1 We began our deep dive into the wonderful and strange world of intervals earlier, and we also looked at how the way the interval sounds acoustically, and what its function is and how it's written can be different things depending on the musical context. Now, what I want to do is to get into one of the most fundamental concepts of Western music, which is the concept of a key or a key signature.
- 4.2 Now, we've looked at the A minor and C major scales. We've looked at how they're related to each other—A minor being the relative minor to C major, and inversely, C major being the relative major of A minor—relative in the sense that they share the same key signature, which is to say the same number of sharps or flats that are needed in order to play a scale in that key.
- 4.3 Now, if we were content to just write pieces that consisted of seven notes played over and over and over again, running up and down the keys and so on, which you could do indefinitely if you wanted to, but it might be quite boring and homogenous, and it might not afford you many possibilities to write a longer piece. In order to have a broader range of motion in a piece of music and a broader range of contrasts and harmonic effects, we need to be able to move around a little bit harmonically.
- 4.4 To a degree, you can do that, as we looked at earlier today, with accidentals— sharps and flats—which allow you to change the quality of an interval. But it's also necessary to have a key. You can think of a key as being an organization of pitches around a central tone. If you see that there's a Beethoven symphony in C major or in B $\flat$ major or in A major, then you might wonder what exactly that terminology means, and why would you bother putting that in the title?
- 4.5 Well, the A major or the B $\flat$ or whatever it is tells you what the key of the piece is. In other words, what is the central tone against which all the others are heard? This gives us a sense of both harmonic stability and a sense of harmonic motion, because you can move harmonically in a piece of music away from home base, if you like, or the tonic, the first scale degree. You can move away from it; you can have a centrifugal movement away from it.
- 4.6 Or you can move back towards it—a centripetal force that draws you back. That allows you, again, to create very long-range harmonic and dramatic effects in a piece of music. If your piece were static—in other words, all you could do were to play the tones of C major all the time—well, you'd be limited. You could do it, though [plays improvised passage on the piano]. I could do that for four hours. It might be nice. There are some genres of music where that harmonic stasis goes on for a very long time, but it's not suited for all styles.

- 4.7 You might want to actually move somewhere else. You might want to have a range of harmonic motion [plays improvised passage on the piano]. Now, what I just improvised there takes me away from C major into different keys, sometimes very fleetingly, sometimes for a longer duration. Maybe I end up in a different key and I stay there for a while, and then I have to work my way back to where I started again. But in any case, in order to allow for that movement, I need to have the possibility of moving between different keys.
- 4.8 First, we have to lay out what these keys are and how they relate to each other. To illustrate this concept, I have to explain about something called the circle of fifths. Now, you might have heard about this already. It's a foundational concept in music theory. Well, it's foundational, but it wasn't actually formalized straight away. But let's have a look at how this works.
- 4.9 If I start on C, and then I continue to move upwards in a sequence of fifths, where each successive tone is a fifth higher than the previous one, if I keep doing that, eventually, I'll get back to where I started again, and it completes a circle. I'm going to have a C, followed by a G—a fifth above the C. Then, if I keep going, I get to D, and then I get to A, I get to E—now, I'm going to have to change octaves because obviously this is getting a little bit high—then I get to B, F#, C#, G#, D#, and then A#, F—this is where the sequence starts to get a bit funny—and then end up back on C again.
- 4.10 Now, you'll notice something in this little sequence here is that if I keep going in fifths [plays sequence of fifths], eventually, I get back to where I started again, in a circle. Having done that, I've played every single semitone. If you take all of these 12 notes and group them together, it gives you a complete chromatic scale. Not in that order, obviously, but that gives you the collection of all 12 semitones that you can play on the keyboard. That is called a circle of fifths.
- 4.11 Now, if I want to build keys, or scales on each one of these tones, preserving the same order: tone–tone–semitone–tone–tone–tone–semitone, that characterizes the major scale as we saw earlier, then I have to keep adding sharps or flats to play those interval patterns. Remember, C major [plays C major scale], no sharps, no flats, just the white notes on the keyboard.
- 4.12 If I go up a fifth and start on G [plays G major scale], I can't play an F; I have to play an F#. That means I have to add an F# to what's called the key signature. The key signature looks like this: on every staff of every page of this piece, as long as I'm still in G major, I'm going to put an F# on the top of the staff there, and it's going to appear on every line. What that means is, unless I include an accidental or cancel out the sharp somehow or modulate to another key, every F is going to be an F#. That just makes it a lot easier because you don't have to constantly write out F# every time you want to play that note.

- 4.13** Now, if we continue the sequence, we go from G to D [plays D major scale], I still have the F $\sharp$ from G, but now there's a second sharp, which is C $\sharp$. If I want to write out a D major scale, I'm going to need a key signature with an F $\sharp$ and a C $\sharp$, and then I can write the notes like this. As you keep going up the cycle of fifths, you keep adding an additional sharp to the key signature. The order is F $\sharp$, C $\sharp$, G $\sharp$, D $\sharp$, A $\sharp$, E $\sharp$, B $\sharp$.
- 4.14** Now, remember, we have these helpful acronyms. How are you supposed to remember this? F C, G, D, A, E, B. The most common one that I've encountered is "Father Charles Goes Down And Ends Battle." Okay, it's not the world's most clever acronym, but for some reason it's stuck: "Father Charles Goes Down And Ends Battle." No doubt there are better ones, but this will do for now.
- 4.15** If you're trying to remember, let's see now. I'm in A major. There's three sharps in A major. What's the order of sharps I need to write? Father Charles Goes: F, C, G. Then you can write the sharps on the staff: F, C, G—that's for sharps. But you run into an issue, which is that if you just keep going up in fifths and adding sharps all the time, you eventually are going to run out of sharps or you're going to need double sharps, and it's going to get seriously weird.
- 4.16** If you're in G $\sharp$ major, then you need how many sharps? Eight. That's more than there are notes. You can do that technically, but it's much better to use an enharmonic equivalent at that point. So instead of writing the key signature of G $\sharp$ major, you would spell it as A $\flat$ major. A $\flat$, conveniently, has four flats. It's a lot easier to read, it's easier to write, and it's simpler. The order of flats for key signatures that use flats is B, E, A, D, G, C, F. Another silly acronym is "Battle Ends And Down Goes Charles' Father."
- 4.17** Now, in order to create a cycle of increasing numbers of sharps in a key signature—remember, you go up in fifths from C to G to D to A and so on. Now, you might want to use key signatures that use flats instead of sharps, in which case you would start on C and go up in fourths rather than in fifths. So start on C, then you go up to F, then to B $\flat$, E $\flat$, A $\flat$, D $\flat$, G $\flat$, C $\flat$, F $\flat$ —again, by that point, you start needing to use double flats, and it starts to get a little bit complicated. Had I pursued the logic all the way to the end, then here in my sequence of sharps in the cycle of fifths, I would actually need an E $\sharp$ rather than an F $\sharp$, and so on.
- 4.18** This gives you the sequence [plays cycle of fourths]. So it also gets you back to C, but you're going in fourths instead of fifths. So C, once again, no flats; F, one flat; B $\flat$, that's two flats; E $\flat$, that's three flats; A $\flat$, four flats, and so on and so forth. So these keys are arranged in cycles. You add a flat each time.
- 4.19** So with this contemporary piano with equal temperament, I can play in any key I want. I can play in A $\flat$, I can play in D $\flat$, I can play in B $\flat$, F, G, whatever. I just have to remember what the key signature is. If I know that, then I can instinctively figure out what notes to play. If I'm in E $\flat$ major, then I remember my little acronym. It starts C, B, E—"Battle Ends And"—that brings me to three flats. I can remember the order of flats

from my little acronym: B $\flat$, E $\flat$, A $\flat$. Then I just have to, every time there's an E, play an E $\flat$. Every time there's an A, play an A $\flat$, and then a B $\flat$.

- 4.20** This is, again, the thing where it can be very counterintuitive to learn it, and you just have to drill these sorts of things. It's a bit of memorization involved. Remembering the key signatures, circle of fifths, the number of sharps or the number of flats in every key, and then you just drill it so that you think, "C $\sharp$ has seven sharps, and B major has five sharps, and A $\flat$ has four flats," and so on. It becomes second nature after a while.
- 4.21** That's a pretty fundamental concept because, again, it allows you to play in different keys, and it allows you to modulate. If you're writing a piece in C major, you can move to another key, and then you can eventually work your way back to where you started again. It gives you a much broader range of possibilities. Now that we've looked at melodic intervals, we've looked at scales and how they're put together, and we've looked at key signatures in the cycle of fifths, now we can look at chords and how chords are assembled.
- 4.22** I've previously given a series of lectures on musical composition for Peterson Academy, and I gave a lecture on chords, on harmony, in that course, but that was very much practically oriented in the sense of, "Here's a bunch of ways that you can take a simple chord and transform it and obtain a whole bunch of different sounds by changing the voicing, by changing the register, by doubling certain tones and so on."
- 4.23** I'm not going to duplicate that material here, but I'm going to talk to you a little bit about how chords are formed in a more abstract sense and the different ways that they can be transformed. Finally, we're going to look at the qualities of chords. Remember that intervals have qualities—they can be major, minor, perfect, augmented, diminished. We're going to look at chords and the qualities that they can have as well.
- 4.24** So first, we have to understand how a chord is constituted. A chord simply means two or more tones that are played simultaneously. It can be any two tones, or three, or four, or five, or however many tones you're able to play at the same time. The two tones could be C and D $\flat$ played together. Or it could be [plays a combination of tones]. Any combination of tones that you can possibly imagine can make a chord.
- 4.25** In the vast bulk of what's called the musical common practice period, or rather the bulk of codified practices that characterize the Baroque, the Classical, and the Romantic eras, and more. Besides, it's not as though this era has, by any means, come to an end. But let's say there was an era in which the basic grammar of music was very broadly and widely applied throughout the Western world, with very few meaningful deviations.
- 4.26** Of course, the world of music has expanded quite considerably. The technical means of music have turned into a vast ocean. So it's less easy to speak in terms of generalizations. But nevertheless, for the sake of tonal music, music written in keys, a triad is the most fundamental harmonic unit. A triad is constituted of the first, third,

and fifth notes of a scale. Let's look at this again. Here's my C major scale— remember that you can assign Roman numerals to each scale degree: I, II, III, et cetera. If I want to make a C major triad, then I take the first scale degree, the third and the fifth—or the root or the tonic, the third and the fifth, and I play them simultaneously. That gives me this [plays C major chord].

- 4.27 So here's my scale: I, II, III, IV, V, VI, VII; do, re, mi, fa, sol, la, si; C, D, E, F, G, A, B. Of course, you can use whatever term you want within reason. We're just going to take the first, the third, and the fifth, and that gives me a triad. The triad is one of the most absolutely fundamental building blocks of Western harmony.
- 4.28 I can construct a triad built on the first, the third, and the fifth note on any scale degree. I can do that on the first scale degree or the tonic. But I can also build a triad on the second scale degree, which is D. That gives me a sequence of triads: I, II, III, IV, V, VI, VII. Now, those are the triads in their most basic, fundamental, undiluted form. Totally simple presentation.
- 4.29 But of course, you can also skip between one and the other. You can move from scale degree I to IV, and IV to V or I to II and then II to VI, and so on. You can have any sequence of triads that you like. What we have here, what I've just played at the keyboard, are a sequence of root position triads, where each successive chord is built on a different scale degree, but always with a root, a third, and a fifth in every chord. These are known as root position triads— root position because the lowest note of the chord is placed on the bottom; there's no change in order of the tones.
- 4.30 Now we come to inversions. I've covered this slightly before in the composition program, but not everybody's going to have seen both courses. Just a brief recapitulation: if I play a triad in root position—that means I have the first, the third, and the fifth, or the tonic, the third and the fifth. If I swing the order around where I put the third on the bottom and then the fifth on top, and the tonic, or the root, on top, then I obtain this [plays E-G-C]. That's called a first inversion, triad. It's written like this: E, G, and C.
- 4.31 So this is the root position. This is the first inversion. Then, of course, there's also the second inversion, which is where the fifth is the bottom note. That gives me this: root position [plays C-E-G], first inversion [plays E-G-C], second inversion [plays G-C-E]. And this gives me many more possibilities. I'm not obligated to put the root every time in the bottom position. Now, here, what I'm doing, this little sequence I'm playing, I'm playing a sequence where sometimes the root is in the bass and sometimes the third is in the bass.
- 4.32 That allows me to have a much smoother and more interesting bass line than if I were obligated to always put the root of the chord in the bass every time. So I can use inversions to put a note other than the root in the bass. Here I've got the third of the chord in the bass [plays a chord with the third in the bass]. And now I've got the root

[plays a chord with the root in the bass]. Now the third [plays a chord with the third in the bass], and so on and so forth. So that opens up the musical possibilities considerably.

- 4.33** One of the things, again, that you do when you're coming to terms with music and the language and grammar, if you like, of music is developing the capacity to recognize instantly these different inversions. So if I play a first inversion triad, you can hear instantly, "Oh, yes, that's not a root position triad. It's in first inversion because I don't have an F as the bottom note." That's one thing that you can do with a triad. You can permute the order of the notes or rotate them rather so that the bottom note goes on top and you keep going.
- 4.34** The other thing, of course, that you can do is you can alter the quality of the chord by using accidentals to change the intervals. We're going to stick with a C major triad for now. I'm going to take the third and lower it using a flat symbol. So now we have a minor third between the root and the third of the chord. You want to get in the habit of listening very closely, not to this chord as just a single thing, but as an entity constituted of three different tones, and you want to hear each tone. You can do that by singing the tones.
- 4.35** The way to remember and to incorporate all of these different concepts effectively is to externalize what you're doing. You have to sing; you have to use your voice. Even if your voice is hideous, which right now is the case for me. So this gives us different qualities of triads: major, minor, augmented, and diminished.
- 4.36** To make an augmented triad, start with a C major triad once again, and I raise the fifth to a G#. That gives me this [sings C-E-G#]. It's hard to sing; augmented intervals are difficult to sing. Hard to sing in tune as well. All of these have C as their root, but there are different qualities of chords. Now, I can also make a diminished triad by lowering not only the third, but also the fifth. That gives me C, Eb, and Gb. That gives me a diminished triad.
- 4.37** So now we've covered some of the basic ways that you can alter or that you can present a triad. So here's our major triad in root position [plays C-E-G], in first inversion [plays E-G-C], and in second inversion [plays G-C-E]. Then we've got a major triad [plays C-E-G], minor triad [plays C-Eb-G], diminished triad [plays C-Eb-Gb], augmented triad [plays C-E-G#]. This is where we start getting into a combinatorial explosion, because if you consider that each of these different chord qualities can also be inverted, and of course, you can play this in any key on the piano, on any scale degree, it becomes truly vertiginous.
- 4.38** A minor triad can also be inverted. I can put any note in the bass. I can put the root in the bass, the third, the fifth, and it's the same with the diminished triad. And of course, the augmented triad. So starting from just a simple chord, suddenly we have tens of thousands of possibilities, perhaps. So it goes very, very quickly. The difficult thing here,

again, is to develop the capacity to recognize chord qualities instantly. So that if I do this or this, or this, or, or [plays various triads], you can say without hesitation, "You just played a diminished triad, an augmented triad, a minor triad, a major triad," and so on. That's without getting into questions of inversions.

- 4.39** If all of that sounds dizzying, you can take some relief in the fact that there are methods to work on this that are actually extremely effective. We're going to talk about that a little bit tomorrow, about some of the ways that you can train your ear to recognize these intervals. Of course, sitting down and playing for hours a day helps as well. It just helps to familiarize yourself with these concepts.
- 4.40** We've looked at the different ways that you can construct three-note chords. There are many, many other possibilities, of course. There are possibilities that aren't triadic. You can have three-note chords that aren't made up of stacked thirds. All of the chords that we've looked at— major, minor, diminished, augmented—are built around two thirds stacked on top of each other.
- 4.41** But you can have other qualities of chords as well. You can have chords that are built on fourths or chords that are based on seconds. Those sorts of sounds are very common in jazz and many other expressive contexts. They're not triadic in nature, but they are three-note chords or four-note chords.
- 4.42** Another thing you can do is you can continue to add thirds to your chord. So remember that the C major chord is built on the root, the third, and the fifth. I can also add the seventh to the chord, and that gives me, logically enough, a seventh chord [plays C-E-G-B]. And you can keep going. You can add a ninth, an 11th, 13th, and so on, and end up with chords like this [plays extended chords].
- 4.43** Now, if I am on C, if I'm in C major, and I move towards the fifth degree of C major, which is G, now we're getting into another concept that we need to understand, which is the harmonic functions. We've looked at how you can take any scale and count off the tones: I, II, III, IV, V, VI, VII. In Roman numeral notation, those are the scale degrees. We've looked at how you can give them solfège syllables: do, re, mi, fa, sol, la, si. But we haven't looked at the harmonic functions yet. So let's explain what those are and how they work.
- 4.44** I need to, yet again, write out my beloved C major scale. Now, I did teach you one term, one harmonic function already, which is the tonic, which refers to the first scale degree, the first tone in my scale. In C major, the tonic is C. Now, if I move towards the fourth scale degree, the harmonic function, the term for the fourth scale degree, is subdominant. If I move to the fifth scale degree—I'm going to start with these three for now—then I'm moving towards the dominant. What we're going to do is we're going to build a triad on the tonic, the subdominant, and the dominant or I, IV, V in C major—I, IV, V, and then back to I.

- 4.45 The term harmonic functions refers to the fact that if I just play you a random chord with no context around it and just leave it hanging there, then that chord is harmonically ambiguous. It doesn't tell me anything. It doesn't give me a sense of movement. It doesn't tell me what the tonic is. It doesn't tell me even what key we're in. I can play a C major triad, and it could be the dominant of F major. Or it could be the tonic of C major, or it could be the subdominant of some other key, and it keeps going.
- 4.46 The important thing to understand is that the most stable scale degree in any key is the tonic. The tonic is—you can think of it as being like home base. You're at home. You haven't moved anywhere. You're just sitting at home. The polar opposite of the tonic, in a sense, the scale degree towards which pieces tend to gravitate as a secondary place, is the dominant. In the case of C major, the dominant is on the fifth scale degree, which is G. That gives you a stable place and an unstable place.
- 4.47 If I just play these two chords, there's this sense that I've moved away from somewhere, and I'm not quite done. I've created an instability. The subdominant, which, as the name implies, lies directly below the dominant, gives me an intermediary degree of instability. So think of the tonic as the most stable harmonic function [plays C major chord]. The subdominant [plays F major chord] is an intermediary degree of instability. The dominant [plays G major chord] is quite unstable. And then we want to get back [plays C major chord] to where we started again.
- 4.48 The amazing thing is you can tell a very simple, archetypal story in music through sounds with those three chords. A stable place, an intermediary or transitional place, and an unstable place. That's enough to tell a story. There's this cliché—it's kind of true—that pop music is based on a three-chord structure. Now, not all pop songs literally have only three chords, but they tend to gravitate around that basic progression: I, IV, V—that's also a blues progression.
- 4.49 I think the reason for that is that the I, IV, V progression is the simplest chord progression you can write that has a sense of movement away from your starting place, and then that allows you to get back to where you started again—avery, very simple little story. What all of this has to do with are the dynamics of centripetal and centrifugal motion away from and towards your center of gravity. These forces, which we might not understand consciously—we might not be thinking about them when we're listening to a piece of music—but they're a large part of what gives it its sense of dynamism and its sense of movement.
- 4.50 Composers who have completely mastered harmonic tension, harmonic movement, can tell extraordinary stories with tones by moving away from or back towards some either permanent or provisional landing spot in the piece. That's why Beethoven can write a 25- or 30-minute-long symphony movement that has a complete sense of coherence dramaturgically, and it takes you on a journey, and you can go very, very far from the starting point and then end up where you began at the end. You can hold your

attention for 30 minutes without necessarily understanding why. A big part of it is because of that coherence on the harmonic scale.

- 4.51** A simple way of thinking of tonic, subdominant, and dominant, again, is in terms of their function within the key. So stable place, intermediary place, and a secondary place. Every one of the seven scale degrees has a different term that assigns it a function within the scale. The others are supertonic— the supertonic is the scale degree immediately above the tonic; that's easy to remember. The third is the mediant —mediant because it's halfway between the tonic and the dominant. If we go to sixth scale degree or A, if we're in a C major, then we end on the submediant. Finally, B, which is immediately below the tonic, is the leading tone, which is a concept we've already looked at.
- 4.52** So then you start to see that this seven-note scale, which seems so simple, is not actually that simple. Every single tone in that scale has a function, a function relative to where you started—on the tonic. So tonic [plays C], supertonic [plays D], mediant [plays E], submediant [plays F], dominant—sorry, I'll start over again. I must be getting tired. Tonic [plays C], supertonic [plays D], mediant [plays E], subdominant [plays F], dominant [plays G], submediant [plays A], leading tone [plays B], and then back to the tonic again [plays C]. Those are theoretical terms that allow us to describe the function of different scale degrees within a key. We're almost done with this.
- 4.53** I wanted to get back to what we were talking about. We looked at a very primordial, very important progression. Now, a chord progression implies that we're moving towards a goal of some kind. It implies that there's a directed motion between one place and another place. In order to have a progression, the movement has to be palpable; we have to have a sense that we're going somewhere.
- 4.54** If you string together chords without thinking in terms of a progression, in terms of a goal-oriented motion of some kind, then you end up with a succession of chords. It's not the same thing. And I, IV, V is a very simple progression, but there are many others, too. One of the things you can do when you're getting a handle on these materials—starting to understand how keys work and how you can join chords together—is to make little progressions and to play them in different keys.
- 4.55** So you can say, "Well, okay, play a I, IV, V progression in B $\flat$ major" [plays a I, IV, V progression in B $\flat$ major]. " Play it in A major" [plays I, IV, V progression in A major]. "Play it in F major" [plays I, IV, V progression in F major]. You can do that in any key, and it's the same progression every time, and the functions of those chords are the same; they're just transposed to a different key.
- 4.56** Of course, you can move between any two scale degrees. You can make any progression you want in principle. You could say, for example, we're going to go from I to VI to II to V. So I to VI to II to V—that's a very, very common pop song type of sequence [plays C major–A minor–D minor–G major chords]. So again, by creating

progressions out of sequences of triads, you can create phrases that have musical coherence and harmonic coherence. Those are just two simple examples.

- 4.57 What you do, again, when you're starting out, you play those sequences in every key. You learn to do them, learn to drill them. It becomes second nature. The advantage of that is it sharpens your ear and it allows you to link up your eyes, your ears, and your hands very rapidly. We'll stop there for now. That was a little tour of triads. It's not an exhaustive one. There's much more we could add to it, but that's probably enough for now. Are there any questions?
- 4.58 Is it possible that music, as the dramatization of deep search into our experience, has led to the birth of more sophisticated feelings or associated values?
- 4.59 Well, music is a mirror of the human psyche, so it's possible to express almost any facet of human experience through music. And what has happened, really, is that that range of expressive capability has just expanded and expanded and expanded. I don't think it's music itself that's driving that expansion; it's the desire of humans to express an ever-broader range, not only of emotions, but also of thoughts and sensations. What I find absolutely incredible about that is that music can always do it. You can express any conceivable interior state, any conceivable character through a piece of music.
- 4.60 The materials of music have sometimes needed to expand in order to encapsulate some of these sensations. You see this certainly with the Romantics. The Romantics were interested in emotional paroxysm in extremes of interior states. This led to Expressionism, which was interested only in the extremes: intense anger, intense happiness, intense horror, intense anxiety. Whether they're positive or negative emotions, Expressionists and the Romantics before them wanted to convey the limit cases—how far can you go in that direction?
- 4.61 In order to convey those extreme emotions, sometimes you need to draw upon unconventional musical resources. When we get into the Bach course, which I'm going to be presenting in a few days, we're going to see examples of this, of how Bach had to sometimes have recourse to extraordinarily sophisticated and unusual progressions and melodic contours in order to convey some of the experiences that are related, for example, in the passion stories, where sometimes a simple triad isn't enough, a simple diatonic scale isn't enough. You need to find something more intense, more stranger, in order to convey the nature of that experience.

Lecture 5: Motion in Music

At a glance — Motion in music.

- 5.1** Music Wants to Move—that's going to be the topic of our lecture today. I've spent my life now trying to work out exactly what lies at the bottom of the musical phenomenon. Different styles of music have different points of emphasis, as we've seen already. Some music is intensely rhythmic. Other pieces seem to be strangely static. Others are full of surprises and non sequiturs and contrast. Others are much more continuous. But there seems to be a constant at the bottom of this phenomenon, which has to do with motion.
- 5.2** Music provides us with an infinite array of possibilities for expressing ideas of movement. Of course, this can be literal, embodied kinetic movement. In fact, that is how perhaps a majority of people engage with the musical phenomenon. But other aspects of music move in a way that is sublimated, which is to say we're not literally moving, we're not literally dancing while we listen to it. But there seems to be, nevertheless, a very profound feeling of forward motion. That motion, of course, can take all kinds of forms. That seems to be something that's absolutely central to the development of this art form through the centuries.
- 5.3** Now, what harmony does, and we started looking a little bit at individual chords yesterday, and we looked a little bit at some simple progressions: I-IV-V, I-VI-II-V-I, and so on. We looked at the basic harmonic functions, the fact that there's a stable function, you might say, an intermediary function that allows you to move, and an unstable function, which allows you effectively to have goal-oriented motion, a meaningful sense of progression using nothing but sounds, nothing but tones on a piano, which is quite extraordinary.
- 5.4** But we haven't quite gone into detail yet about precisely how the mechanics of that work. In order to understand that, we need to talk a little bit about consonance and dissonance. Now, those are terms that you've probably come across before. For many people, dissonance has an emotional connotation of something unpleasant, something cacophonous, something that might strain the indulgence of your ear. But consonance and dissonance, as well as being aesthetic concepts to a degree, can also be defined technically in terms of the musical phenomenon. We're going to look at that.
- 5.5** Consonance doesn't mean pretty or agreeable, and dissonance doesn't mean ugly or disagreeable. Rather, consonance can be thought of in a musical context as something stable, something that doesn't need to move. And dissonance is something that creates a need for movement or possibly a need for resolution of some kind.
- 5.6** To a degree, these are contextual ideas. So the dissonance in one style might, in fact, sound relatively consonant in another style. It comes back to what I said yesterday when I was answering a question, which is that you can think of this as a dance

between the world of natural acoustic phenomena and the world of culture, which is a world of artifice. To some degree, you can explain consonance technically in terms of the simplicity of a harmonic ratio.

- 5.7** Remember, if you take a length of string, a vibrating string of some kind, stretch it between two posts that hold it in place, and then divide it into exact multiples— divide it in half, divide it in thirds, divide it in fourths, and so on—divide it into exact ratios— 1:2, 3:2, 4:3, 5:4, and so on—you obtain a series of simple harmonic intervals. The simpler the ratio, the closer the sound is to the fundamental of that string, which is to say the tone that it produces when the string is undivided, when you're hearing the full length of the string.
- 5.8** It's easier for the ear to relate those simple ratios, those simple intervals, to the fundamental. Let's try this again. I have to keep returning to this point because it's a fundamental property of acoustics, and it informs our experience of music, certainly, and also the music that we produce. If you take all of those ratios in sequence and present them one after the other, you obtain the harmonic series.
- 5.9** The harmonic series is going to start with the fundamental tone. We're going to start on a C here [plays C]. If you divide that string in half, that gives you a ratio of 2:1, and that produces the octave [plays C-C]. Frequential tones, this octave, the upper note is exactly double the frequency of the lower note. It's a very simple ratio. The only thing simpler than that is 1:1, which is a unison, two notes sounding at the same pitch. That gives us this octave [plays C-C]. All of these successive octaves stand in that very simple relationship to each other.
- 5.10** We're not in the arena of culture; we're not in the arena of artifice here. This is a natural phenomenon. Next comes the fifth [plays C-G]. Now, this is something that you can train your ear to be sensitive to because any harmonic sound, that is to say, a sound in which the various frequency components stand in a certain ratio to each other, let's say, and have a certain periodicity to their frequencies, any harmonic sound is going to be made up of some combination of these overtones.
- 5.11** But you don't experience that directly when you're listening to it. What you hear is a synthesis, and that synthesis is in the particular distribution of intensities and of dynamic curves that it contains—because some partials, some overtones will die away faster than others, some will be more intense than others—the particular way that those sounds evolve over time and their frequency distribution, their intensity, creates timbre. That's how we're able to tell the difference between instruments, between a piano and a trumpet, and so on.
- 5.12** If I take these natural harmonics [plays C-C-G-C-E ascending], especially if I play the lower partials louder than the upper partials, what you hear is an enriching of the sound, but you don't necessarily hear it as a chord [plays C-C-G-C-E ascending]. That's how they blend together and create almost a single synthetic sound.

- 5.13** So if you just take these simple ratios, then you can use that to create consonances that are very stable. But music abhors constant stability. I could just sit down and play that sound for four hours, and it would go nowhere. There'd be no story. Nothing would happen. You can try to imagine a narrative that would be like that, where you're just sitting there eating cupcakes or something, and everything's fine.
- 5.14** So we need a stable entity of some kind, whatever that is—it can take the form of a pattern, it can take the form of a sound— and we need a destabilizing force applied to that so that we can move. This is why we have these progressions that want to take us somewhere.
- 5.15** We've looked at concepts such as the circle of fifths. We've looked at how that gives us this whole network of different keys with different amounts of sharps and flats, and that gives us the capacity to move around in music and create a more interesting sense of motion. In theory, you can get extraordinarily far from your starting place and still work your way back there. This is why the symphonies of Beethoven are so long and complicated and emotionally absorbing. It's because of the sophistication, the complexity, and the daring with which he's able to get progressively farther and farther from his starting point, but without ever completely losing his way.
- 5.16** Let's return to consonance and dissonance. We've looked at how consonances are formed. We've looked at the fact that they are technically made out of stable things—simple ratios, sounds that don't particularly have a need to move.
- 5.17** What can you do to create a dissonance? Well, the dissonance is something that, again, pulls our ear. This is a contextual phenomenon to a large extent. I could sit and play very beautiful seven-note chords on the piano that may technically be dissonant chords, but if I play them in a certain way, they might not necessarily need to move all that much. It's partly to do with context, and it's partly to do with matters of style, and also what comes before and what comes after.
- 5.18** If I just sit here and play something like [plays dissonant chords]. Both of those are very dissonant chords, but by carefully manipulating rhythm, timbre, register, and dynamics, you can make something that doesn't create the impression necessarily of an unpleasant or discordant sound, but rather that creates a certain atmosphere, let's say.
- 5.19** Now, in a harmonic context, let's try to set up a situation where we have the need for a resolution. I'm going to go back to my program here, and I'm going to write a simple triad—we're going to start that again. You remember how triads are made? It's constituted of the root, the third and the fifth. In this case, we're going to be in C major. This is going to be the first scale degree. So it's a triad built on C, E, and G [plays C-E-G].
- 5.20** Now, I'm going to make a very small change to this chord. I'm going to take the G and add a sharp. That's going to look like this. Now, as you might remember from yesterday, if you take a triad, raise the fifth up a semitone, you obtain an augmented

triad [plays C-E-G#]. Now, it's possible to imagine chords that are considerably more dissonant than this one, this augmented triad. But notice the difference in quality and the difference in apparent stability [plays C-E-G, C-E-G#]. Now suddenly, we're going somewhere. That's what dissonance is.

- 5.21** Now, that G#, that can resolve in any number of ways. For example, I could take it up another semitone, and here we start to see a little melodic progression [plays C-E-G, C-E-G#, C-E-A]. That's called chromatic voice leading—chromatic voice leading because the top note of the chord is proceeding stepwise up by a semitone each time [plays C-E-G, C-E-G#, C-E-A, C-E-A#, B-D#-B, B-D#-A, B-E-G]. Now that I've done that, I've landed on a mediant chord, an E minor chord in C major. That gives me a feeling of moderate stability. It's more stable [plays B-E-G] than the augmented triad [plays C-E-G#], which really wants to go somewhere, but it's not completely stable because I'm not on the tonic; I'm not in C major.
- 5.22** Music is this continuous dance between provisional stability, provisional stasis, and an experience of foreignness, an experience of not being where we started. And there are all these subtle degrees. So if I play that progression again [plays C-E-G, C-E-G#, C-E-A, C-G-A#, B-F#-B, B-D#-A, B-E-G], I can eventually get back, but because of the nature of this little story I've been telling through sounds, I don't feel that this has been completely resolved. There's a strong possibility that I'm going to keep going and end up somewhere else, and you can just keep doing that in a piece of music.
- 5.23** So how do we know when we've arrived at a signpost, let's say? We've arrived at somewhere intermediate? Imagine that you're driving from Paris to Prague or something like this. You're going to stop at various points along the way. You might go through Switzerland, or you might go through Germany, or you might go through Italy, even. There's all kinds of routes you could take, and you might stop at different cities along the way. You haven't reached your destination, but you've reached a stopping point temporarily. The way that this is signaled in music is through cadences.
- 5.24** Cadences are a form of punctuation—we've addressed this a little bit in the composition class, but we need to get into a little bit more detail with this idea here. Cadence is a term that means a close. It relates to, among other things, the French word "cadenas," which is a padlock. It's something that closes. It's a signal to your listener that we've arrived at a pause. Just like the period at the end of a sentence or a comma or a semicolon, it's a form of punctuation, if you like. It allows us to create a continuous sense of movement, but a movement that is articulated by different arrival points.
- 5.25** Yesterday, we looked at a simple progression, which was I, IV, V, and I want to look at that again. I'm going to draw a simple bass line here. Remember those Roman numerals? I, IV, V. If I stop there, at least temporarily, if I go from I and I move towards V, towards the dominant, what that gives me is called a half cadence or an imperfect

cadence—imperfect because I've arrived somewhere else, but I haven't arrived back home. It's a temporary or provisional resting place [plays a sequence of chords].

- 5.26** That characteristic, I, IV, V, motion [plays C–F–G], or even just directly I to V [plays C major chord, then G major chord], very often signals a moderate degree of break in a phrase. As you might imagine, the opposite movement from V back to I is a perfect cadence. So, movement from I to V gives you an imperfect or half cadence; movement from V to I, or dominant to tonic, gives you a perfect cadence. Notice how the feeling of suspension is always there when you move from I to V directly [plays C major chord, then G major chord]. When you move from V to I [plays G major chord, then C major chord], you have a feeling of closure or finality [plays G major chord, then C major chord].
- 5.27** One thing that people, students, coming to grips with music theory in the early stages, have to work on is cadences. And so what you do is you pick a random key—it could be an A major or a B $\flat$ or A $\flat$ or a D or whatever, and you practice playing cadences. This process of drilling them, first of all, helps your ear to become accustomed to the function of these progressions, and it also helps you to attain a certain degree of fluidity.
- 5.28** Now, there are other types of cadences. In fact, there's many of them. Cadences change throughout the centuries. If you listen to medieval music, Renaissance music, you're going to hear a variety of cadences that sound very little like these. But that, nevertheless, within that context, within that cultural and stylistic context, create a palpable feeling of closure.
- 5.29** One of the other effects you can produce with a cadence is called the deceptive cadence. This is a cadence that doesn't do what you're expecting, but takes you somewhere else. The deceptive cadence, or interrupted cadence, as it's sometimes called, ends with a motion from V to VI. Remember, we're in C major here. That means moving from G to A in the bass, from the dominant to the submediant [plays G major chord, then A minor chord].
- 5.30** Now, what you might be expecting to hear if you're coming to the end of a phrase is this [plays G major chord, then C major chord]. And what the deceptive cadence does is it gives you a little trick [plays G major chord, then A minor chord]; we haven't ended up where we're expecting to go.
- 5.31** That allows you to do a couple of things. One is that you could just keep going from there. Now you're in the relative minor, you're on A minor, and you could say, "Well, that's the new key for a while. I didn't go where you expected me to go; I'm going on this supplemental journey." Or you can use the deceptive or interrupted cadence as a way of prolonging the eventual return back to the tonic again [concludes the phrase in C major]. So it can form part of a larger phrase unit that gives us a little bit of a more elaborate formula.

- 5.32** One of the things to be careful about in music, generally, is avoiding things that are too obvious or that happen too many times with a minimum of variety, a minimum of change. Just as how you would be justifiably exasperated if a movie you were attending did exactly everything you were expecting with absolutely no surprises and always took the shortest route to the exit, as it were. It's the same in music— we have a variety of different effects that allow us to create feelings of stability, instability, and movement between things. That's the deceptive cadence, which is a move from the dominant to the submediant.
- 5.33** One other type of cadence we need to look at is the plagal cadence: P-L-A-G-A-L. The plagal cadence is associated with early music, which is to say, medieval and Renaissance music. What characterizes it is a sense of close, but a somewhat weaker sense of close than V to I. Instead, the plagal cadence moves from IV to I, so the subdominant to the tonic. That looks like this [plays F major chord, then C major chord]. All of these different cadences have different functions that orient our ear differently and create different expectations and different degrees of release or tension.
- 5.34** So we've been looking at how you can join these chords together, how you can make simple progressions, how you can signal the end of a phrase or a pause in a phrase through cadences. Now we come to a concept called voice leading.
- 5.35** Voice leading is a very important concept to understand because you have to remember that European art music for centuries was based exclusively around the voice, which is to say that all music was conceived in some manner or another in terms of an actual voice. Even when pieces were played on instruments, they were generally adaptations of vocal music.
- 5.36** So if you wanted to stack two or more sounds together, you would be thinking in terms of individual voices, where every tone in the chord is one component of a polyphonic texture. It's one voice that has its own trajectory independent from the others, which gives you a fusion, if you like, of the horizontal and the vertical—the horizontal being the melodic dimension and the vertical being the harmonic dimension. Horizontal, one sound after another; vertical, two or more sounds heard at the same time.
- 5.37** Why this is important is that it's easy to just sit at the keyboard and play block chords like this [plays a sequence of chords]. You can chain chords together like that without there necessarily being any particular notion of a voice. What you're hearing there are block sonorities that you can shove together, like blocks of granite, effectively. There's not a sense of organic connection between them where every note is leading to some other note, but it can actually be quite disjunctive, like this [plays a sequence of disjunctive chords].
- 5.38** It's not that the chords themselves or the harmonic functions are necessarily completely disjunctive. It's that the gap between one tone in one chord and another

tone in another chord is so great, so disjunctive, that you can't relate them with your ear as belonging to the same voice. So we're going to look at some basic principles of voice leading. What voice leading aims to do is to take the principles of combining voices in vocal polyphony and apply them not only to voices but also to instrumental situations. That has a further benefit, which is an ergonomic benefit.

- 5.39** If you look at old harmonic treatises—treatises on thorough bass, for example, from the 17th century, it's quite interesting. There's a shift that goes on in the world of music theory—particularly in Germany, but not only— where at that time, if you wanted to learn how to harmonize a bass line, if you wanted to learn how to join chords together and make smooth progressions, a lot of it was actually predicated on ergonomics, on shapes that fit the hand, the natural shape of the hand. It had an abstract dimension, but it also had to do, very pragmatically, with limiting lateral hand motion on a keyboard so that you're not having to do this all the time.
- 5.40** One of the basic principles of classical voice leading—in other words, how do you move from one tone to another in a set of different voices so as to obtain a smooth result—one of the basic principles is you move as little as you need to. So if there's a tone in common between two chords, then you don't move it; you don't move it to another octave. You keep it where it is, and the other fingers move as little as possible.
- 5.41** Let me give you an example. I'm going to write out a very, very simple little progression here, just two chords. For the sake of clarity, I'm going to also point out that vocal polyphony tends to be articulated in terms of four voices, the four basic human voice types, which are soprano, alto, tenor, and bass. You'll often see that abbreviated as S, A, T, B: soprano, alto, tenor, and bass. You'll see references to S-A-T-B choirs—that means two of each voice type. Generally, soprano and alto are female voice types, and tenor and bass, or sometimes tenor and baritone, are male voice types. That's called a mixed choir [plays C-F].
- 5.42** That's my bass voice. The soprano, or the highest voice, is going to stay put—it's going to play a C, and it's going to stay on C. That's going to sound like this [plays C-C, F-C]. That's not a complete harmony; we need to fill in the gaps a little bit. My alto voice is going to be on the G, just below the C of the soprano [plays G-C]. It's going to move up a whole step to A. My tenor voice is going to move from E to F. That's going to look like this [plays E-G-C, F-A-C, E-C, F-C]
- 5.43** Now, you can start to sensitize your ears so that you can hear that not in terms of one big chord, followed by another big chord, but rather in terms of voices. The tenor is doing this [plays E, F]. That's pretty smooth. The bass is doing this [plays C, F]. The alto is doing this [plays G, A]. The soprano, which has a common tone between I and IV, the C [plays C, C], isn't moving.
- 5.44** A couple of things to observe there. The first is that we have a four-voice texture: S, A, T, and B. There's our S, there's our alto, tenor, and bass. But triads only contain three

notes. That means you have to double one of the tones. The doubling can take the form either of a unison, where you have two voices singing the same tone in the same octave at the same time, or you can take one of the three notes of the chord and have it present in another voice in a different octave.

- 5.45** In the case of the first chord here, which is [plays C–E–G–C], the bass and the soprano are both singing a C [plays C–C]. We consider that to be a doubled tone. In order to produce the most stable and the simplest result, the order of priority is you double the root or the fifth, and generally speaking, you're not meant to double the third. Sometimes in exceptional circumstances, it will happen, but it's the absolute last priority. And the reason is that the third is harder to tune, and it's a less stable interval than the root and the fifth.
- 5.46** Therefore, I've decided to double the root in this case. In both cases, in fact, because the bass and tenor voices in the second chord are both going to sing F [plays C–E–G–C and F–F–A–C]. And notice how these inner voices, the alto and tenor, just move a tiny bit [plays E–G, F–A]. And the soprano on top [plays E–G–C, F–A–C].
- 5.47** That's exceptionally easy to play because your hand hardly has to move at all, which makes it both ergonomically comfortable and easier on the ear because it's very easy for the ear to hear the individuality of each voice. Whereas if I were to have all sorts of large leaps in there and disjunctive movement and dissonant melodic intervals all over the place, then it would be much more difficult, both for a singer—you can do this experiment yourself. It's very difficult to sing large intervals. It's difficult to sing dissonant intervals, and it's difficult to have multiple skips or leaps in a row.
- 5.48** If you can keep the voice leading smooth, it's easier to sing, it's easier for the ear to relate the chords together, and it's also easier ergonomically. It's quite interesting to me that those two things are connected, that what's easy to sing also ends up being what's easy to play on a keyboard, even though, mechanically speaking, there's nothing in common between the two.
- 5.49** So well-written vocal music tends to translate extremely well to keyboard music, interestingly enough. The ranges are not that dissimilar. If you're dealing with the four voice types, you've got a range of maybe three octaves, something like that, maybe three and a half octaves, which is a very comfortable range to play on the keyboard as well. Those are some basics of voice leading.
- 5.50** Generally speaking, the ear is drawn to the outer voices, primarily. This is the same reason why when you listen to a song, you are going to be primarily focused on the voice. The voice is usually, although not always, front and center in the texture. If it's a high voice, a soprano, then it's going to sit on top, and you're going to hear it without any difficulty. Your ear tends to be drawn to the highest note in a sonority. That's where you're going to be focusing your attention. That tends to be where the melodic interest is.

- 5.51** Likewise, you're also going to be drawn to the bass because, in tonal or classical harmony, the bass determines the function of the tones that lie above it. The bass gives you a strong sense of movement, and everything that happens above it is related by the ear, acoustically, to whatever is happening in the bass.
- 5.52** Now, the inner voices will tend to attract somewhat less attention. They matter, obviously, for all kinds of reasons, but it's not where your ear is going to be focusing the most necessarily. In actual practice, in actual compositions, what you'll find is that as a very skilled composer, we'll never neglect the inner voices. They'll always have nice lines for them to play— that's regardless of if it's a keyboard piece, a vocal piece, or something else. But you want to attend to all the details in a texture; they're all important. A skilled composer, again, will make sure that the inner voices are not boring. They're not just moving in circles doing nothing, but there will be an interesting, distinctive line in every single part.
- 5.53** So we've talked about the fact that you need to double a tone in four-part vocal writing. If you're writing triads, primarily, then you have to double a tone in order to complete the four-voice texture. We've looked at some basic principles of voice leading, about how you aim to have common tones remain in one voice, if possible, to keep things smooth, and the inner voices should not move more than is necessary.
- 5.54** There's a couple of other principles we have to look at. The first is that, in general, you want the voices to remain stratified in their own space, so to speak, so that the soprano doesn't impinge upon the alto line, so that we can always hear the distinctness of each individual line. The voice leading principle there is avoiding the crossing of voices, so you don't have the soprano dipping below the alto, and you don't have the alto singing higher than the soprano.
- 5.55** Of course, you could do that technically. There's a certain amount of overlap between all the vocal ranges, which makes it possible. Again, with all of these things, it comes back to what I said in the first lecture on the first day: you have to look at these as descriptive rather than prescriptive. Anything I'm saying here, you can easily find countless examples of composers not doing what I'm saying to do. You can think of this again as a normative practice that is based around ideal voice leading, let's say, but expression isn't so straightforward.
- 5.56** There will be instances where for technical reasons, for expressive reasons, the composer may decide to go against those principles. Notice I don't say rules, but rather principles. And provided that there's a reason for it, an expressive reason, and it works musically, then you accept it, but you have to be doing it consciously and have an understanding of what the normative, let's say, process would be.
- 5.57** So we avoid crossing voices in order to make it so that the distinctness, the individuality of each voice, is not impinged upon. You will find examples of voice crossings, sometimes as a special effect, sometimes because, for one reason or another, maybe

for reasons of range, or maybe because the composer wants to preserve a certain melodic contour in the tenor or in the alto, where you may occasionally see that, but it's a general principle to avoid doing that.

- 5.58** For example, in this instance, I'm going to color-code the voices so you can see what I'm doing a little bit more clearly. You can imagine the soprano doing this [plays C, G (below), A]. Then we're going to write an alto voice, and the alto is going to be written in red. If I just played on a piano where it's a matter of hammers striking strings, obviously, the effect is a little bit harder to hear. You would have to hear singers attempting to do this.
- 5.59** I've got my alto moving up [plays G, B, C] and my soprano doing this [plays C, G (below), A]. That gives me [plays G-C, B-G, C-A]. You would have to look at the diagram on the board to see that, in fact, the alto is moving above the soprano and the soprano is moving below the alto. In reality, there would be almost no reason you would want to do that in the first place because it's quite simple for the soprano simply to go C, B, C, like this [plays C, B, C]. But nevertheless, it's a mistake to be avoided in voice-leading exercises. One of the basic principles of voice leading in terms of texture is to preserve the autonomy, the independence of the voices. This is why we don't cross the voices; we try to keep them relatively smooth most of the time.
- 5.60** There are another couple of rules to observe. One is that you avoid having an interval of more than an octave between any voice pairs. You won't see a gap of more than an octave between the note that the soprano is singing and the note that the alto is singing, with one exception, which is that you can have a gap larger than an octave between the tenor and the bass.
- 5.61** All of these principles, all of these practices, again, it's not useful to think of them in terms of rules that have been dreamt up by a pedagogue or by a theorist, but rather as principles that serve a particular function. The reason you can have a larger gap between the tenor and the bass is so that you can preserve the independence of the bass line and have very wide-ranging basses that are interesting to hear. Also, simply because if you try to do that in the inner voices, you're very likely to end up with voice crossings.
- 5.62** So generally speaking—and this also relates back to an acoustic principle. Remember that when we looked at the harmonic series, the intervals between successive partials or overtones get smaller as you go up. As the ratios get more complex, as you get farther from the fundamental, the intervals get smaller.
- 5.63** That seems to be something like a natural acoustic law, which is that the voices on the lower end of the spectrum, in the bass, tend to sound better if they're more widely spaced, and the voices on top tend to sound better if they are more closely spaced. So you'll see a larger gap between the bass and the tenor. As you get into the middle

voices, the voices get closer together. It seems to be easier for the ear to assimilate and to fuse those different voices together with that texture.

- 5.64** So we've looked at principles of voice spacing. You can see here, for example, that I've got a distance of a perfect fourth between my soprano and alto [plays G-C]. There's a third, minor third between tenor and alto [plays E-G]. And an interval of a 10th between bass and tenor [plays C-E]. So we have both harmonic intervals here, or vertical intervals, and melodic intervals when I move from one sound to another.
- 5.65** Now, another principle that serves also to preserve the independence of voices is the avoidance of forbidden parallels. Forbidden parallels have a very lengthy history. Here again, we come to issues of artifice. There's no natural or acoustic reason why you wouldn't have certain types of parallel motion between a voice pair in a four-voice texture.
- 5.66** As you move through the different historical eras and listen to music of different centuries, you'll find that the tolerance for or avoidance of certain parallels changes over time. There were certain parallels that were considered to be very dissonant and unpleasant in the medieval era that are completely commonplace today. But in their most fundamental form, students beginning to learn about voice leading and joining chords together, learn to avoid what are called parallel fifths and parallel octaves, primarily. That means that if two voices are moving together, then you don't have two successive intervals of a fifth between the voices.
- 5.67** Now I've got this [plays G-D, A-E]. It doesn't sound horrible, but in a continuous four-voice texture, what that does acoustically is if you have these parallel consonant intervals like that, like an octave or a fifth, what happens is the voices start to lose their independence. They start to fuse together acoustically in a way that disrupts the fluidity of the texture. It's something that you might not even necessarily notice right away, but it does have the effect of subtly damaging that texture somehow, impinging upon its fluidity and the independence of the voices. It's not that they sound bad, exactly, but it's not ideal.
- 5.68** When you're learning to do these kinds of exercises, you're taught that consecutive fifths in two different voices are to be avoided. And likewise, consecutive octaves are to be avoided. That would look like this [plays D-D, E-E]. So no consecutive fifths and no consecutive octaves.
- 5.69** Once again, if you really go looking, you can find subtly hidden examples of that in the works of classical composers. Beethoven, for example, was quoted as once having said that there's absolutely no reason to avoid consecutive fifths. They don't sound bad. The justification is that it sounds bad, but in certain contexts, it's very difficult to even hear.
- 5.70** So you always have to look, with these things, at: What is the purpose of this principle? Why does it exist? Then you can see for yourself, you can train your ear to recognize that phenomenon and decide if it indeed sounds disturbing or if it somehow does not

create the effect you want. So always try to understand the reason behind the principle, and then if you need to break it, you'll know why you're doing it, and you'll be aware of it.

- 5.71** So those are a few basic principles of voice leading. We've looked at the need to keep it smooth as much as possible unless you're aiming for a special effect. We've looked at the need to be very cautious about the spacing of voices, avoid having the voices cross, and then we've also looked at the avoidance of forbidden parallels.
- 5.72** That's just the beginning because we haven't really talked about seventh chords and the resolutions of dissonances and all the different ways you can join various different chords together. We haven't looked at chromatic voice leading. It's an extremely vast subject, but I think this is at least a fairly concise introduction to the subject.
- 5.73** In practical terms, what that means is you can try this out as an experiment—sit down at a piano keyboard and just move one finger at a time. Try to keep it smooth and see if you can create little progressions by moving from I to VI or I to IV to II to V, and so on. And what beginner students will do is they'll take a little bass line like that [C, E (below), F, G, C], and they'll have to harmonize it in four voices, or rather add the missing three voices: the tenor, alto, and soprano above that bass line, bearing in mind all of the different voice leading principles we've looked at previously.
- 5.74** So that's a pretty big topic. We're going to leave it there for now in terms of voice leading, and I'm happy to take any questions.
- 5.75** As someone trains himself to become more sensitive to nuances in music so that he can understand it better, does this heightened sensitivity also translate into his persona in usual life?
- 5.76** That's where art is extraordinarily interesting, because what happens is you develop your senses to a very, very high degree of sensitivity. You become hyper-aware of phenomena such as the joining of two chords together. Then that does spill out into everyday life as well.
- 5.77** There is a Canadian composer called Murray Schafer, S-C-H-A-F-E-R, for example, who was very active in the 1970s and 80s in Canada, who pioneered the concept of the soundscape, which was developing an intense awareness of all the sounds that are going on around you at all times and how that affects you psychologically and how that affects you also physiologically and thinking about what our actual sonic environment is like. To a large degree, that also comes from people like John Cage. But to a large degree, that's a result of this sensitivity brought to bear upon the acoustic phenomenon. You start to notice things.
- 5.78** It completely conditions you. I think it's so useful, it's so enriching, it's so fascinating to concentrate on these phenomena. See what your senses are able to distinguish. Find out what the threshold, the perceptual threshold is. What can you hear? How far can

you push that? Then see what happens to the rest of your life as well. You become much more attentive.

Lecture 6: Structure in Music

At a glance — Structure in music.

- 6.1 We've been looking at the way we can create a palpable sense of movement using the progression of tones, voices, voice leading, harmony—the ways that this creates either a sense of stasis or movement. We talked about the technical definition of dissonance, both its acoustic reality and also its more aesthetic or cultural dimension.
- 6.2 What I want to do in this lecture is talk now a little bit more about other ways of creating a sense of movement in music. What we're going to do here is to start with the very small and zoom out to the very large. We're going to look initially at questions of duration, and then we're going to look at musical forms.
- 6.3 In one of my earlier lectures in this series, we looked at rhythm. We looked at the rhythmic values, how they're notated, how we start with a large unit and divide it by halves successively. We also looked at different ways of putting rhythms together, how you can use time signatures to create a sense of pulse and a pattern against which a performer or a composer can push and create interesting rhythmic sensations.
- 6.4 But we also have to look at duration as a phenomenon in a little bit more detail. For this, we're going to look at both pulsed and unpulsed rhythms, what that means technically, and we're also going to look at how you can use that simplest of devices, which every musician has, a metronome, in order to practice different rhythmic patterns and also to understand the way subdivisions work. We're going to start with the very small, and then we're going to look at musical forms, which is the largest scale on which we can imagine phenomena of movement and of change.
- 6.5 This is also a very important thing to come to grips with if you're studying music theory because you need to understand formal articulation in a piece of music. When you understand that, when you understand that forms in music aren't just there in order to give you some abstract, idealized pattern that you have to follow, but you can think of them as an eternal push and pull, an eternal subtle balance between repetition, transformation, and contrast, all of which are necessary in order to, again, create a sense of movement in a piece of music.
- 6.6 Let's start with duration. I want to start by talking about the distinction between pulsed sounds, sounds that can be related to an underlying pulse grid— if you're working on a digital audio workstation, then you'll be accustomed to this concept because those platforms, whether it's Pro Tools or Cubase or whatever you're using, the rhythms are quantized, which means they're expressed in terms of milliseconds.
- 6.7 What you can do, of course, with those programs, which are pretty incredible, is you can make it so that the pulses in a rhythm are exactly equidistant to each other in terms of milliseconds. You can have this absolute precision. You can have a pulse grid,

and if your drummer is slightly out of time, then some producers will go in, cut every little individual beat apart from the others and make sure they line up with the grid, which I don't recommend you do because it really kills the rhythm of the music, but that's a technological possibility we have now.

- 6.8** You can think of pulse as being something regular and iterative. You can think of unpulsed sounds, the sounds that we do not hear in relationship to a grid. Now, this brings us to a perceptual limit case, which is that we're able as humans to distinguish between values that are short but that are slightly different from each other. As the pulses get longer, as the ratios get more complex—and this relates to the acoustic phenomenon we were talking about earlier—it gets almost impossible, and in some cases, literally impossible, to distinguish between two long, but very similar, but not identical durations.
- 6.9** Let me demonstrate what I mean. If I play you a pulse like this [snaps a short rhythmic pulse], that gives me a simple periodicity. Imagine I play or snap out a pulse like that, and then the second pulse is a little bit longer than the first. I'm going to do this [snaps a longer pulse]. Now, anyone can easily tell that the second pulse is longer.
- 6.10** You can try this as an experiment and see what the threshold is. If I tap out a sound that's one second long and then one that's two seconds long, that's pretty easy. If I do one that's one second long and then another one that's three seconds long, even easier. You can try these different ratios. How about 2:3? Can you still tell? As you get into longer and more complex ratios, let's say 9:10 [sustains note for 9 seconds]—approximately 9 seconds; it's probably a bit fast. In 10 seconds [sustains note for 10 seconds], it's very difficult to tell the difference.
- 6.11** That's when we're no longer dealing with rhythm, at that point. We're dealing with durations. We don't hear them in relationship to any audible pattern or ratio to each other. It's too complex. Duration can take many different forms in written music. There are some notations that you'll see where the duration is not explicitly given. We're going to look at a few of those and see what they look like.
- 6.12** If we stick with the conventional rhythmic values that we've looked at already: a quarter note, a half note, a whole note, and so on. Then, once again, we're in the realm of rational durations, but that doesn't tell me what the actual speed is or what the actual length of time that I have to spend on each note is. In order to know that, I need to have a tempo.
- 6.13** A tempo is expressed in terms of the number of beats per minute or BPM. A tempo marking looks like this. Here at the top of this staff, you see the notation quarter note or crotchet equals 60. That means that in one minute, I'm going to have 60 pulses. In other words, each pulse is going to last for one second. That's a very useful tool for a musician because it allows you to do, among other things, a quick calculation of how long a piece is going to be. You look at the number of beats per minute, you count the

number of beats, and you can estimate that this piece is going to be approximately two or approximately three minutes long.

- 6.14 The other thing is it gives performers and musicians an exact reference point so that the composer can say, "I want this piece to be played at this metronomic speed: 60 beats per minute." Any performer who plays the score can pick up a metronome and find that speed without any difficulty.
- 6.15 The other advantage of it is it allows us to think in terms of what is actually possible to play at a certain speed on a given instrument. Because I can turn on a metronome, I can check for myself if a given figuration or a given rhythmic value is far too fast to be played. That's also a very important thing for composers or anybody dealing with a performance situation to understand is you want to know what the pulse speed is, and then from there, you can work out what is the range of figures that are possible in that tempo.
- 6.16 If my tempo is quarter equals 60, which is approximately like this [demonstrates tempo at quarter note equals 60 bpm], then I can subdivide that pulse in all kinds of different ways. I can go [demonstrates subdivision into eighth notes at 60 bpm] in eighth notes. I can divide it into 16th notes [demonstrates subdivision into 16th notes at 60 bpm]. And I can divide it into irrational values like quintuplets [demonstrates subdivision into quintuplets at 60 bpm] in equal groups of five in each pulse. I can also keep going. You can divide it into thirty-second notes [demonstrates subdivision into thirty-second notes at 60 bpm].
- 6.17 There does come eventually a point where you reach a practical limit in terms of the figuration speed that's possible. Musicians have to understand what the limits are. You look at the piece of music and you see, "Okay, this is the smallest pulse that I have, and I have to be able to execute it in a tempo that is consistent."
- 6.18 That can also help you to figure out how to play a piece of music so that the pulse speed or the tempo isn't always speeding up and slowing down, unless that's a particular effect that the composer is asking for. Generally speaking, you want those things to be constant so that there's an audible relationship between the underlying pulse speed and the figurations, the different subdivisions that you hear rhythmically within that.
- 6.19 But not every rhythmic value, as we saw yesterday, is a rational rhythmic value. We've looked at tuplets or triplets, quintuplets, septuplets, and so on. But there are other forms of notation that you're going to encounter that break with this regularity. One of them is the whole category of ornaments.
- 6.20 We don't just have structured, rational rhythmic values in pieces. An easy demonstration of this is that there are a lot of pop singers who are constantly embellishing the lines that they're singing. A good example of that would be the singer Mariah Carey, who can never just sing one line where there's one note connected to

another, but there always has to be a florid ornamentation in the line at all times. That's a very particular vocal technique. It's extremely old. It's more present in some than others, but it's called ornamentation.

- 6.21 The ornamentation breaks us out of the regularity of the pulse grid. The way to notate that is, among other things, with what are called grace notes. Grace notes are notes that are not assigned an exact rhythmic value or a proportional rhythmic value, but rather sit outside of that framework. We're going to look at what that looks like.
- 6.22 A grace note is a smaller note head like this, with the stem going up and a little slash through it—you can see that notation there. The grace note typically precedes the main beat; it's played before it. Often, grace notes are connected to the main note by a little slur or a little tie like that. The grace note doesn't have a definite rhythmic value. It's not counted towards the 4/4. Even though this grace note is, at least on paper, it looks like an eighth note with a slash through, it is not counted as an eighth note.
- 6.23 You could have a figure like this, for example. Now I've got a very simple rhythm in 4/4, which, if I were to subtract the grace notes, I would have this [plays four equal quarter notes], just a very simple succession of four equal pulses. That gives me a periodicity—these are pulsed as opposed to unpulsed sounds.
- 6.24 Now, I can make that subtly ornamented, let's say—this is a very basic form of ornamentation—by having a grace note precede each pulse [plays four equal quarter notes with preceding grace notes]. That little ornamentation adds a supplementary degree of rhythmic interest and also melodic interest, since it consists of two different tones, the B and the C.
- 6.25 Depending on the style of music and the tempo, the grace note might be articulated; it might be played in very different ways. If you're in a slow tempo, it might sound like this [plays slow ornamental grace notes]. It might even be slower than that. What you're doing there is you're taking a main note and embellishing it. If the tempo is very fast, then you might have something where you almost can't quite distinguish the two notes [plays fast ornamental grace notes], where they almost sound like they're being played simultaneously.
- 6.26 Grace notes and ornaments, generally, have a very, very broad range of applications in virtually every musical style you can think of. But you can go farther than that. Now, I used the term melisma earlier to describe vocal flourishes. So that if you don't want to sing something straightforward and banal like this [plays C, D, E, D, C], you might want to ornament it using runs of grace notes that might look something like this [plays ornamented C, D, E, D, C], for example.
- 6.27 That's not a perfect demonstration of what I'm talking about, but it at least gives you the idea that there is a distinction between what you might call a structural level of a melody, which is just the basic tones, the basic movement, if you like, the basic

direction of the line, and then the capacity to add little ornamental figures that make the connection between tones a little bit more fluid.

- 6.28** The way this might be notated, you could have groups of grace notes connected by a beam with a little slash, and that would sound like this. So first, without the ornamentation [plays C, E, E, C]. And now with [plays ornamented C, E, E, C]. One more time [plays ornamented C, E, E, C].
- 6.29** Now we're again in a phenomenon where we're not thinking in terms of pulse exactly, but rather of something much more fluid. We still know that there's a pulse there, but we're able to break the regularity of that pulse using grace notes, using ornamentation. If the music gets extraordinarily embellished with these grace notes, you can end up in a situation where the pulse essentially disappears. You don't really hear it anymore. You can end up with something that sounds improvisatory and completely free [plays highly ornamented passage] and so on and so forth. In other words, you can get very, very, very far from any regularity of pulse.
- 6.30** This concept of push and pull, of polarities, and of a balance between two opposing forces is absolutely central to music. We looked at, in harmony, the fact that you can have consonances and dissonances, and that you need both. You can't have a piece that consists of nothing but one perfect consonance after another—you could, of course, but it's not going to necessarily be very interesting.
- 6.31** Likewise, you might not want to have a piece that consists of completely homogenous dissonances without any flow, without any regulation of the degree of dissonance. That would result in a very gray sound where we have this impression of constant agitation, but no particular sense of direction. Likewise, in rhythm, there's a continuous push and pull between the regularity of pulse and either ornamentation or local disruptions of the regularity of the pattern.
- 6.32** My sense, having studied music my entire life, is that these polarities are fundamental. They're fundamental to our experience of music, our perception of music, and it can take a very broad range of different forms. To a large degree, again, they're contextual. The composer will make it clear, if they're skilled, within seconds, what we're dealing with. Are we dealing with something pulsed? Are we dealing with something unpulsed? Are we dealing with music with a lot of dissonance? Are we dealing with relatively consonant music?
- 6.33** Once we have a context, an expressive and a musical context that's established, then you can start to play with that. You can start to move around within that space, and that creates a feeling of a trajectory. What's amazing is that it operates on every level simultaneously—it operates on the level of rhythm, on the level of texture, on the level of harmony, melody, and so on.
- 6.34** So we've looked at the world of the very small, of these little pulses and rapid melismas, rapid ornamentations, which there are many, many kinds. If you were to

open a Baroque manual on keyboard playing, for example, you would often find page after page of detailed descriptions of ornamental figures. That's a whole science in and of itself. If you play early music, then you have to learn the whole repertoire of these different ornamental figures, and how to apply them, how to improvise them when you're executing a piece of music.

- 6.35** Let's zoom out now to the largest scale. Form also gives us a notion of periodicity and a notion of ratios, let's say, between different durations, but it's on a much, much larger scale. It's on a scale that you don't apprehend consciously, but that nevertheless articulates and differentiates your experience of a piece of music.
- 6.36** I've already laid out in the composition course some of the basic categories of musical forms and how they operate. But I attempted to present those in a much broader manner so that a creative person can think in terms of, let's say, what do you want the trajectory of a piece of music to look like? What do you want the sensation of that trajectory to be like? Do you want it to be bumpy and unpredictable? Do you want it to be smooth and continuous? Do you want it to be highly dynamic? Do you want it to be much more meditative in nature? And so on. That's a more general or a more applied way of thinking in terms of musical forms.
- 6.37** But let's get specific about what those forms are in Classical, Baroque, and Romantic music and how exactly they work. Remember how we talked about binary and ternary rhythms? Those basic divisions exist in musical forms as well, on an obviously much, much larger scale. Perhaps the simplest directional musical form is binary form, that is to say, an AB form. We're going to look at what that consists of exactly.
- 6.38** Binary form, again, is the absolute simplest way that you can create this sense of a push and pull between the same and the different, between something continuing and something changing. Stasis and contrast. So ABA forms in classical music look like this—sorry, AB forms in classical music look like this.
- 6.39** We talked about the fact that harmony allows us to differentiate between a stable place and a secondary place that is less stable towards which we move. You can have a meaningful sense of harmonic motion using the opposition: tonic and dominant. That is one of the fundamental polarities of classical music: tonic-dominant opposition. When you have that as a central feature of your harmonic language, it inherently creates a strong sense of teleology because you can have a movement towards a place or away from where you started, and then you can move back towards it.
- 6.40** What gives the classical forms their intense dynamism, their sense of forward motion, is the fact that you start on the tonic—you have all sorts of adventures, you modulate, you go towards different keys, you visit different scale degrees— but there's a magnetic attraction back to your starting place. That's why, again, if you're in C major or you're in D major, that means that the action, so to speak, begins and ends on that tone. It's like

a magnetic pull towards which everything is attracted, or you might be repelled away from it for a time.

- 6.41** That trajectory can be very, very, very long-reaching. It can be extended as long as you want, as long as your ability to imagine a complicated trajectory can be stretched. But in order to have that dynamism, that sense of forward motion, that sense of directed motion, then again, there needs to be strong magnetic poles, as it were, a strong sense of opposition between these scale degrees.
- 6.42** In the Western tradition, that has coalesced around the tonic-dominant opposition. Now, if you look at really early music, early medieval music, for example, that opposition isn't there. It starts to coalesce towards the end of the Renaissance and becomes much stronger in the Baroque era. In a lot of more recent music, if you look at some jazz, even some popular music, that polarity is severely weakened. You can think of this not as a heaven-sent rule, but rather as a cultural construction that mirrors the dynamism of the culture that produced it.
- 6.43** So this tonic-dominant opposition gives us, in an exploded form, the basic outline of an ABA or an AB form in a piece of music. That looks like this. Again, for the sake of simplicity, we're going to argue that we are in C major, once again. We're going to start by establishing the tonic, and then we're going to gradually move towards the dominant. Once we've done that, we will have made a move from I to V; we will have modulated. We're no longer in C, but we've moved towards G—so I to V, and we'll call that our A section.
- 6.44** Now, you'll notice that at the end of my A section, there's a little repeat bar line, the double bar line with the two dots, which indicates to the performer that you have to start over and repeat the whole section again. We're going to move to I to V, and we're going to jump back to the beginning and do the whole thing a second time.
- 6.45** Then, once we've done it a second time and we're on V, we need to round out the form by proceeding to a B section. The B section moves from V back to I. It's difficult to come up with anything simpler than that, but there are tens of thousands of Baroque pieces that follow that exact, precise pattern over and over and over again. You see it today still— hundreds of years later, we still have countless pieces that are built upon this basic pattern.
- 6.46** If I were to improvise some simple little phrase, I could show you what that sounds like [plays A section, I to V]. Now, I've moved from I to V. I've ended here. I've made that little trajectory. Now, what I have to do is I have to start on V again, or rather keep going [plays B section, V to I]. Now, this is not a great piece. The point is to show you what that trajectory sounds like. Move away from where you started, move back again.
- 6.47** Okay. Now, that's a very, very rudimentary form, a very effective one. The contrast there is not necessarily one of melody or texture; the contrast is simply a dramatization

of the movement towards this other place. That's a binary form, a simple binary form with a repeated A section and a repeated B section.

- 6.48** Now we come to ternary forms. Ternary forms are slightly more complicated. In a ternary form, we might have an A section, a contrasting B section, and then a return to C at the end—sorry, a return to A at the end. In terms of musical analysis, typically a return of a section that we've heard previously is demarcated with a little asterisk. So we have AB, and then the second occurrence of A at the end. That's a nice little symmetrical form. It allows you a few more expressive possibilities.
- 6.49** The B section in an ABA form takes the form of a little contrasting episode. Of course, the composer can decide exactly what form that contrast takes. The contrast can be on the level of harmony. You might start in C, and then maybe you move to the parallel minor of C, which is C minor—we have a contrast there in terms of the harmony; we move from major to minor. Then we go back to the A section at the end, and we return back to C.
- 6.50** Very often in ABA forms, the "reprise" or the repeat of A at the end is slightly varied; it's not identical to the initial appearance. That gives you, again, this very nice little balance between something that's the same, something that holds the form together, and something that provides variety and a sense of change, and you need that balance.
- 6.51** Let's see if I can quickly improvise a little demonstration of that idea [plays phrase in ABA form]. I've moved from C major [plays C major chord] to C minor [plays C minor chord]. Then all I have to do is repeat my initial idea, and then I've created a little symmetrical ABA form. The level of contrast there is not very great because I've mainly just moved from major to minor. But of course, you can also have any degree of contrast you want. In principle, the B section in an ABA form can be different in every respect from the A section. It could be different in terms of tempo, in terms of key, in terms of rhythmic figures, in terms of the texture, melody, and so on. Everything could be very different.
- 6.52** So that's a very simple example that gives you a tiny little trajectory, A to B and back again, or A to B. Now, what that gives us is the embryo of some of the more sophisticated classical forms. We'll start with one of the simpler ones, which is the rondo. The rondo in itself is a classical form, but it's derived from much earlier forms. It's derived from old poetic forms, and it's also derived from a form called the ritornello, which is very widespread in the Baroque.
- 6.53** What all these forms have in common is a continuous alternation of refrains and episodes—refrains being the fixed element and episodes being the contrasting element. In its simplest form, rondo looks like this: ABACA, and so on. You can continue that series as many times as you want. You can have any number of episodes. You could go ABACADAEA, et cetera, the idea being that in between each repetition of A, there's a little contrasting episode.

- 6.54** You can think of that as an expansion of the ABA principle into a longer series of contrasting episodes. That's the rondo. But in its essence, what that really is is a circular form in which you can keep returning to a central idea again and again and again. To a degree, that simple alternation between refrains and episodes, again, is the skeleton; it's the embryonic skeleton of the contemporary pop song with its verses and choruses.
- 6.55** We looked at ABA forms—those are also known in the Germanic tradition as Lied forms, L-I-E-D, Lied, or the plural form Lieder, being German art song. German art songs, as practiced by Hugo Wolf, Franz Schubert or Schumann, many composers who wrote songs in that tradition, very often are patterned in that ABA form. You might think of them as formal archetypes that can take a lot of different forms.
- 6.56** Now, eventually, those binary and ternary forms in the Classical era, animated by the intense teleology, the tendency to move away from the tonic towards the dominant, became expanded into a much more complex form called sonata form. You have to think of sonata form again, not as something that just appeared out of the blue, but as an extension of this principle, and as something that is almost an inevitable consequence of setting up your musical system in terms of a fundamental polarity or a fundamental opposition between tonic and dominant.
- 6.57** All of these forms are effectively ways to play out and dramatize that opposition. Again, you're not thinking of that consciously when you're listening to these pieces, but nevertheless, it's orienting and forming your experience of the work. Let's look at how sonata forms operate.
- 6.58** First, just a couple of words about where you're likely to find sonata forms. In the Classical era, the first movement of a symphony, of a string quartet, of a piano sonata, is very often written in what's called a sonata-allegro form. This is a dialectical form in which you have different contrasting ideas that are initially presented as a simple opposition and that are later brought into dialogue with each other, and eventually, you end up with a synthesis.
- 6.59** If that idea sounds familiar, it's because it's a very ancient idea. As I've mentioned previously—I'm going to restate it here because it's such an important point— that is the dialectical principle; that is the Socratic ideal of a discussion between two people with opposing viewpoints. Person A thinks one thing. Person B has a different point of view. They have a discussion. They hash it out. Person A moves a little bit towards the position of person B and vice versa. Ideally, in an ideal scenario, you end up with a synthesis of the best features of both arguments.
- 6.60** Now, that's an Enlightenment idea also. It ties in with the search for rationality, order, and logic. You present this again, initially as an opposition. Then you combine the ideas, you fragment them, and you put them into a dialectical opposition, and a synthesis arises out of that. Then you restate the ideas at the end of the piece again, but within a unified framework this time.

- 6.61** I'll show you how that works, how that plays out in terms of an actual piece. In a sonata form, you may begin with a first theme. The first theme is typically presented or exposed on the tonic. Then you have a little transition— the transition allows you to move, usually towards the dominant, not always, but very often. Then there's a little double bar line, and you repeat that, and that's called the exposition—the exposition allows you to lay out the basic thematic materials and the essential harmonic trajectory of the piece. There's your exposition.
- 6.62** Once you arrive at the dominant, typically, although not always— we're talking about idealized, almost platonic forms here. In reality, virtually every sonata form ever written deviates from this to some extent. It's very rare that you'll find a piece that just follows this as though it were a recipe, but look at this as an idealized form. The first theme: transition, move towards the dominant. Then you have a second theme, a contrasting theme, one that has a different character from the first one. The first theme in a stereotypical idealized sonata form might have, for example, an assertive character. The second theme might be more ornamented. It might be more introverted, it might have a very different form of expression.
- 6.63** For example, if I wanted to improvise a first theme, maybe I'm in F major this time [improvises the first theme]. That might be my opening theme. Now, it's got that very articulated march-like rhythm. It's very simple. I might move towards the dominant, end up on C [plays C major chord], and once I'm on C, I might have something very, very different [improvises the second theme]. Something like that.
- 6.64** Again, take all of this with a grain of salt. This is a demonstration of the principle. In reality, you're going to find many exceptions. For example, if you look at the music of Haydn, you'll find that he often uses only one theme—he states a theme, moves towards the dominant, and states the same theme again. If you look at Mozart, you'll find that there's a profusion of different melodic ideas— there's a first theme group where he might lay out three or four themes, each of which might be loosely interrelated, or they might be fairly different from each other, and then he moves to the dominant.
- 6.65** So in reality, composers, of course, will take this basic framework and treat it any number of ways. But this is the basic shape of it. The fundamental concept is a dialectical one. Once we've been through the exposition, we've had the A-theme, we've transitioned towards the B-theme or towards the dominant. Then we have a middle section, and this middle section is usually called the development.
- 6.66** Why do we call it development? The German term is *Durchführung*, which means to lead through, a little bit of a different concept. Effectively, what it means is you create a trajectory. You create a harmonic trajectory that takes us on a journey away from where we started towards somewhere else. You might modulate to multiple different keys. But the main thing to retain from this is not so much the modulations and the

different keys that you pass through, but the idea of instability, of getting far from your point departure.

- 6.67** During the development, all sorts of things are going to happen to these materials. They're going to be fragmented, they're going to be superimposed, they're going to be played in different combinations, and then you'll have little episodes and little transitions and various things that happen. That part of the piece is quite free. The composer can give full rein to their imagination and hit you with one surprise after another. Remember, these forms are intensely dynamic, they're very full of contrast, and they're always moving.
- 6.68** Once we've had our development section, which is not repeated, unlike the exposition, then typically that will lead us back to the recapitulation. The recapitulation is a restatement of the materials that we heard in the exposition, but with one major difference—remember how we said that the exposition starts on the tonic, moves towards the dominant? So we've got a movement away from our point of departure. Then we have this whole adventure that happens in the development. In the recapitulation, we present the same thematic materials as we heard in the exposition, but without modulating, without transitioning to the dominant.
- 6.69** So these ideas that were initially pitted against each other, posed with their contrasting qualities clearly laid out, are synthesized into something unified. And so then we have our first theme. Usually, we keep the idea of the transition, except that the transition just leads us back to where we started again. We're not actually moving to the dominant, but we stay on the tonic, and then we hear the second theme again, and without moving to V. So these oppositions, which create dynamism, which create movement, are reconciled and synthesized at the end of the piece.
- 6.70** I find it so phenomenally interesting that that basic principle became the central motor of the high classical style. It's a very narrative idea; it maps onto stories. It's simple, simple, archetypal story-like forms, but expressed entirely through sounds, entirely through tones and rhythms. It's, in a sense, the most abstract possible embodiment of that principle of goal-directed motion, opposition, contrast, synthesis, all within this form.
- 6.71** So that's a brief introduction to some of the basic categories of classical forms that you'll likely encounter. In short forms, for example, in Baroque or Classical dance suites, what you'll have is a succession of pieces, each of which is oriented around a particular dance rhythm. You might have a bourrée, and then a gigue, and then an allemande, or there are all kinds of dances you see in the Baroque: minuets and so on, sarabandes. You might have five or six of those pieces strung together in what's called a suite.
- 6.72** The dance suite is, again, one of the most typical and one of the most commonly found ways of articulating a piece in the Baroque. Dance suites tend to be made up of short,

little movements that contrast with one another. Within those movements, you're likely to see lots of binary and ternary forms, a very simple form.

- 6.73** Then, as we move into the Classical era, the narrative ideas, which are becoming more and more popular and more present through the genres such as opera, start to infiltrate instrumental music and animate it—again, always through a form of sublimation. There's no literal movement; you're not literally dancing when you listen to it. You're not literally following a story. Those concepts, those patterns of movement, are sublimated into pure sound.
- 6.74** The next time you're listening to a piece of music, no matter what the genre is, no matter what the era is that it comes from—if it was written yesterday, if it was written 600 years ago— try to think about it in terms of the formal articulation, in other words, the division into parts, where you can clearly hear a section coming to an end, or you can clearly hear a contrast between one material and another, between one form of rhythmic movement and another, one theme and another, and you can start to understand the formal articulation of a piece.
- 6.75** Just as how our limbs and fingers are articulated—we have joints—it's the same in musical forms as well. That, of course, gives us flexibility and movement and possibilities for contrast. All of these things have a theoretical basis. But what you want to do is to link up your ears, your perceptions, with these different concepts so that you understand them, not just in an abstract sense, not just in an intellectual sense, but you can relate them to your perception.
- 6.76** That was our journey from the world of the extremely small, from very tiny little iterative pulses all the way up to large-scale forms, sonata forms that can take 30 minutes or longer to play out. We're going to stop there for now. I'm happy to address any questions you may have. Yes.
- 6.77** Hello. Music has always been a profoundly human experience. Now, during this class, you've broken it down to almost a science, almost mathematical formulaic. I was wondering: in the age of AI, if you feed it all of these principles and patterns, will AI be able to replace the human creator of music, to provide the same experience to listeners as previous composers? Or in your opinion, is there something profoundly human about music that cannot be artificially created, and what would that be?
- 6.78** Well, here's one simple distinction: humans have a physical body, and machines do not. So that's, I think, an insoluble technical problem. If you're going to try to have machines duplicate some aspect of human art. All of these things that I've been describing, these different forms of movement in music, can be embodied kinetically in terms of dance, and they can also be sublimated. But I think for these concepts to make sense, for them to affect us, then you have to imagine it on some level in terms of movement.
- 6.79** This might be the limits of my own imagination, my inability to imagine what lies over the horizon, because things are moving quickly. It's anybody's guess what things will

look like in five years, let alone 10 years or 20 years from now. Who knows? But from my present-day vantage point, I have trouble understanding how something disembodied can replicate a phenomenon that is an embodied one. That's the first thing.

- 6.80** I personally am not particularly concerned about the advent of AI as it pertains to creative endeavors. The other reason for that is that music isn't a thing that you go and buy. It's not a commodity. It's something that you do. It's a process. That's a very intensely personal thing. Each of us has a personal relationship to the music that we're interested in, an autobiographical relationship to it that can't be duplicated any other way.
- 6.81** On top of that, if you have any engagement with music, either as a listener or as somebody who performs or composes, you'll know that it's a process, and there's no goal at the end of that process; it's something that is continuous and ongoing. That's why you can read interviews with very old composers who will tell you that they get the sense that they're just starting to figure out what music's about. There are so many artists where they say exactly that at the end of their life, that "I would need four more lifetimes, I need five more lifetimes." There's so much more to explore. It never stops. There's no end to that.
- 6.82** The other thing is—and I think this is very interesting—what AI is going to reveal to us is that some of the things that we think are uniquely the result of our particular form of intelligence are actually maybe not ultimately all that distinctive or personal. Those are going to be the things that are very easy to model. Something that we might imagine as being a unique expression of our subjectivity might turn out, in fact, to be very easy to replicate because it's actually less complex than we thought. Other things, I believe, are going to be very difficult and perhaps impossible to model in that way.
- 6.83** What might happen—this is a hypothetical—what might happen is it might bring us closer into contact with the things that are uniquely ours, that only humans can do. We'll see. It'll certainly be disruptive, but probably not in the ways we imagine.
- 6.84** You talked about the structures and that they repeat: ABA, and so on. Is this the same for every type of music, or do the structures tend to work better for a certain type of music, and for others, a different structure works better?
- 6.85** Right. The forms that I've just shown you are forms that are prevalent in the high classical style. If you think about 10,000 years of human music making, the high classical style is one step along that development, and it was a fairly short-lived one, actually. That style brings you, roughly speaking, from the end of the 18th century to maybe the beginning of the 20th century, something like that. So it's not a very big span.
- 6.86** As it happens, there were tens of thousands of symphonies and sonatas and concertos and suites and so on that were written during that period. There's an immense body of

work, most of which is forgotten, some of which we've retained and have entered into what we call a repertoire, where orchestras and chamber ensembles and soloists have a certain— what would you say—selection of pieces that have, for one reason or another, survived to the present day and are still being played.

6.87 Sometimes that selection process is profoundly unjust. There are, no doubt, countless very interesting pieces languishing in libraries that are no longer played. But nevertheless, that's a process that is inevitable because you can't, as a culture, retain everything you produce. This particular form of dynamism of teleology that you see in the classical forms, as I tried to lay out in my lecture, is a consequence of the polarity, the opposition between tonic and dominant. And not every musical culture has that as a central feature of their music.

6.88 For example, if your music is highly predicated on rhythm and minimally involved with pitch, then you're going to see very, very different musical forms, different musical phenomena coming out of that. Likewise, if your music is not particularly oriented towards harmony, if you have a virtuosity of melodic elaboration, and that's the focus, then you're likely to see much less dynamic forms. Or you might see forms that heavily rely on external factors, such as the music being embedded in a social ritual of some kind. That's a little bit different.

6.89 I think one of the particularities of the classical style is that those ritualistic dimensions, the dance dimension, let's say, are all sublimated. There is a ritual involved in putting on a suit or whatever you do, going to a concert, sitting there and listening for 2 hours, and then going home—I suppose that's a form of a social ritual, but it's not the same thing.

6.90 So, depending on the broader culture, the role of music within that culture, whether or not it's embedded within some form of ritual, it might have a more or less greater degree of abstraction to it. Some of the forces that animate music might be more abstracted or sublimated as opposed to physically embodied. So no, there's a very broad range. These are forms, again, that find their expression most commonly and are most strongly associated with the European classical tradition. But I believe that they have aspects to them that are archetypal and that stem from things from the ancient world that have persisted.

6.91 But again, I keep coming back to this point: it's really a conversation between irreducible archetypes—God knows how old those are—and the aleatoric dimension of the evolution of our culture, which has all kinds of things about it that are over-determined or very complex and impossible to predict, and that change over time.

6.92 You were mentioning the polarity between the tonic and the fifth. I think it was medieval music that you were saying wasn't as reliant on that polarity. I was curious about what they were using instead, or maybe even a musical example of what that would sound like.

- 6.93** Okay. We looked yesterday at the phenomenon of the leading tone, remember? How the leading tone is immediately below the tonic? [plays B, C] The leading tone, again, gives you that sense of movement, and it stems, at least in part, from the V-I progression. So if you have that leading tone as a central feature of your scale, then it's going to have that strong inclination towards movement and resolution.
- 6.94** We looked at how the relative minor [plays A minor chord], if you do nothing to alter the scale and present it as a natural minor scale [plays natural A minor scale], it doesn't have a semitone below the tonic; it has a whole step. That little difference [plays G, A] changes everything. There's much less of a sense of tension there or movement.
- 6.95** That's why you have this variant of the minor scale—there are two variants, one of which is [plays melodic A minor scale] you raise the sixth and the seventh degrees in order to have that semitone below the tonic. Or [plays harmonic A minor scale] you might only raise the seventh degree.
- 6.96** Medieval music has a broader range of different scales or modes that it uses. Today, we think in terms of the minor or the major scales, but in medieval music, there were at least six different modes that were used, each of which had a different composition of intervals. Some of them have a leading tone, and some of them don't, but it's not predicated on that opposition. You might have a mode that sounds like this [plays D Dorian mode], or one that sounds like this [plays A Mixolydian mode]. There are many, many different examples, but we don't have that distillation into major and minor modes.
- 6.97** I would have to play you some examples of early motets, like 15th-century music, to illustrate the concept. But I might refer you to the History of Western Music course in which I do that, and you can hear the difference. The music is much less directional, and it has to do fundamentally with the use of different modes and with that fundamental lack of opposition.

Lecture 7: Musical Exercises

At a glance — Musical exercises bridging theory and practice.

- 7.1** Now we're going to attempt to bridge the theoretical concepts that we've been looking at and the world of actual practice of musical performance and practical skills. We've been taking a tour through concepts such as motion as expressed in music, the different ways that you can do that. We've looked at all these different parameters of how you can do that in melody, how you can do it in harmony, how you can do it rhythmically, and also how you can differentiate and articulate time, both on the micro level with individual short pulsations, and also on the macro level with musical forms that allow us to have this wonderful dialectic between the same and the different.
- 7.2** We mentioned at the outset of these lectures that one of the peculiar aspects of music theory, as it's conventionally taught, is that it pertains to everything except for playing and listening to music. It's an abstract set of ideas that we're learning: how these systems work together, how the notation works, and so on.
- 7.3** But remember, I said that in different countries, the body of materials that's covered under music theory is constituted somewhat differently. We talked about the fact that music theory, as I'm describing it, is somewhat of an Anglo-Saxon concept, which also has a certain existence in the German system. The French and Italian systems are very different. They're much more focused on solfège and écriture or music writing, which is, again, they cover some of the same technical aspects, but in a more practical application.
- 7.4** What we're going to do today is we're going to look at some simple techniques that anybody who is interested in music, wants to perform music, wants to learn about it, can try out to sharpen their skills. A lot of what we're going to cover basically falls under ear training. But I don't particularly like the term ear training because it sounds fairly rebarbative. It's not necessarily something that you would enjoy doing, or at least not from the term itself.
- 7.5** What we're going to look at—these are really elementary skills that can be not only beneficial, but are necessary for musicians. In a sense, it doesn't really matter if you're working in commercial music, popular music, the fine art musical tradition, or something else. These skills are, I think, fairly universal.
- 7.6** Now, again, this is where questions of individual aptitude for music come into the equation. I believe anybody can improve on these exercises that we're going to look at. It's true that some people will pick them up more easily than others. Don't let that stop you. If you're finding this difficult or if it's taking longer to get into it, I've worked with so many students over the decades, I can say with 100% confidence that absolutely anybody can improve their skills in these areas. It might take a little longer; it might be

a little bit more painful for some than for others. Ultimately, that doesn't matter. What matters is your capacity to work on these things and make measurable progress.

- 7.7** What this does is it covers all the basic aspects of musicianship that can make you a flexible and competent musician. Of course, as with anything, you can take this as far as you want to, or you can learn the basic skills, and that might be enough. It falls under a few different types of exercises, and they involve rhythmic dictation, melodic dictation, harmonic dictation, and also rhythmic clapback.
- 7.8** Fundamentally, what all of these different skills have to do with is developing the capacity to externalize a musical idea or to reproduce a musical idea using your voice, using your hands to clap out a rhythm, or using an instrument. So it's very, very important to develop these skills because as a musician, what you want to do is to link up your eyes, your ears, and your fingers, or your feet, or whatever you're using to play your instrument. Those three things have to work together.
- 7.9** Let's start with some basic rhythmic dictation. Some of these things are going to be exercises that you can do by yourself. Some of them are things that you're going to need either a book or an online tool to help you with them. Fortunately, it's now possible to go online and find all kinds of exercises like this without having to sign up or have a subscription or anything.
- 7.10** What you want to do is you can go on to either a website called musictheory.net, which is a fantastic resource, has all kinds of ear training exercises, and they're graduated from very simple things all the way up to much more advanced concepts. Or, alternatively, you can just practice by getting some sheet music at whatever level is appropriate for you and practicing sight-singing—this is where you're away from the instrument and you attempt to reproduce with your voice what you see on the page. All of that, of course, presupposes that you've spent the time to familiarize yourself with musical notation.
- 7.11** As you start doing these things, you will inevitably find that it is extraordinarily slow and painful and clumsy, and you're going to feel incompetent, just as you would feel if you were beginning to learn German and you walked into a coffee shop and tried to order a coffee—every sentence is going to have 15 mistakes in it. That's fine; that's completely what you expect. Don't expect anything higher than maybe a 15% success rate at best with some of these exercises at first. That's completely to be expected.
- 7.12** So let's start with rhythmic dictation. What happens in a rhythmic dictation is that a teacher or an online assistant, or whatever it might be, will set up a pulse, a regular recurring pulse, and then they'll dictate a simple rhythmic figure on top of that. Then the student has to write down the rhythm that they've just heard. This is a very simple exercise.
- 7.13** For example, I can just start clapping a simple 4/4 rhythm. What I'll do is say in advance what the meter is, so you know what the time signature is. I start clapping a pattern,

maybe for one bar. Then, with my voice, I'm going to sing this very simple rhythm. Then the student has to write that down, using what they know about rhythmic values.

- 7.14** For example, I might go [claps 4/4 time signature and sings contrasting rhythm over it], something like that. Simple rhythm. Then, what you would do—I'll go back to my app here—remember, I said this is going to be in 4/4, so we're going to start by writing out our time signature. Then I have to first remember what I've heard—it's partly a mnemonic exercise because you have to be able to remember the figure that I just played. Then you think, "How would I express that in terms of musical notation?"
- 7.15** Now, in some instances, you'll get to hear it more than once, which is helpful because one of the hardest things at first is remembering what you've just heard and then being able to mentally convert it into rhythmic values [claps 4/4 time signature and sings contrasting rhythm over it]. Once you've done that, you might need to hear it two or three times, then you convert it into notation. It's going to look like this.
- 7.16** In order to do this exercise accurately, you need to be able to remember things like dotted rhythms. Remember, we spoke about how if you take any rhythmic value and add a dot to it, you increase its value by half. So a quarter note with a dot added to it is equal to three eighth notes. Then that gives you that characteristic unequal rhythm that we talked about previously. The unequal rhythm being this [sings long pulse, followed by two short pulses].
- 7.17** You notice how one of the pulses is longer and one of them is shorter. Over time, this will become completely instinctive, where you will hear that figure and immediately think, "Ah, yes, that's a dotted rhythm, and you know how to write it down." At first, it's a conscious intellectual effort to relate these things, but after a while, you start to absorb the patterns. They become second nature, and you don't think about them anymore.
- 7.18** As you go through these rhythmic dictations, you'll get into exercises that are progressively more difficult, and they might include a wider variety of rhythmic figures. For example, you might have something like this [claps 4/4 time signature and sings contrasting rhythm over it].
- 7.19** Now, that had a simple—you might have noticed—a simple triplet rhythm. Then you think back, "How do I write down a triplet?" And you have this. Now, the way I performed that was, I suppose, slightly ambiguous in the sense that it would have been hard to tell if the last pulse that I did with my voice lasted for two beats or one. So you could legitimately write that down as a half note at the end or as a quarter note, followed by a rest; either one would be acceptable.
- 7.20** So that's an example of what rhythmic clapback looks like, or rather, rhythmic dictation. You hear a pulse, you relate it to a pattern, and then you try to write down the figure. That's a very good place to begin with ear training. It's forcing you to pay

close attention to what you're hearing, remember it, and reproduce it. That's an example of rhythmic dictation.

- 7.21** We also have melodic dictation. Melodic dictation is the same basic idea, except you also have pitch to deal with. You're going to hear a little melodic fragment, and you're going to try to remember it and write it down. In most instances, when you're doing an exercise like this, you'll get a chance to hear it at least two or three times, especially if you're beginning with this kind of exercise; it's necessary to hear it multiple times.
- 7.22** What you want to do is begin by focusing on the direction of the line. Even if you can't tell the exact interval, even if you can't name the notes, focus on: Is it going up or is it going down? What's the basic shape? From there, you listen again a second time, you see if what you remembered was accurate, if it really corresponds to what you heard. Then the third time, you try to work out what all the intervals are.
- 7.23** In other words, you start from a bird's-eye view, a broad impression of the shape of the line, and then progressively, you fill in the details. It's like starting with one of those connect-the-dots drawings where you at first just see a field of dots, and then you start to gradually fill it in. You can see a bit more detail in it.
- 7.24** You might have a simple line like this—usually, the teacher in that scenario would start by giving you the first pitch. Let's say that it's middle C [plays C]. If you're beginning, you're going to mostly have stepwise or conjunct melodies that have no skips or that might have only one skip in them or one leap. Let's try a little line [plays C, D, E, F, E, D, E, D, C]. Very simple. Hear it again [plays C, D, E, F, E, D, E, D, C]. Try to reproduce it. Try to sing it [sings C, D, E, F, E, D, E, D, C], hopefully better than me.
- 7.25** This is such a primordial skill for musicians. If you're a jazz musician, if you play the trumpet or the piano or the double bass and you're improvising, you can't play a line until you can hear it first. You have to be able to hear inwardly what you're trying to do, even if it's just a split second before you do it, but you have to be able to audiate it. You have to be able to hear it inwardly before you can reproduce it.
- 7.26** If you can't imagine a line, if you can't imagine what the interval sounds like, you won't be able to hit it, and that's particularly crucial on instruments where the intonation is not fixed. If you're a singer, for example, or if you play the violin, the violin has no frets. The violin just has a fingerboard, and there are various positions, of course, that you learn, but there's no fixed place to put your hand on the fingerboard. You have to be able to hear the intervals inwardly with precision before you play them. If you can't do that, your playing will be out of tune, even if you have an excellent mechanical understanding of the positions on the violin.
- 7.27** If you can't hear the interval, if you can't reproduce it on some level imperfectly with your voice, then it's going to be very difficult. Let's hear that little phrase again [plays and then sings C, D, E, F, E, D, E, D, C]. What you'll find is that if you can sing it back, you're much likelier to retain it; it'll be much easier to write it down.

- 7.28** Then we're going to go to our sheet of music paper here. Now, remember that when I started the exercise, I said what the first note would be—the first note was middle C [plays C]. That gives us a starting point. Now I can put a middle C on the treble staff. Then I can go from there, and I can think, "Okay, it goes up [sings C, D, E, F]; that's just a stepwise progression. If I just write a series of adjacent notes, then that should give me the line.
- 7.29** Then I have to go, "Okay, what happens next?" [sings C, D, E, F, E, D, E, D]. Okay, remember the shape? It's not that many notes, not that many distinct pitches in this line. Then the last note sounds like it's the same as the note that we started on [sings C, D, E, F, E, D, E, D, C], and then you write it down. There's our melodic dictation.
- 7.30** Now, again, that's an exercise that can be as simple or as difficult as you care to make it. That line that I just played is 100% conjunct. There are no leaps anywhere in that line. But if I were to play something with larger intervals or a line that was more disjunctive, then that would significantly increase the level of difficulty. That's something that you want to work towards progressively.
- 7.31** Another thing that can, of course, make it very difficult is if you have chromatic intervals. In other words, tones that do not belong to the natural diatonic scale, but rather chromatic tones. For example [plays C, D, E, F, F#, D, G, G (below), C]. That's far more difficult. But there again, what you do is you practice with your voice [sings C, D, E, F, F#, D, G, G (below), C] until you can reproduce the sounds reasonably accurately. It might speed it up to sit at an instrument so that you can play it again and try reproducing it. That's a progressive exercise you can do. That's melodic dictation.
- 7.32** We've looked at rhythmic dictation, and there's also harmonic dictation. Harmonic dictation has to do with reproducing chords, either at an instrument or with your voice, or simply identifying the quality of a chord based on hearing it once or twice.
- 7.33** Remember that when we were looking at triads and how triads are built—these three-note chords that are built on stacked thirds—we looked at the different ways that you can alter a triad and obtain major, minor, augmented, and diminished chords. If we're going to use A flat this time as our reference point, what you want to do is to develop the capacity to recognize instinctively if what you're hearing is a major or a minor triad.
- 7.34** This is a little bit more tricky because it's harder to relate it to something like a shape. If I just play you an isolated chord, then you're getting into qualitative aspects of sound. There's not anything else to compare it to. You're just hearing a sound, and you have to identify it. [Plays A $\flat$ major chord], there's my A $\flat$ major triad. [Plays A $\flat$ minor chord], there's my A $\flat$ minor triad. [Plays A $\flat$ augmented chord], there's my augmented triad. [Plays A $\flat$ diminished triad], and there's my diminished triad.
- 7.35** That's an exercise that just takes a lot of repetition and gradual familiarity with these different types of chords. You can speed things up to an extent by sitting down at a keyboard and figuring out how to form these different chords. Choose any random key

on the piano—maybe it's C# [plays C#]. Can you build a triad starting from that note? You think, "I need to have a major third" [plays C#–F], and then I have to have a minor third on top [plays F–G#]. You practice building triads like that, different types of triads. You practice playing them all throughout the range of the instrument.

- 7.36** You can do this on other instruments, too. It's a lot easier on a keyboard because you can actually visualize the patterns; you can see them on the keys. On an instrument like a guitar or a harp, for that matter, you can do that, too. There's a visual dimension, but it's not quite as instinctive as it is on a piano keyboard. I want to get into some example exercises for chord dictation. But I also want to look at recognition of intervals because, in a sense, this is one of the most fundamental skills. When we had that little melodic phrase [plays C, D, E, F, E, D, E, D, C], remember that it was all stepwise. So the range of intervals that you have to recognize in there is very, very limited.
- 7.37** But what you also want to do is to practice recognizing intervals, not just the size of the interval, but also, as with these triads, recognizing their quality. For example, if I play a random note on the keyboard, let's do a G [plays G], and then I play a second note after it, so I'm playing a melodic interval, then you have to try to identify what the interval is [plays G–C]. There's simply no substitute in this case for singing and playing so that your ear can become accustomed to the intervals.
- 7.38** One trick that works amazingly well is to associate the interval with a particular song or piece of music that's familiar to you. If there's a song that you've heard 100 times and you can effortlessly sing it, then you'll have a much easier time recognizing it when it occurs in an exercise like this. For example, one exercise that's often given is to recognize the sound of a perfect fifth [plays C–G]. Can you think of a song that starts with that interval, with that sound? Then you remember that, you file it away in your memory, and if you have to reproduce the perfect fifth, you think, "Oh, yes, that's the opening of—" whatever it might be.
- 7.39** With tricky augmented or chromatic intervals like this [plays G–C#]. That is a tritone. The tritone is a famously difficult interval to sing. It's called a tritone because it covers three whole tones [plays G, A, B, C#], very, very tricky to reproduce with your voice [plays and sings G–C#]. Conveniently, those are the first two notes of the theme song from The Simpsons [plays G–C#–D]. There you go; it works every time [plays G–C#–D]. That's the mnemonic aid or trick you can use to remember these intervals. Then you just go through them.
- 7.40** We're going to look at the different intervals that you're going to need to familiarize yourself with. We're going to start on F. Starting from the beginning, we have a minor second. That's written as a lowercase m with a 2. We have a major second, so uppercase M and 2. Then we have the minor third in ascending order, major third, perfect fourth, so P4, augmented fourth, perfect fifth, and augmented fifth or minor sixth—actually, I'm going to spell that as a D $\flat$ to make it a little simpler.

- 7.41** Now we have a sequence of melodic intervals that you can practice singing. [Plays and sings F–G $\flat$]—that's the minor second. Major second [plays and sings F–G]. Minor third [plays and sings F–A $\flat$]. Major third [plays and sings F–A]. Perfect fourth [plays and sings F–B $\flat$]. Augmented fourth or tritone [plays and sings F–B]. Perfect fifth [plays and sings F–C]. Minor sixth [plays and sings F–D $\flat$]. You can just drill those.
- 7.42** What you can do is just pick a note anywhere on the keyboard and a random interval. You say, "I'm going to start on this note, and then I have to sing a perfect fourth, an ascending perfect fourth, starting on that note." [plays and sings C $\sharp$ –F $\sharp$]. Now, remember, intervals can go up or down. They can be ascending intervals or descending intervals. You want to make sure that you're including both in your exercise. So I could start on an A [plays A] and say, "I have to sing a descending tritone" [sings and plays A–D $\sharp$]. You have to reproduce it with your voice. Or a rising major third [sings and plays E–G $\sharp$].
- 7.43** If you can get to a point where you can do that effortlessly, starting from any pitch, then you'll be well on your way to developing fluency with these sorts of ear training exercises. A lot of musicians, funnily enough, never get there. They don't work on these skills very hard. That's going to make things harder and slower for you. It's not that you can't do anything in music without those skills, but it certainly makes things a lot easier.
- 7.44** It makes it faster. You can retain things more easily. You can reproduce things more easily. You can explain your intentions to musicians that you might be working with more easily, and so on. If you want to sit down at a keyboard and start to play or improvise, developing the capacity to externalize your idea, even if it just means hearing the next sound before you play it inwardly, then it's going to be much, much easier for you to sit down and extemporize something.
- 7.45** We've looked at the melodic intervals—some of them anyway, the ones that you're at least likely to encounter as you first start working on these exercises. We've looked at melodic dictation and rhythmic dictation, and we've started to talk also a little bit about chord dictation or harmonic dictation.
- 7.46** The exact nature of a chord dictation exercise is going to depend somewhat on the teacher and on the particular pedagogy that they're following. But very often, what you do is you start by hearing a chord—and it might be arpeggiated or played one note at a time—then you try to reproduce, you try to sing back all the tones in the chord [plays and sings F–A–C–A–F]. In other instances, the chord might be played just completely straight like this [plays F major chord]. Then you have to try to separate it out into its component tones, hear them inwardly, and then try to reproduce them with your voice [plays F major chord and sings F–A–C].
- 7.47** Again, you can do this exercise by yourself to an extent, by sitting at a keyboard and saying, "I'm going to start on B $\flat$ [plays B $\flat$]. I'm going to play a diminished triad [plays B $\flat$ –D $\flat$ –E]," and you sing it back. As I understand it, there really is no substitute for

sitting and doing this until you get to a point where you can link up what you're hearing with your voice. It all goes through the voice.

- 7.48** Now, because of the particular nature of this exercise, because of the fact that it's partly physiological, and it's partly got to do with your perception and your ability to distinguish sounds, you're going to find that there's a practical limit to how long you can spend on these exercises. That limit is going to, of course, vary from person to person. Some people will have a higher tolerance for them. Some people will learn them more easily. Others will find it very slow going. It doesn't matter that much.
- 7.49** But what you want to do, ideally, is to not torture yourself to death with these exercises, because as I mentioned at the outset, they can be rebarbative and they can be tedious to work on. You want to set this up in a way where you could actually conceivably enjoy it, where you're going to stop at the point where it's going relatively well and you're not getting frustrated.
- 7.50** I've had students make excellent progress by working on ear training exercises 10 to 15 minutes a day. That might not sound like much, but anything that's difficult, if you're actually sitting and concentrating on it and giving it everything you have, 10 minutes is actually—you can do a lot in 10 minutes if you're really focused. But it's not a major commitment in terms of time, even though the focus and the concentration might be challenging.
- 7.51** What I would recommend doing is just sitting with a timer. Let's set a timer for 10 minutes, try doing some of these exercises, and see how it goes. If you're having the absolute time of your life, then you can do a bit more than 10 minutes. But if that's enough and you'd rather do something else now, you can stop. That's the deal.
- 7.52** What also seems to be very effective is to have multiple different working sessions over the course of the day. And they can be very short. They could even be as short as maybe five minutes. But if you do it three times, then you're starting to condition yourself through regularity and through repetition, and that can be quite powerful. So it's not the amount of time that you spend practicing; it's not the amount of time you spend sitting at the piano, croaking out these out-of-tune intervals, but rather how effectively you're working and training yourself to listen, to hear. There's an art to listening.
- 7.53** Remember that listening and hearing aren't the same thing. Hearing is an indifferent physiological phenomenon, like a glass falling onto the floor and shattering. It's just something that happens in the physical world—airwaves hitting your eardrum. Hearing has nothing to do with culture or understanding. Listening requires culture, intention, and understanding. What you're learning to do is to move away from passive hearing, where the vibrations are hitting your eardrum, but you're not necessarily receiving, you're not necessarily understanding, you're not differentiating, and you want to learn how to hear. That's an art, and you can spend your life on that.

- 7.54** Those are a few practical exercises that you can do. Again, depending on how far you want to go with them, you can get into things that are more challenging. But I want to show you another couple of exercises that you can do that will improve your fluency with reading notation. A lot of the things we've looked at so far have had to do primarily with training your ear to recognize and differentiate intervals and rhythms, and then to be able to reproduce them. Those exercises largely bypass notation; you're not necessarily needing to deal too much with the language of symbols and note heads and rhythms and so on.
- 7.55** With some types of exercises, however, what you want to work on is your fluency with reading sheet music and instinctively recognizing and being able to reproduce rhythms that you see on paper. That will start with a rhythmic clapback. This is where you will have a sheet of rhythmic exercises. You'll have the time signature, and you have to reproduce them: you have to clap them back or sing them back.
- 7.56** If you look on the board there, I've got four rhythmic exercises: one, two, three, four. In actual fact, I'm the author of these exercises, but I'm going to pretend that I didn't write them and that I have to sit down and reproduce them. The first thing to do is to check the time signature and make sure you understand the pulse groupings within that time signature. We've got an exercise in 6/8, one in 4/4, one in 3/8, and one in 3/4 — four different time signatures.
- 7.57** You'll remember that we talked about binary and ternary beat patterns and compound meters. Remember that 6/8, although it has six eighth notes in it, it is actually articulated into two pulse groups, each of which is subdivided into three [counts six eighth notes with emphasis on beats 1 and 4] like that. You have to have that pattern in your ear before you start working on this.
- 7.58** Then you're going to think, "Okay, I've got two pulses, and I can subdivide them" [counts six eighth notes with emphasis on beats 1 and 4]. If you don't do that first, if you don't know what the pulse is, and if you don't know what the order of weak and strong beats is in that meter, then you're going to have a really hard time with these exercises. So make sure that that's clear. And then you can read and start to read it at a comfortable tempo [demonstrates rhythm as written on the board], et cetera. As you get more advanced, you can, of course, get more fluent with that, and you can experiment with faster tempi.
- 7.59** Now, some of these exercises are relatively straightforward. Some of them are a little bit more difficult. In exercise number two, you'll notice that I've got quarter notes, eighth notes, there are some rests in there, and there's also a triplet. So that means you're going to have to switch between different pulse speeds many times over the course of this exercise. The first thing, again, remember that we're in 4/4, so we have to remember what that rhythm is like [counts 4/4 time signature]. Then we're going to start to try to reproduce the rhythm [demonstrates rhythm as written on the board].

- 7.60** These are not particularly easy exercises because number two contains what's called a syncopation, and a syncopation is a beat that doesn't line up with the underlying pulse grid. Another way to think of it is as an offbeat. If you're doing [demonstrates syncopation], that thing. Syncopations are a fundamental part of dance styles around the world and different popular genres as well. We might understand them instinctively without necessarily being able to reproduce them accurately. Then you just work through the exercises. Again, these can get extraordinarily difficult. You want to start with very, very simple, straightforward ones as you start getting familiar with notated rhythms.
- 7.61** So next, we have an exercise in 3/8. So 3/8 is a ternary rhythm, so that means we're going to have three pulses per bar, three eighth notes per bar, and I have to try to reproduce this rhythm. So choose a tempo, one that's comfortable. Think about the pulse, think about the time signature [demonstrates rhythm as written on the board]. Then you try it again if you need to. You might have to do them 10 times. You might want to slow them down if it's difficult.
- 7.62** As you get into particularly advanced rhythmic clapback, you'll find that there are going to be examples where the time signature changes. The whole piece isn't in 3/4 or 4/4, but the meter will change partway through the example. Number four is an instance of that. In number four, we start off in 3/4, and four bars in, it switches to 2/4, and then it switches to 3/8, and then back to 4/4 again. The challenge there is that the fundamental beat unit changes between 2/4 and 3/8, where we have to move from thinking in terms of a quarter note to thinking in terms of an eighth note pulse.
- 7.63** If I'm moving from 3/4 to 3/8, then I have to do [demonstrates a pattern of 3/4 followed by 3/8]; that's switching from a pattern in 3/4 to a pattern in 3/8. The exercise from beginning to end is going to sound like this [demonstrates rhythm as written on the board]. There, not only did I have to deal with triplets and rests, but I also had to deal with this shifting meter. All of it takes practice, but that's a very good method anyway for learning to read rhythmic notation and achieve fluency with it.
- 7.64** The next exercise, which we're not going to do just yet—we're going to do it after the interval—is in many respects the most challenging of all. But the thing is, as these exercises get more difficult, you're going to find that they stretch your musicianship and they're amazingly effective. I really recommend anybody who wants to work on their basic musicianship skills to practice all of these different exercises: externalize with your voice, sing what you hear, what you see on the page, reproduce it, tap out the rhythms through whatever means work best for you.
- 7.65** There are also some resources that you can use to work on these in a more concerted way. I mentioned that you can go on musictheory.net and you can find lots of examples where you can just click on something and you'll hear a chord or a short little melodic phrase or rhythm, and then you have to reproduce it or write it down.

- 7.66** But there are also a number of resources that can be incredibly helpful, such as sight-singing exercises that were devised by composers and teachers to try to help students achieve greater fluency. One of the best such books I've ever seen, although it's a little bit on the advanced side, is the *Vocalise-Étude* of the French composer, Gabriel Fauré, F-A-U-R-E.
- 7.67** Fauré was the director of the Paris Conservatory in the 19th century and a very influential French composer, teacher of multiple generations of major French figures, including Maurice Ravel, among others. He was an amazing composer and also a gifted pedagogue. As it happens, he was in charge, at one point, of writing little entrance exams for students who wanted to get into the singing classes. He composed a series of little pieces called vocalizes, that were wordless pieces that could be sight-read by singers with the piano accompaniment. You can get hold of a book like that.
- 7.68** Edition Peters in London has a beautiful edition you can buy of these pieces. You can take them home, try to bash through them at the piano or try to sing the lines and expect that the results are going to be horrible because it's difficult. That's part of the point. Then, like anything, you start at 2% proficiency, and you try to get a little bit better each time. You can look at the vocalizes of Gabriel Fauré. There are many, many, many useful books for harmonic, melodic dictation, and so on. You can take them and either listen to a CD or recording online, or there'll be a bit of music notation, and you have to practice singing it or reproducing the rhythms.
- 7.69** That's my introductory foray into the wonderful world of ear training. You can spend years and years on that, by the way. In some respects, your ear is never definitively formed. You can always continue to improve, no matter how experienced you are as a musician. Before we move on to the next and final lecture in this series, does anybody have any questions, concerns, or complaints about the exercises we've looked at so far?
- 7.70** In the cases when people would spontaneously like to engage in making a song, see what they could do, how do they know when they've done a good job at it?
- 7.71** You've probably seen there's a comic book in French called *Astérix*. There's a character in there who's a pretty terrible musician who always has a harp with him. Every time he starts playing, you see all these jagged, discordant music notes coming out of the instrument. Everyone runs for cover. So that's one way: you just check the reactions of the people in the room. Unless they're especially indulgent as a group of friends, usually you can tell if your efforts are meeting with a positive perception or not.
- 7.72** But more seriously, how can you tell? Well, you might not be able to tell if you haven't trained your ear. You might not be able to tell the difference very easily between an interval that's been accurately reproduced, sang, and one that's completely inaccurate. I've noticed this with people.

- 7.73** You can listen to ordinary people whistling or humming a song, and you can see to what extent they've managed to perfect these skills. Because a lot of people, when they're whistling or singing—I've noticed this many times—they'll just randomly modulate; they can't hold the pitch. So they'll start off in C major, singing some little tune, and then by the time they're done, they're in F# major because they don't remember the tonic; they don't remember where they started from. The intervals get progressively more and more out of tune, and then they get farther away from where they started.
- 7.74** Other people, even some completely untrained musicians, will be exactly on pitch all the way through. It's amazing. With these sorts of things, I wonder, I think it is partly physiological. I think there are individuals who have an easier time remembering and reproducing intervals and pitches accurately than others. There's a faculty called perfect pitch that some people have—"l'oreille absolue," absolute pitch, some people call it—which is the ability to recognize a note by name simply by hearing it. You can hear a pitch and then say, "Oh, that's an A flat, that's an A, that's a G," and so on.
- 7.75** I'm not an expert on this topic. There's a bit of literature about it now, about what exactly that is, where that skill comes from. It might be hereditary, but it appears that you can increase the likelihood of developing perfect pitch if you start as a child at an instrument. It certainly seems to help to be in a musical environment.
- 7.76** I've seen claims that it's possible to acquire something like absolute pitch later in life, although very often it seems to be more of a trick than an actual instance of perfect pitch, where you can memorize a reference pitch—some people get very good at doing this—they can remember a frequency somehow and then use that as a reference point to think, "What would be a perfect fourth higher than that?" Or, "What would be a fifth lower than that?" And then they can find all the notes, which is actually a very impressive skill. It's not quite the same as absolute pitch, but it's very impressive nonetheless.
- 7.77** But absolute pitch can also be a detriment in some respects. So remember that before the development of—well, the widespread adoption of things like tuning forks and ease of travel amongst different parts of the world, if you were living in some village in Italy and you wanted to send a piece to another orchestra or ensemble or choir in Greece or somewhere else, that other place, even if it was just 10 miles away, their tuning standard might not be the same. There was no reliable means to establish a consistent system of pitch and tuning from one town to another, let alone from one country to another.
- 7.78** But now there's basically a universal standard where the pitch A is equivalent 440 cycles, and that's the case basically throughout the world, with some exceptions. There are some orchestras that tune slightly higher than that—442, sometimes 443—but you can think of that as a universal standard that's being adopted.

- 7.79** Different musical traditions and eras, for example, if you go back into the Baroque, the instruments were not tuned to the same pitch as the instruments of today. They were tuned almost a semitone lower, sometimes more. So the pitch has tended to rise over time. The basic reference of tuning has risen somewhat.
- 7.80** So if you're playing early music on period instruments, on authentic instruments from the period—imagine it's a recorder sonata or a piece for harpsichord or something—then to somebody with perfect pitch, it's going to sound wildly flat because it's at least a semitone lower than it would be if it were played on a contemporary concert grand piano. So that can be disturbing to people with absolute pitch.
- 7.81** They also have difficulty with intervals other than the tempered sounds of the equal-tempered scale. If they have to listen to, let's say, Bulgarian wedding music, they're going to find that disturbing because the intervals are not the same. Somebody without perfect pitch might not even notice.

Lecture 8: Musical Appreciation

At a glance — Musical appreciation, with a recap.

- 8.1** So what remains to be done? Let's recapitulate. We've looked at the notated basis for music, which has allowed musicians and performers to write their musical ideas down in a fixed form and communicate them to other people in other parts of the world. We've looked at the particular language of symbols that has allowed us, through a long process of historical refinement, to fix such nebulous things as pitches to an exact standard, to come up with a concept for notating rhythm that is accurate and precise and flexible.
- 8.2** We've looked at the different ways that we can impart instructions to performers as to what character they have to bring out in a piece of music, the different rhythmic figures that allow us to do that. We've examined the organization of pitches into patterns called scales and keys, key signatures. We've examined phenomena such as the circle of fifths. We've also looked at some practical exercises that will allow you to work on your musicianship to whatever level you desire. That's a grand tour through the fundamentals of music theory, both in applied and in more abstract forms.
- 8.3** I think what remains to be done now is to try to put those different skills together. I said at the end of the last lecture, which was focused on ear training and practical exercises, that there was another level that you could try to get to of musicianship. We looked at rhythmic dictation, rhythmic clapback, melodic dictation, harmonic dictation. We looked at all the different intervals and some ways that you can try to remember them and sing them back accurately.
- 8.4** What you can also do is to develop your capacity to hear more than one thing at a time. This is just one of the most amazing things about music is you can have a coherent framework that allows you to integrate the horizontal and the vertical dimensions—the harmonic and the melodic—into a unified, coherent framework.
- 8.5** And because of our capacity, this extraordinary technology that we've refined over centuries of writing down lines with precise pitches and rhythms, we can have this beautiful thing called polyphony, which allows us to layer multiple different voices, multiple different independent rhythms and melodies into a piece of music.
- 8.6** And through the miracle of things like counterpoint, which codify and, to a degree, regiment these different possibilities, we can have these extraordinary musical textures that are made up of multiple different independent parts. So the next step is to learn to deal with polyphony itself, to learn to deal with that multiplicity of more than one thing happening at the same time.
- 8.7** I don't know of a better way to work on that skill than to work with this book. If you've seen my other courses, you'll recognize this volume. It's always on my piano, no matter

where I'm traveling. I'm also going to talk about it this weekend—I'm doing a course on Bach, where this is going to be the subject of one of the lectures.

- 8.8** So, what this book is—Choralgesänge—a collection of over 300 four-part chorales. The chorales of Bach are short pieces drawn from his cantatas, passions, and oratorios. They consist of Lutheran hymn songs, Protestant plainsong, in other words, that Bach has taken and to which he has added three additional voices. So remember, we talked about the four different fundamental voice types? S, A, T, B: soprano, alto, tenor, and bass. Well, that conveniently gives us a four-part harmonic texture if we have those four voices singing together.
- 8.9** What's ingenious about these pieces—I'll get into the musical significance of them in the Bach course, but for now, we're going to stick to a more practical orientation. What's amazing about these pieces is, first of all, the hymn melodies, the plainsong that Bach started with, were in many cases quite old, sometimes 200 years old by the time he was working on them. They were from another musical era. They don't have the same characteristics as the music of Bach's day.
- 8.10** Remember, we were talking about—and there's a question, actually, a very perceptive question about the difference between medieval music, Renaissance music, how it tends to be less sharply directional, less teleological in its orientation than the music of the high classical style. Well, the melodies in this book, many of them, are not exactly tonal in the way we understand it now, but rather modal. They use scales that are not exactly the same as the major and minor scales.
- 8.11** So they exist in another musical era. But by the time Bach was working on them, the harmony of that period had moved towards a higher degree of what you might call codification. The rules had changed somewhat. The types of harmonic progressions that you would hear were not the same as they would have been 200 years ago. There was an incompatibility, in a sense, between these older melodies and the harmonic language of Bach's day. So there's a very interesting tension in these pieces between these archaic melodies and the then relatively contemporary style within which he was harmonizing them. But we'll leave that whole question aside for now.
- 8.12** The thing that is incredibly useful about this book is that it gives you an inexhaustible supply of four-part writing at the absolute highest level of the art. I haven't seen four-part settings from any other composer that can equal these, and there's hundreds and hundreds of them. So this is an extraordinarily useful resource for anybody who wants to improve their musicianship.
- 8.13** I've made this point before, but I'm going to say it again: this can enhance your musical skills more than anything else I've personally encountered, at least within this musical tradition. So what you can do is you can get the Choralgesänge— there are multiple different editions of it—you can open them up, and if you're very advanced, you can try

to sight-read them at the keyboard, although that's challenging, because you have to be able to play four different independent lines at the same time. It's very hard to do.

- 8.14** Or what you can do is play them just a few beats at a time, maybe just one bar. Work on it really slowly, pull it apart, look at each individual voice, try to sing it, try to play it at the keyboard, and then try to play a second voice that goes along with it, and I'll explain what I mean.
- 8.15** I'm going to take number 188 here from this collection, which is called "Jesu, der du meine Seele." I chose this one not at random, but because it's going to be the subject of one of my Bach lectures, so that allows me conveniently to tie it together with that. So we're going to look at the first couple of bars of this four-part setting, which you can see on the screen behind you. I'm going to point out a couple of things about how it's notated and how it's written.
- 8.16** Because it's a chorale, the principle is that everybody singing— soprano, alto, tenor, and bass—is singing the same words. For the most part, in a texture, that is where everybody is essentially singing the same rhythms. Now, you'll see some little passing notes, some little ornamental passages in some of the voices. So the rhythms are not exactly identical, but it's written in such a manner that you can have everybody singing the same syllables on the same beats, even if some of the notes are stretched out or prolonged with additional little passing notes.
- 8.17** This gives us four complete independent melodic lines that together are coherent both horizontally and vertically. So they work harmonically, and they also form coherent melodic lines. So the text for the first couple of bars here is, "Herr! ich glaube, hilf mir Schwachen." We're going to isolate those two little bars. That can be roughly translated to, "Lord, I believe. Help me in my weakness."
- 8.18** Let's start by pulling apart the four individual lines here. Here's the soprano [plays soprano line]. These hymn melodies would have been familiar to everybody who was a churchgoer at that time. So the exercise and the essence of Bach's artistry is that he could take these very familiar melodies and elevate them to the highest level of the art form through these harmonizations. Let's look at the alto line [plays alto line]. Okay, let's hear the tenor [plays tenor line]. And the bass [plays bass line].
- 8.19** Four independent lines. If we put them together [plays all vocal lines together], we get that. Part of the point of this exercise is to learn to hear that not as an undifferentiated texture, a block of sound, but as four independent voices, each with their own melodic character. Let's try that again. Here's the soprano [plays soprano line]. Here's the alto [plays alto line].
- 8.20** Now, what you can do is you can go through this and choose any random pair of voices. You can say, "I'm just going to isolate the soprano and the alto," or "I'm just going to isolate the altos and the tenors, or just the bass and the soprano, or the bass and the alto voices, and play them together." I'm going to take just the sopranos and the altos

[plays soprano and alto lines together]. Two beautiful melodic lines. Here's the bass and the tenor [plays tenor and bass lines together]. So it's one thing if you're going to do that at the piano.

- 8.21** It's another thing if you have to do it with your voice. An exercise you can do is to try to sing each individual line as best as you can manage. Now, you won't be able to do them literally on pitch because you don't have a two-octave voice range or a three-octave voice range, which you would need to sing something like this, but you can transpose it up or down to the most convenient octave that fits your range, and then try your best to reproduce these lines.
- 8.22** Here's the soprano [plays and then sings soprano line]. Now, here's the alto [plays and then sings alto line]. I'm not doing the words. I should try. I'm going to try [sings alto line, then plays soprano and alto lines together]. Now, where it gets really tricky is, you're going to try to sing one of the lines and play another line at the same time [plays soprano line while singing alto line]. That's hard because you have to do two things at the same time, and you have to have a coherent line in both.
- 8.23** And then what you do is you go through all the voice pairs, right? So then you try to sing the soprano; you switch them around: sing the top and play the alto. Now, I won't be able to do that because it's out of my vocal range, but maybe if I put it down an octave [sings soprano line an octave lower and plays alto line]. It gets a little bit tricky depending on the range of your voice.
- 8.24** This is one of the best exercises I know to develop this skill: understanding polyphony. If you sit down and do this or try to do it, you'll find that it's dizzying. It's like a form of musical gymnastics. You can keep going where you play three of the lines at the keyboard and sing one of them, or you can even play all of them at the piano and sing one of them in unison at the same time. There's all kinds of different ways you can do it.
- 8.25** But you could spend your life with this book and never get to the end of it. It's pretty well inexhaustible. What you will discover as you work through it, you start getting familiar with these little pieces, is that no matter how complicated the harmony gets, no matter how ingenious these four-part settings are, the individual lines are always eminently singable. They're always perfectly conceived in terms of the human voice. And not only that, they fit under the fingers incredibly well as well.
- 8.26** That's one of the extraordinary things about Bach. There's an abstraction to his writing, where it survives transposition from any instrument to any other instrument, pretty much intact. You can play it on anything you want. You can sing it, you can play it on a keyboard, you can have four saxophones play it, you can have a string quartet play it, you can play it on a synthesizer—it doesn't matter; it survives. It's got this extraordinary abstraction and adaptability.

- 8.27** You'll also notice if you spend some time with these pieces that the individual lines are extraordinarily elegant. Let's look at the bass line for this little hymn song [plays bass line]. He truly does go the extra mile because that's a way more interesting bass line than you strictly need to have for a four-part setting of a piece of plainsong. And amazingly, pretty much every line has an equivalent degree of musical interest.
- 8.28** So you really can learn everything from these pieces. They're extraordinary aids for composers trying to understand polyphony, harmony, voice leading, and how to set up an interesting texture, and how to set texts. And in terms of coming to grips with music theory, well, what a lot of students do—this is pretty universally taught—is to either analyze these pieces and figure out what the chords are, assign them a function, a Roman numeral function, in keeping with what we were looking at earlier, the different scale degrees: I, II, III, IV, V, VI, VII, and how they fit into the key of the piece, that's a harmonic analysis that is widely taught.
- 8.29** Or you can isolate a particular line from this piece—it could be just the bass, and you ignore all the other lines, or you might just take the soprano more commonly— and then you try to write the remaining three voices. In order to do that, of course, you need to understand voice leading, which is what we talked about earlier today, the different principles involved in leading a group of different voices together with a coherent texture.
- 8.30** Now, the other thing that these pieces allow you to do is to come to a better understanding of cadences, which we talked about earlier. Remember, we looked at the different types of cadences and how you can have a perfect cadence, an imperfect cadence or a half-cadence, and plagal cadences, and so on.
- 8.31** I want you to look at this phrase, and we're going to try to determine how it ends, what type of cadence? Remember that the cadence signifies the end of a phrase; it's a form of musical punctuation [plays opening phrase]. The end of that phrase is a perfect cadence. How do I know that? Well, I can look at the bass line here—if you look at the screen behind you, on the word "Schwachen," on the end of the second bar there —"hilf mir Schwachen"—you have a bass progression from D to G [plays D-G (below)]. Because this piece is in G minor, if I have that V to I progression in the bass, D to G [plays D-G (below)], that tells me this is a perfect cadence.
- 8.32** It's a cadence that ends on G in a piece that is in G. Let's hear that again [replays the phrase]. Hear that more slowly right at the end [plays final cadence]. There's your perfect cadence. One more time [replays final cadence].
- 8.33** Let's keep going [plays next phrase]. That's a half cadence. This is a cadence from I to V [plays D-G], from D to G [plays D-G]. The first cadence, remember, gives us a perfect cadence [plays V-I cadence: D to G]. And the second cadence is a half cadence [plays I-V cadence: G to D], so a less definitive feeling of close.

- 8.34** And then what you do is you go through the piece, you look at all these different phrase endings, and you try to determine what cadence they are. What is the quality of this close? [plays another phrase] There's another perfect cadence [plays another phrase]. These two phrases that I've just played are very similar to the opening of the piece. They're basically variants of the opening two phrases.
- 8.35** In the following phrases, things get a little bit different [plays next phrase]. Now, what you might have noticed is that this phrase seems to end in F, which is a little strange because we're meant to be in G minor. But this is, again, the strange characteristic of these melodies—they wander; they're not as sharply oriented towards a tonic. They're not as sharply directional as a piece that was originally written in this era would have been, the era of Bach. Remember, these melodies are much older than that.
- 8.36** That means that these pieces will wander sometimes from key to key and then back again. Remember, we started here [plays phrase in G minor]. Then in this phrase [plays phrase ending in F major], we end up somewhere else. That's a perfect cadence, but it's not a perfect cadence in G; it's a perfect cadence in F. Now, let's hear the next phrase [plays phrase ending in B $\flat$ major], perfect cadence in B $\flat$. Now we're winding back home. This is now a half-cadence in G minor [plays phrase ending in G minor]. And the final phrase [plays final phrase ending in G major]. If you've done your ear training exercises, you've noticed that the piece doesn't end in minor; it doesn't end in G minor, but in G major.
- 8.37** Why is that? We started like this [plays phrase in G minor], and we end with this chord [plays G major chord]. That has a term which is a Picardy third, and what a Picardy third is, or "Tierce de Picardie," is simply the habit that composers had in the Baroque, and to a degree also in the Classical eras, of ending a piece in a minor key with a major third rather than a minor third.
- 8.38** The reason for this, historically, is that the minor third is a slightly more dissonant interval than the major third. Especially if you were playing in those older tuning systems before equal temperament, the minor third could be quite a chaotic interval. So, in order to end the piece with a reasonably stable sensation of close, you would end with a major third rather than a minor third.
- 8.39** Let's try to put this piece together, and I want you to pay attention this time, not only to this four-part texture that we've been pulling apart, but also to the qualities of the cadences and the way the piece moves around. We have a perfect cadence in G, a half-cadence in G. Then the same thing repeated. Then we move to F, we move to B $\flat$, then we eventually move back to G. It moves around quite a lot [plays the chorale]. And thus concludes the music theory lesson. Any questions?
- 8.40** So first of all, thank you for breaking down the class. It was really insightful. Now that we've learned quite a bit about music theory, I'm just curious, what's your piece of advice for someone to be able to take in all the theory while still experiencing music as

an experience? Or on another note of that question, for someone who hasn't really been exposed to classical music, for example, or things like that, how would you suggest that they can learn to appreciate or experience the music properly, if that makes sense?

- 8.41** I don't know about properly because music is an intensely personal phenomenon, and different people have completely different relationships with this art form. So it depends. It depends on what you want that relationship to be like. But it's an interesting question because when you're learning these things, it can be dizzying, and it can take you outside of the fluency that you thought you had previously, where you would sing a melody and not be thinking about it—you're not thinking analytically in terms of intervals, in terms of cadences, in terms of phrase structure. You're just reproducing it.
- 8.42** The way that a native English speaker might not need to preoccupy themselves with questions of syntax and word order, you're not thinking about the parts of speech when you are having a conversation with somebody. You're not going verb, object, and so on, adjective. It's the same in music. You can learn these things—you can learn the terminology, and it will certainly help, but in the act of making a piece of music or performing it, you probably aren't thinking analytically.
- 8.43** If you're playing a piece for somebody, all of that falls away, sinks into the background, and you're focusing on what you're communicating, what you're trying to express to the person you're playing for. So some of this can be thought of as skills that you work on that you attempt to absorb so that they become second nature, and then you don't think about them.
- 8.44** You can, of course, work on them continuously. I personally start every working day by playing something at the keyboard. I do that all the time. That's like a form of warming up. It's like a form of exercise. But the idea is to not have to think about it all the time, not to be reaching for the terms, not to be thinking analytically.
- 8.45** The exercises that we've looked at help you to become familiar with the language of music so that you can write it down, you can communicate it, and you can read it. You can understand what you're hearing. But at a certain point, it becomes instinctive, it becomes habitual, just like learning a foreign language. The exact way that that plays out, of course, is entirely individual, but it will eventually sink into the background.
- 8.46** I don't really know how to formulate this question, but I noticed that different people with different personality types tend to like different types of music. Let's say people who care about a lot of abstract thinking and so on, they like abstract music. People that are a bit more, let's say, living simple lives, they also usually listen to very simple music. Is that a common thing that as your personality changes or develops, or let's say as you develop as a person, you start listening to other music, the taste of music, what sounds good and what doesn't sound good changes?

- 8.47** Obviously, I'm not a psychologist, so I'm not going to venture into waters that are not mine. But I can make one comment about this, which is the result of just observing my own relationship with music. I'll start there.
- 8.48** I did take the OCEAN test once to see what my personality was like, the Big Five personality traits. You can do these tests that will give you a score on all of those different parameters. I did that test, and one of the things that it revealed was in terms of openness, I'm way off the charts, like 99.9% or something, which I suppose wasn't surprising, but it explained one thing, which is that my relationship with music is one of continuous expansion. I'm voraciously curious about the musical phenomenon. I'm constantly hearing new things.
- 8.49** I'll approach any genre, any historic period, any geographic region with dispassionate interest at first, which is to say I won't go at it with any a priori judgment. I won't even be thinking about whether I like it or I don't like it. I'm not interested in that question when I first start listening. I'm just thinking, "Okay, what is this? Let's give it a listen." Then you start listening. It might take three or four times to really get a sense of what it is—still not thinking about whether I like it or not. I don't care about that question very much.
- 8.50** What you find is that if you're extremely high in openness, you're going to be naturally curious about lots of things, and you're going to want to continuously expand into waters that you don't already know. But obviously, statistically, not everybody is on the extreme outer edge of openness.
- 8.51** It also makes it so that I have a very high tolerance for innovation, novelty, exploration, and experimentation in music. I'll listen to things that 99% of the population would find unlistenable, but I'm interested in lots of things, and I have to nourish that curiosity and that desire to explore. But I recognize that on that dimension, I'm not typical. That seems to be a function of personality, at least to a certain extent.
- 8.52** I know people who have other personality traits, but they're not particularly high in openness. For them, it can be very, very difficult and just frankly, unappealing to sit down and listen to a bunch of things they're not familiar with, even if it's only slightly outside of what they're used to. You can push them to a degree and say, "Try this, listen to this, see what you think." Sometimes they will, but it's not really in their nature.
- 8.53** Also, the other thing is a lot of people have a relationship with music that seems to be—I find this very interesting—seems to be predicated on nostalgia, where a piece of music will evoke a particular time in their life. It's like, "Oh, yeah, that time I was 16. This is what I was listening to then." They find it enjoyable to go and listen to that again because it helps them to maybe relive something, a pleasant memory or a part of their life that they would like to remember.

- 8.54** I don't really have that relationship with music exactly, partly because I'm a professional and I'm listening to so many things all the time, but that seems to be where it's at for a lot of people, where they want to stick with things that have that biographical resonance for them, things that remind them of an experience they had or a time in their life. Then you can get stuck there.
- 8.55** There's guys who just listen to oldies on the radio all the time for decade after decade. They just want to hear the Eagles or whatever. I understand that. It's not my experience of it, but I think it's very easy to get caught on things that work, that you enjoy, and that you know what to expect. I would never in a million years judge that because God knows life is hard, right? You don't necessarily want to torture yourself with more unfamiliar, uncomfortable experiences than you need to.
- 8.56** But the corollary to that, though, is that you might also be cheating yourself of the opportunity to expand into things that might really enrich your life. So I'd always try to encourage people to get into the habit of listening to things that you're not familiar with. We all know what the gaps in our culture are, the things that we know we might find interesting, but we don't take the time. I certainly do. I've got a long list of books that I'd really like to read that I haven't.
- 8.57** The advantage of music is that it takes a lot less time to listen to, even a very long piece of music, than it does to read, say, a 600-page novel. The time commitment is actually not that considerable, unless you want to sit down and listen to the entire Ring cycle by Wagner, which will take you a long time. But otherwise, you can get a pretty good idea of a lot of different musical styles by sampling them; you just dip into them. You listen to one movement of a piece. You don't even listen to the whole thing. Don't listen to all four movements of a Mozart symphony. Just listen to one movement. That could take you three minutes. That's a pretty low cost of entry time-wise, and it's something that you can pretty easily fit into your life.
- 8.58** So to sum up the question: to a degree, it's a function of personality, definitely. Some people will naturally just be voracious listeners and always wanting to hear everything. Others might have to push a little bit in order to develop that as a habit.
- 8.59** I'm just thinking about the construction or composition of some of these, like the four-part harmonies, specifically, because it seems like it would just turn into this maddening Rubik's Cube of three of them working in one not. Do you have a sense of what the typical process is? Do they start with the soprano line and then the bass, I guess, would create the fundamental support for it, and then the alto and tenor would fill it in? Or is it more this intuitive process of hearing it as complete harmony altogether once you've amassed a level of experience and expertise with it? Yeah, I'm curious about the process of composing itself.
- 8.60** We don't know exactly how Bach did it because he was in the habit of destroying his sketches. He didn't save them. There's only a few fragmentary things here and there

where there's little pencil indications on some of his ink scores. But for the most part, we have very little insights into his process of composition.

- 8.61** But as far as how students typically learn to do this, you break it down into steps. There's no other way to do it because it's a very complex task. The first step is to isolate the top line—you don't look at Bach's four-part setting. You separate the top line out, which is the original Lutheran hymn. Then you look at it as a phrase, and you look at the cadences. You start there. You look at the ends of the phrases.
- 8.62** That's easy to do because if you look at this sheet music here, at this score, you'll see that every now and then, you come to the end of a phrase, and there's a little symbol above the note—as you can see here, that little hat over the G, which is called a fermata. The fermata indicates a pause or the end of a phrase. So all the four-part chorale settings of Bach make this very clear by having a fermata at the end of the phrase where the cadences are.
- 8.63** So in this case here [plays opening soprano line of "Jesu, der du meine Seele"], and then there's a fermata on the last note [plays G] telling you that's the end of the phrase. If you wanted to make a four-part setting, you would first observe where the cadences are, and based on the trajectory of the line and where it seems to be going harmonically, you make a decision. Am I going to end with a half-cadence, a perfect cadence, a plagal cadence, a deceptive cadence, or something else?
- 8.64** Often, there aren't that many possibilities. There aren't a lot of viable possibilities for interpreting this in terms of a cadence. You can try them at the keyboard, and you'll quickly see that the perfect cadence is the one that fits. So then what you do is you start by identifying the cadences. You say, "This is going to be a perfect cadence in G, and this is going to be a half-cadence in G."
- 8.65** You don't write a single note. You're just identifying the cadences and the tonality, the key of the piece. You take note of any modulations where you move to a different key. You observe, "Oh, here we seem to have a perfect cadence in F. We're moving towards F there. Then we seem to be moving to B $\flat$," and so on and so forth. So you start with a little harmonic and phrase analysis of the piece.
- 8.66** Then, once you've determined that, you just write the bass at the cadential points. If your line is doing this [plays opening soprano line of "Jesu, der du meine Seele"], then you just write those two bass notes that allow you to affirm what the cadence is [plays D–A, G–G], and you do that throughout the piece. So then what you've got is this little fragmentary bass that only has a couple of notes at a time.
- 8.67** The next step is to write a complete bass line. Based on that, what you've done is you've chosen your points of orientation. You know how the phrases end, you know what the harmonic context is, you know what key you're in, you know if it's a half cadence or a perfect cadence and so on.

- 8.68** Then you try to write a line that works with the soprano, observing all the principles of voice leading that we looked at earlier. So you avoid forbidden parallels, you avoid crossing voices, you try to make it so that you're mostly moving by step with some leaps to make the line interesting as necessary, but mostly in conjunct motion, and so on and so forth.
- 8.69** Once you've got a complete bass line that you're satisfied with, then you can start to think about the inner voices. You might begin by only harmonizing the cadences and then filling in the rest of the lines. You start by understanding the structure, the function, the harmonic function, and the phrase structure. If you don't understand that, you won't be able to write a note. You have to have a sense of what key you're in and so on.
- 8.70** Then, once you have a sense of that, you start dealing with, as you said, the whole Rubik's Cube of fitting the voices together. At first, it's extraordinarily difficult because at any given juncture, there are 30 possibilities. But what you learn, just as in chess or indeed with Rubik's cubes—my daughter's really into those. I learned that there's actually a method, by the way, for solving them. There are competitions where people can solve a Rubik's Cube in 10 seconds. It's completely mind-boggling.
- 8.71** Anyway, what you have to figure out is that any move you make has consequences downstream of that note. If you set up a chord in a particular way, then it locks you into the next set of possibilities. So if you choose something that doesn't work, then even if that chord is by itself coherent, it's going to lead you into something where you get completely jammed up, and it doesn't work. You have to think multiple moves ahead.
- 8.72** Yeah, it's a complex task, but thousands of music students learn to do it. It's hard, but it's possible. One of the things that studying music has taught me is that there's a very big difference between extremely difficult and impossible. Extremely difficult is doable.
- 8.73** You said earlier that you're very high in openness. I also took the test, and I'm also very high in openness—I think it's like 96 or something. I was wondering if you have some artists or types of music that you've really enjoyed a lot that are really outside the box.
- 8.74** Oh, yeah, sure. Definitely. When I was first starting to study music, the stranger and the more out there, the better, as far as I was concerned. It was all kinds of things. I got very interested, particularly, in French and German avant-garde music of the 1950s. There were a bunch of composers from that era that this post-war generation, born in the 1920s, that I really gravitated towards.
- 8.75** There was an era, historically, that is completely singular in the history of music, which is the immediate post-war. There was this sense among the young artists at that time that civilization had narrowly escaped a complete cataclysm; it had very nearly just completely fallen to the ground. They were trying to come to grips with this.

- 8.76** One of the things that occurred to them was, to paraphrase Theodor Adorno, it's impossible after Auschwitz to write poetry in a 19th-century style. Once these things have blown up in the world, you don't see the world in the same way. A lot of the younger composers, born in the 1920s, who experienced the war firsthand and experienced the aftermath of it in all the big European capitals, were faced with this problem: Do you return to the aesthetics of the late 19th century?—which for them was a world that was gone; you can't go back there again. Or do you move forward with something else? And if you move forward, then what is that something else?
- 8.77** They developed these approaches that were marked by a technological utopianism, let's say, this belief in the possibilities of the future, the possibility that you could build maybe a new and better society out of the rubble of the old. For most of human cultural history, the pace of change was extremely slow—you wouldn't have really noticed much difference from one generation to the next, culturally and musically speaking. Things would be largely stagnant for centuries at a time, where if there was any change in the technical fabric of music, it would happen so slowly that you wouldn't really notice it over your lifetime.
- 8.78** It starts to speed up dramatically in the 20th century, and it produced all of these amazing things—where this had never happened before—where suddenly there's electronic music. And suddenly you can experiment with entirely new forms, entirely new types of harmony. You can produce microtones on instruments. You can do all kinds of crazy things. There's this excitement in the air about all of that, and also, along with it, this desire to turn away from a catastrophe into something else. So there is an extraordinary creative ferment in the air.
- 8.79** I was very interested in the music of that period, the composers such as Karlheinz Stockhausen, who is quite well known in Germany and internationally as well. The Beatles were huge fans of his work. In France, a figure like Pierre Boulez, B-O-U-L-E-Z, was another musician that I was very interested in. As far as pop music went, anything that was exploding with creative potential, where everything was new, stimulating, and unexpected, that's what I gravitated towards. So there's my answer.
- 8.80** Why isn't the grace note just written like a regular note with a link to the next one?
- 8.81** Because it's not played the same way. There's a couple of distinctions—some of the distinctions are functional or technical in nature. The grace note allows you to put an accent on a dissonant tone, for example, and immediately resolve it. This has to do with the grammar of music and the rules for resolving dissonances.
- 8.82** But you could start a piece like this [plays ornamental B $\flat$ -D leading to A, G], where the upper note that I started on is an accented passing tone or an accented dissonance [plays ornamental B $\flat$ -D leading to A, G], and it's resolved on a weak beat. And in the Baroque here, you might have written that as a grace note, so it was not to have the dissonant tone appear to fall right on the beat. So there's a technical reason for that.

- 8.83** But also in terms of performance, it's not the same thing. A grace note or a string of grace notes implies a rhythmic flexibility that can be very difficult to obtain if you write it with a precise rhythmic value.
- 8.84** The third part of it is that in Baroque music, the ornaments were not always written in the score. They were often, in fact, not given, and it was expected as a part of the performance practice of the day that the performer would embellish this skeletal melodic line with ornaments, and the performer would know how to do that because they would have absorbed this whole repertoire of ornamental figures. So there are historical, technical, and aesthetic reasons for the use of grace notes.